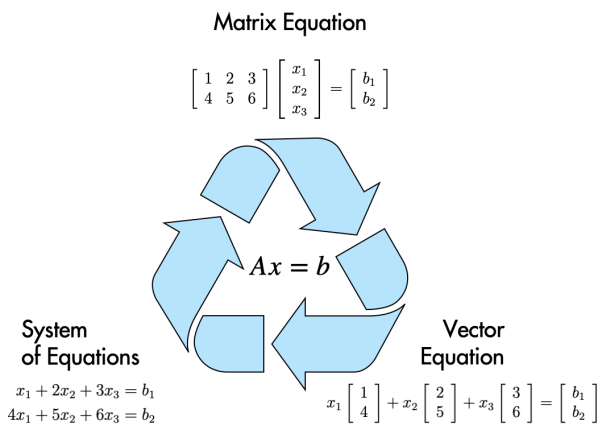
The Linear Algebra Piece of CIS-2166

Fall 23

David Dobor



At the top corner of this picture is a matrix equation of the form $A\boldsymbol{x} = \boldsymbol{b}$. Spelling out its meaning, you will get the system of linear equations shown in the bottom left corner. And if you pull out the dependence on x_1, x_2, and x_3 as coefficients of vectors that are the columns of A, then you get the vector equation in the bottom right corner, which asks this question: What linear combination of these columns equals $\boldsymbol{b}$?

A matrix equation, a system of equations, and a vector equation are really saying the same thing.

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Part I

Linear Systems

1 Row and Column Pictures of Linear Systems

Example 1. To solve the system of linear equations $\begin{aligned} x_1 + 7x_2 &= 6 \\ 3x_1 + 5x_2 &= 2 \end{aligned}$, you might proceed as follows:

1. (*eliminate*) Keep the first equation intact and add -3 times the first equation to the second. This eliminates x_1 from the second equation:

$$\begin{aligned} x_1 + 7x_2 &= 6 \\ 3x_1 + 5x_2 &= 2 \end{aligned} \quad \rightarrow \quad \begin{aligned} x_1 + \ \ 7x_2 &= 6 \\ -16x_2 &= -16 \end{aligned} \quad \rightarrow \quad \begin{aligned} x_1 + 7x_2 &= 6 \\ x_2 &= 1 \end{aligned}$$

2. (*write down the solution*) The new second equation tells us that x_2 must equal to 1. Substituting $x_2 = 1$ into the first equation gives $x_1 = -1$. So the

$$\text{solution to the system} \quad \begin{aligned} x_1 + 7x_2 &= 6 \\ 3x_1 + 5x_2 &= 2 \end{aligned} \quad \text{is } \boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

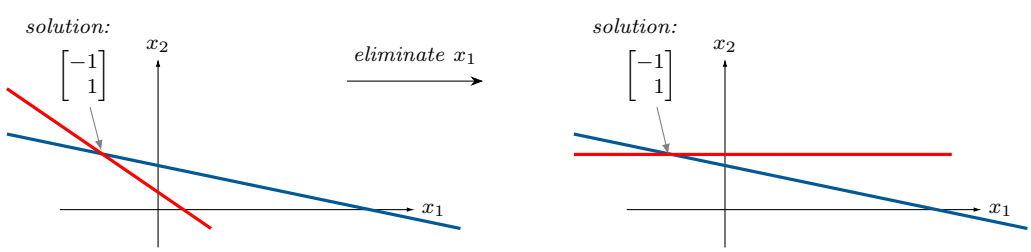


Figure 1.1: *Row Picture*: Adding a multiple of one row to another doesn't change the solution. Eliminating x_2 from the second equation (red line) amounts to pivoting the red line around the solution, $(x_1 = -1, x_2 = 1)$.

The Row Picture

row picture: equations as lines, planes

The so-called"row picture" of the system recognizes that each equation represents a line. The solution to the system is the intersection point of the two lines. Figure 1.1 shows the row picture of the system in Example 1.

The Column Picture

column picture: $\boldsymbol{b}$ as a linear combination of A's columns

The so-called"column picture" takes a different point of view of the system. If you write the system in its matrix form, that is,

$$\text{write} \quad \begin{aligned} x_1 + 7x_2 &= 6 \\ 3x_1 + 5x_2 &= 2 \end{aligned} \quad \text{as} \quad A\boldsymbol{x} = \begin{bmatrix} 1 & 7 \\ 3 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 6 \\ 2 \end{bmatrix} = \boldsymbol{b},$$

then to find $\boldsymbol{x}$ that solves $A\boldsymbol{x} = \boldsymbol{b}$ is to answer this question: *"What linear combination of A's columns produces the right-hand side $\boldsymbol{b}$?"*

$$A\boldsymbol{x} = \begin{bmatrix} 1 & 7 \\ 3 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad \text{is written as} \quad x_1 \begin{bmatrix} 1 \\ 3 \end{bmatrix} + x_2 \begin{bmatrix} 7 \\ 5 \end{bmatrix} \quad \text{must equal} \quad \begin{bmatrix} 6 \\ 2 \end{bmatrix} = \boldsymbol{b}.$$

So, what are the x_i's that make $x_1 \begin{bmatrix} 1 \\ 3 \end{bmatrix} + x_2 \begin{bmatrix} 7 \\ 5 \end{bmatrix} = \begin{bmatrix} 6 \\ 2 \end{bmatrix}$? Only $\boldsymbol{x} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$ makes this happen:

$$(-1) \begin{bmatrix} 1 \\ 3 \end{bmatrix} + (1) \begin{bmatrix} 7 \\ 5 \end{bmatrix} = \begin{bmatrix} 6 \\ 2 \end{bmatrix}.$$

Figure 1.2 shows the column picture view of this system.

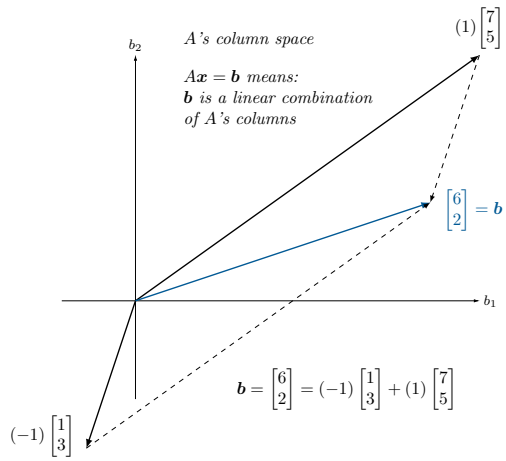


Figure 1.2: *Column Picture*: To solve $A\boldsymbol{x} = \boldsymbol{b}$ is to find the coefficients x_1 and x_2 that make the linear combination $x_1 \cdot$ column-1-of-$A + x_2 \cdot$ column-2-of-A equal to the right-hand side $\boldsymbol{b}$. For the A of Example 1, $x_1 = -1, x_2 = 1$ are just right to make this happen.

Example 2. This system has two equations and three unknowns: $\begin{matrix} x_1 + x_2 - x_3 = 1 \\ 2x_1 + x_2 + x_3 = 6 \end{matrix}$; its matrix form is

$$A\boldsymbol{x} = \begin{bmatrix} 1 & 1 & -1 \\ 2 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 6 \end{bmatrix} = \boldsymbol{b}.$$

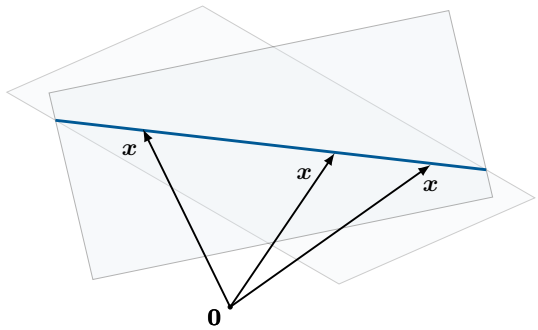
☞ A's columns are 2-tall vectors. We denote the space of vectors with 2 components by $\mathbb{R}^2$.

A's rows are 3-wide vectors. Our notation for vectors with 3 components is $\mathbb{R}^3$. A's job is to send the 3-tall vectors of the row space to the 2-tall vectors of the column space.

Unlike the 2-by-2 system of Example 1, this system has infinitely many solutions. Let's see how this is revealed by its row an column pictures.

The Row Picture

A's rows are planes in 3-dimensions. The intersection of those planes is a line[1]. A has infinite solutions because any point on the line solves $A\boldsymbol{x} = \boldsymbol{b}$.



The row picture of the system

$$\begin{bmatrix} 1 & 1 & -1 \\ 2 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 6 \end{bmatrix}.$$

The two equations determine planes that intersect along the blue line, and any point along that line solves the system.

Figure 1.3: *The row picture* of the 2-by-3 system of Example 2. Any point on the blue line is a solution to the system.

This system has infinitely many solutions. $\boldsymbol{x} = \begin{bmatrix} 3 \\ -1 \\ 1 \end{bmatrix}$ is one solution, $\boldsymbol{x} = \begin{bmatrix} 5 \\ -4 \\ 0 \end{bmatrix}$ another.

All these solutions are points in the 3-dimensional space $\mathbb{R}^3$, and they all lie along the blue line in Figure 1.3. We'll have a clean algebraic way of writing them down when we look at a systematic way of solving such systems, using what's called Gaussian elimination.

The Column Picture

The column space of the system is spanned by the columns of $A = \begin{bmatrix} 1 & 1 & -1 \\ 2 & 1 & 1 \end{bmatrix}$. Those columns are 2-tall and so they live in the 2-dimensional space $\mathbb{R}^2$. Figure 1.4 shows what the column space looks like.

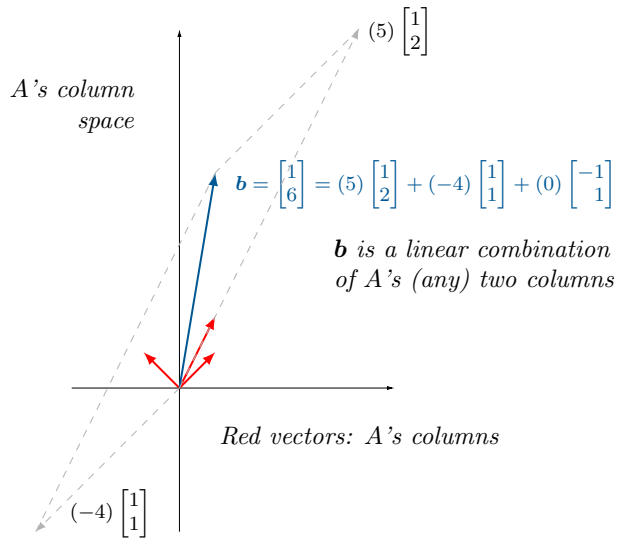
[1] This is true for the system in this example. However, if the system is such that the planes are parallel, then the system has no solutions. In the case when the two planes coincide, the system again has infinite solutions.

Figure 1.4: *The column picture* for the 2-by-3 system of Example 2.

A is 2-by-3:

$$A = \begin{bmatrix} 1 & 1 & -1 \\ 2 & 1 & 1 \end{bmatrix}.$$

A's columns live in the 2-dimensional space $\mathbb{R}^2$. Moreover, the columns do *not* lie on the same line. So any two of A's columns can be used to express *any* $\boldsymbol{b}$ that is also in $\mathbb{R}^2$.



To solve $A\boldsymbol{x} = \boldsymbol{b}$ is to answer this question: *"What linear combination of A's columns produces the right-hand side* $\boldsymbol{b}$*?"* This particular system is solvable for any right-hand side $\boldsymbol{b}$.

Example 3. Here's a system with three variables and one equation:

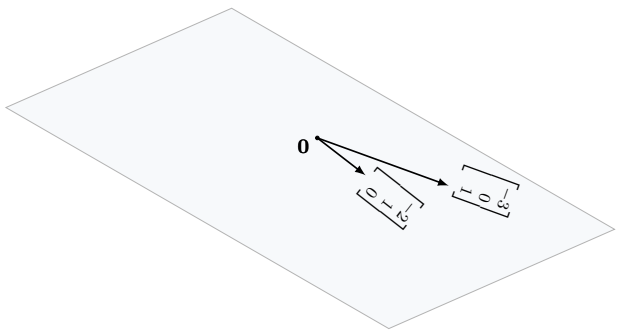
$$x_1 + 2x_2 + 3x_3 = 0.$$

As in Example 2, we have more unknowns (variables) than constraints (equations) imposed on how the unknowns can vary, so we expect an infinite number of solutions here too. This time, the solutions fill the entire plane in $\mathbb{R}^3$, not just a line.

You can get the solutions by setting any of the two variables to any values and calculating the value of the third variable, which will then be uniquely determined. For example, if we set $x_2 = 1$ and $x_3 = 0$ then x_1 must equal -2. Similarly, if we set $x_2 = 0$ and $x_3 = 1$ then x_1 must equal -3.

The two vectors $\begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix}$ generate the plane of solutions to $x_1 + 2x_2 + 3x_3 = 0$. By

"generate" we mean that we can write all solutions to $x_1+2x_2+3x_3=0$ as $c\begin{bmatrix}-2\\1\\0\end{bmatrix}+d\begin{bmatrix}-3\\0\\1\end{bmatrix}$, where c and d are understood to be some numbers.



The row picture of the system

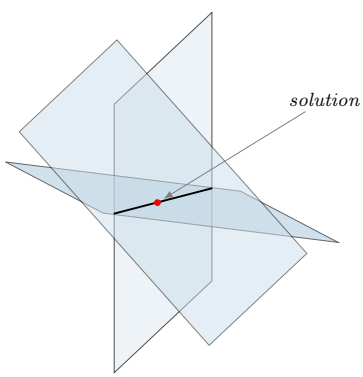
$$\begin{bmatrix}1 & 2 & 3\end{bmatrix}\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix}=\begin{bmatrix}0\\0\end{bmatrix}.$$

Any point on this plane that goes through the origin is a solution to it.

Any combination of the two vectors shown here is a solution.

Figure 1.5: *The row picture* of the 1-by-3 system of Example 3. Any point on the plane solves it.

Example 4. One last example: the 3-by-3 system $\begin{aligned}x_1+x_2-x_3&=1\\2x_1+x_2+x_3&=6\\x_1-x_2+2x_3&=6\end{aligned}$ has a unique solution, $\boldsymbol{x}=\begin{bmatrix}3\\-1\\1\end{bmatrix}$.



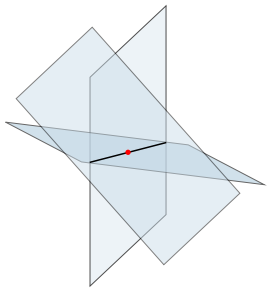
The row picture of the system

$$\begin{bmatrix}1 & 1 & -1\\2 & 1 & 1\\1 & -1 & 2\end{bmatrix}\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix}=\begin{bmatrix}0\\0\\0\end{bmatrix}.$$

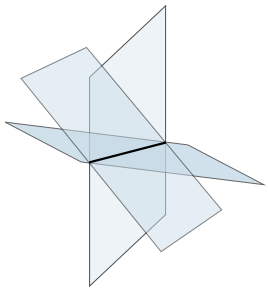
The only solution to it is the single point where the three planes intersect.

Figure 1.6: *The row picture* of the 3-by-3 system of Example 4. Each equation is a plane in $\mathbb{R}^3$, and the three planes intersect at a single point.

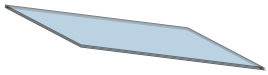
Generally, a 3-by-3 system will have either 1 solution, no solution, or an infinite number of solutions. The various ways the planes' positions can relate to one another is shown below.



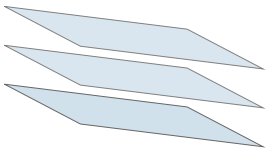
A single solution: Two planes intersect along a line and the third plane cuts through that line.



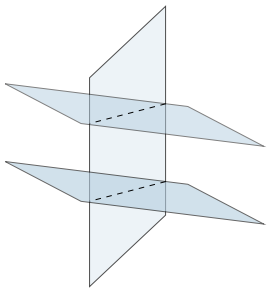
An infinite number of solutions: All planes intersect at a single line.



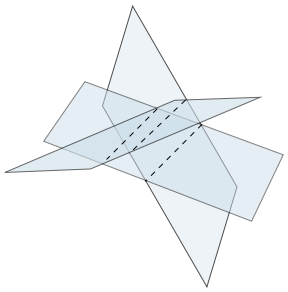
An infinite number of solutions: The planes coincide.



No solution



No solution



No solution

We drew only some of the possibilities here. Others a possible. For example, two planes coincide and the third is parallel to them – the system has no solution in this case. Or, two planes coincide and the third intersects them along a single line – the system has infinite solutions.

As you look at all the different ways the three planes can be configured in $\mathbb{R}^3$, a clear picture emerges:

> A system has either no solution, one solution, or infinitely many solutions. There's no way to have any other number of solutions.

1.1 Takeaways & What's Next

In this lecture, we saw examples of a few small systems. We could guess the solutions to some of them, but we don't yet have a systematic way of finding the solutions - that's coming next when we discuss Gaussian elimination.

We emphasized two different viewpoints of a system of equations: the row view and the column view.

A's rows represent lines or planes (generally, subspaces) in $\mathbb{R}^3, \mathbb{R}^2$, or $\mathbb{R}$ (generally, in $\mathbb{R}^n$). Solutions to the system depend on the relative positions of the spaces determined by the equations in the system.

Next time, we'll see a methodical approach to finding the solutions to a system. The method is called Gaussian elimination.

Gaussian elimination manipulates the rows of A without changing the solutions to the original system, simplifying A so that solutions can be easily read from it. This is precisely what we did in Example 1 when we pivoted the red line in Figure 1.1 without changing the solution. Generally, the Gaussian elimination steps can be thought of as repositioning the planes determined by the system of equations in a way that doesn't change their intersection point (the solution). So keep the row picture in mind as we work through the Gaussian algorithm.

1.2 Exercises

Exercise 1.1. Write the following problem in matrix form and solve it: "X is twice as old as Y and their ages add to 39."

Exercise 1.2. In Example 3 we claimed that the vectors $\begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix}$ are just two solutions to the system $x_1 + 2x_2 + 3x_3 = 0$. We also claimed that this system has infinitely many solutions, and that all of them lie on the plane drawn in Figure 1.5.

Pick any two numbers, c and d, that make you happy and verify that indeed

$$\boldsymbol{x} = c \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + d \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix}$$

is a solution to the system. As you vary c and d, you get all the points on the plane in Figure 1.5.

Exercise 1.3. We never drew the column picture of the system in Example 4. What does the column picture say for that system? Can you draw it?

1.3 Answers to Selected Exercises

1.1 *Remark:* Finding the answer is not the point of this exercise – we know you can figure out the answer to this question. You are asked to convert the question into a matrix form and then solve the resulting small system of linear equations.

1.2 Indeed, take, say, $c = 3$ and $d = -4$. Then

$$\boldsymbol{x} = (3)\begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + (-4)\begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 3 \\ -4 \end{bmatrix}.$$

This $\begin{bmatrix} 6 \\ 3 \\ -4 \end{bmatrix}$ lies on the plane of solutions (the one in Figure 1.5): Plugging in these coefficients into the system $x_1 + 2x_2 + 3x_3 = 0$, we get

$$A\boldsymbol{x} = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix}\begin{bmatrix} 6 \\ 3 \\ -4 \end{bmatrix} = 1 \cdot (6) + 2 \cdot (3) + 3 \cdot (-4) = 0,$$

verifying that it is indeed a solution. Any other c, d pair also produces a solution that lies on the same plane (and there are no other solutions).

2 Column Space and Null Space

A matrix is a function. It may seem strange at first to think of a rectangular grid of numeric entries as a function, but it is.

In a perhaps more familiar notation for functions, people often use symbols like f, g, and so on. For example, the symbol f in $f(x) = 2\,x$ stands for a function that takes one number as input and outputs another number equal to twice the input.

2.1 Matrices Map Vectors to Vectors

Matrices work the same way. The difference is that matrices generally take vectors instead of numbers as input and output vectors. The input and output vectors need not be of the same size.

A matrix is a linear transformation

Instead of the word "function," it's common practice to use the term "transformation" when referring to matrices – the two terms are synonymous. Moreover, matrices are *linear* transformations. We'll define linearity precisely soon. The key takeaway from our upcoming discussion is this:

> **Key Point** Every matrix is a linear transformation, and every linear transformation can be represented by some matrix.

Example 5. The 2-by-3 matrix $A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$ takes 3-tall vectors like $\boldsymbol{x} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ to 2-tall vectors like

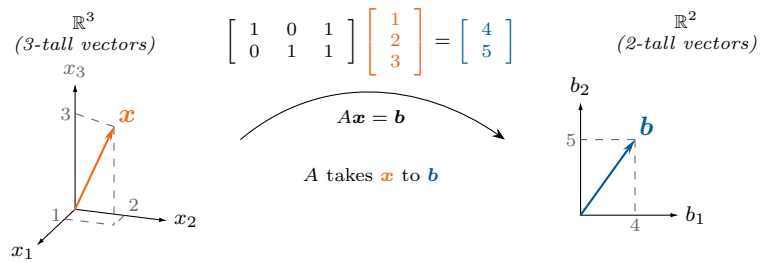


Figure 2.1: a 2-by-3 matrix A is a linear transformation from $\mathbb{R}^3$ to $\mathbb{R}^2$.

$\boldsymbol{b} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$. True: multiply A times $\boldsymbol{x}$ to get $\boldsymbol{b}$.

Figure 2.1 shows A in action: A squashes the 3-dimensional vectors of $\mathbb{R}^3$ into plane $\mathbb{R}^2$ vectors.

Any 2-by-3 matrix takes vectors from $\mathbb{R}^3$ to $\mathbb{R}^2$. Generally, any m-by-n matrix A is a transformation of $\mathbb{R}^n$ into $\mathbb{R}^m$ We write this as:

$$A : \mathbb{R}^n \to \mathbb{R}^m \qquad \text{(where } A \text{ is } m \times n\text{)}$$

2.2 The Column Space of A

We think of $A\boldsymbol{x}$ as a linear combination of the columns of A. The entries in $\boldsymbol{x}$ are the coefficients that "weigh" A's columns.

Here's what we mean. Looking back at the A and $\boldsymbol{x}$ of Example 5, we have

$$A\boldsymbol{x} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = 1 \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 2 \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} + 3 \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \end{bmatrix} = \boldsymbol{b}$$

Thus, the $\begin{matrix} 1, \\ 2, \\ 3, \end{matrix}$ entries of $\boldsymbol{x}$ are the coefficients by which we "weigh" A's columns. And these entries of $\boldsymbol{x}$ make up "just the right combination" of numbers by which to weigh A's columns to produce the right-hand-side $\boldsymbol{b}$.

All linear combinations of A's columns make up the colum space of A.

If instead of $\begin{matrix} 1, \\ 2, \\ 3, \end{matrix}$ we use arbitrary $\begin{matrix} x_1, \\ x_2, \\ x_3, \end{matrix}$ we get all linear combinations of A's columns:

$$A\boldsymbol{x} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_1 \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} + x_2 \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} + x_3 \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

By varying the $\boldsymbol{x}$'s, we get 2-tall vectors that make up what we call the *column space* of A.

To solve the system of equations $A\boldsymbol{x} = \boldsymbol{b}$ is to answer this question:

Does the vector $\boldsymbol{b}$ belong to A's column space?

In other words:

Can we find a vector $\boldsymbol{x}$ such that the linear combination of A's columns with $\boldsymbol{x}$'s entries as the coefficients produces $\boldsymbol{b}$?

Example 6. We write the system $A\boldsymbol{x} = \boldsymbol{b}$ with $A = \begin{bmatrix} 1 & 1 & -1 \\ 0 & 1 & 1 \end{bmatrix}$ and $\boldsymbol{b} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$ as

$$\begin{bmatrix} 1 & 1 & -1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \end{bmatrix},$$

with the understanding that it corresponds to the system of equations

$$\begin{cases} x_1 + x_2 - x_3 = 4 \\ \qquad x_2 + x_3 = 1 \end{cases}$$

The solution to this system is $x_1 = 1, x_2 = 2$, and $x_3 = -1$. In our column-picture view, this means that these x's are the coefficients in the linear combination of A's columns that gives rise to the right-hand side $\boldsymbol{b}$:

$$x_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + x_3 \begin{bmatrix} -1 \\ 1 \end{bmatrix} =$$

$$1 \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 2 \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} + (-1) \cdot \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$$

☞ Notice, however, that $x_1 = 1, x_2 = 2$, and $x_3 = -1$ is <u>not</u> the only solution to this system. Another triplet of x's that solves the system is $x_1 = 0, x_2 = 5/2$, and $x_3 = -3/2$. In fact, there are infinitely many such triplets. We'll see a systematic way of writing down all solutions to linear systems in the next lecture.

2.3 The Null Space of A

If a and x stand for numbers, and $x \neq 0$, then the only way to make $a\,x = 0$ is to have $a = 0$. Matrices behave differently: you can have $A\boldsymbol{x} = \boldsymbol{0}$ and still have both A and $\boldsymbol{x}$ be non-zeros.

Example 7. The matrix $A = \begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix}$ can act on a non-zero vector $\boldsymbol{x}$ and output a zero vector. For

example, if $\boldsymbol{x} = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$, we get $A\boldsymbol{x} = \mathbf{0}$:

$$A\boldsymbol{x} = \begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix} \begin{bmatrix} 3 \\ -1 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + (-1) \begin{bmatrix} 3 \\ 6 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \mathbf{0}.$$

Neither A nor $\boldsymbol{x}$ are zeroes (or even have any 0 entries), but their product is equal to $\mathbf{0}$.

this A is singular

This will not be true for any matrix, but notice that A's columns (and rows) are multiples of one another: the second column is 3 times the first. Matrices like A, that is, matrices that take non-zero vectors to zero vectors are called *singular* or *degenerate*.

The set of all vectors $\boldsymbol{x}$ that A takes to the $\mathbf{0}$ vector is called the nullspace of A. In other words,

Definition The nullspace of A, denoted by $\mathrm{Null}(A)$ or $\boldsymbol{N}(A)$, is the set of all $\boldsymbol{x}$'s that satisfy

$$A\boldsymbol{x} = \mathbf{0}.$$

Example 8. The matrix A of Example 7 crushes any multiple of the vector $\boldsymbol{x}$ into the zero vector $\mathbf{0} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$: vectors $c\boldsymbol{x} = c \begin{bmatrix} 3 \\ -1 \end{bmatrix} = \begin{bmatrix} 3c \\ -c \end{bmatrix}$, where c is any number, are all taken to the $\mathbf{0}$ vector when A is applied to them. Vectors $c\boldsymbol{x}$ lie along the orange line in Figure 2.2.

collinearity

Vectors that are multiples of $\boldsymbol{x}$ are said to be collinear with $\boldsymbol{x}$ because they all lie on the same straight line with $\boldsymbol{x}$. If A takes $\boldsymbol{x}$ to $\mathbf{0}$, then it takes every vector collinear with $\boldsymbol{x}$ to $\mathbf{0}$, too.

non-singular matrices

Other matrices never take non-zero vectors to zero vectors. We call such matrices *non-singular*. For example, the matrix $A = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$ of Figure 2.3 is non-singular.

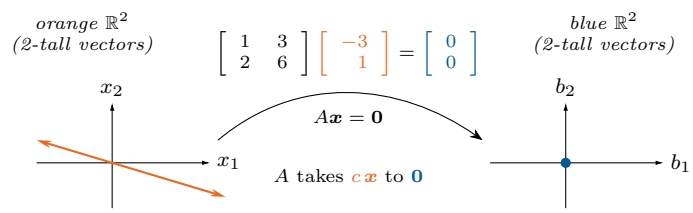


Figure 2.2: a *singular* 2-by-2 matrix A maps $n = 2$ dimensional vectors to $m = 2$ dimensional vectors, mapping every vector along the orange line to the zero vector represented by the blue dot in $\mathbb{R}^m$.

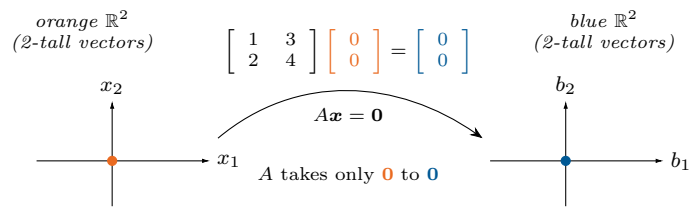


Figure 2.3: *non-singular* matrices always take non-zero vectors to non-zero vectors (the zero vector always maps to the zero vector under any A).

Key Point To solve the system of equations $A\boldsymbol{x} = \mathbf{0}$ is to answer this question: *What $\boldsymbol{x}$'s are in the nullspace of A?*

If A is non-singular, then the *only* solution to $A\boldsymbol{x} = \mathbf{0}$ is $\boldsymbol{x} = \mathbf{0}$.

If A is singular, then, in addition to $\boldsymbol{x} = \mathbf{0}$, the system $A\boldsymbol{x} = \mathbf{0}$ has non-zero solutions.

Example 9. One last example. Consider the $A\boldsymbol{x} = \mathbf{0}$ system with a single equation $x_1 + 2x_2 + 3x_3 = 0$, which in its matrix form looks like this:

$$\begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0.$$

Certainly, $\boldsymbol{x} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ is a solution, but there are more $\boldsymbol{x}$'s in the nullspace.

For example, these vectors: $\begin{bmatrix} -5 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix}$, and many more

are also in A's nullspace. In fact, and entire plane in the 3-dimensional space on which A acts gets squashed into $\mathbf{0}$. Can you draw this plane? Any two of the above three vectors span this plane.

In the next section, we'll look at a streamlined way of finding A's nullspace. That is, we'll see how software packages solve the $A\boldsymbol{x} = \mathbf{0}$ system of equations. We will call $A\boldsymbol{x} = \mathbf{0}$ a *homogeneous system of equations.*

2.4 Exercises

Exercise 2.1. What is the column space of $A = \begin{bmatrix} 1 & 0 \\ 4 & 3 \\ 2 & 3 \end{bmatrix}$? What must be true about $\boldsymbol{b}$ for $A\boldsymbol{x} = \boldsymbol{b}$ to be solvable?

Exercise 2.2. Johnny needs to solve a homogeneous system of linear equations, $A\boldsymbol{x} = \mathbf{0}$. He knows that $\boldsymbol{x} = \mathbf{0}$ will always be a solution, but his 11-by-5 matrix A seems to look nothing like what he has seen in his homework.

Johnny wants to know if there is a *non-zero* $\boldsymbol{x}$ that solves his $A\boldsymbol{x} = \mathbf{0}$ system, where A is

shown below.

$$A = \begin{bmatrix} 1 & 0 & 3 & 0 & \frac{1}{2} \\ 0 & 2 & 0 & 5 & 0 \\ 2 & 0 & 4 & 0 & 1 \\ 0 & 2 & 0 & 6 & 0 \\ 3 & 0 & 5 & 0 & \frac{3}{2} \\ 0 & 2 & 0 & 7 & 0 \\ 4 & 0 & 6 & 0 & 2 \\ 0 & 2 & 0 & 8 & 0 \\ 5 & 0 & 7 & 0 & \frac{5}{2} \\ 0 & 2 & 0 & 9 & 0 \\ 6 & 0 & 8 & 0 & 3 \end{bmatrix}$$

(a) A takes ____ dimensional vectors to ____ dimensional vectors.

(b) If there is a $\boldsymbol{x} \neq \mathbf{0}$ that solves Johnny's system, please write it down. If such $\boldsymbol{x}$ does not exist, please briefly explain why.

3 Intro to Gauss-Jordan Elimination

When solving systems of linear equations, we usually write them as matrix equations. One advantage of this $A\boldsymbol{x} = \boldsymbol{b}$ form is that it puts attention on the linear transformation A that takes the unknown vector $\boldsymbol{x}$ to the known vector $\boldsymbol{b}$. Example 10 solves a 2-by-2 system using the method of Gaussian elimination.

3.1 A Small $A\mathbf{x} = \mathbf{b}$ System

Example 10. The $A\boldsymbol{x} = \boldsymbol{b}$ form of the system $\begin{cases} 2x_1 + 7x_2 = 5 \\ x_1 + 4x_2 = 3 \end{cases}$ is

$$\begin{bmatrix} 2 & 7 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 3 \end{bmatrix}.$$

To solve it, Gauss and Jordan wrote down the so-called augmented matrix $[A \mid \boldsymbol{b}]$ and proceeded with row reductions:

$$\left[\begin{array}{cc|c} 2 & 7 & 5 \\ 1 & 4 & 3 \end{array}\right] \xrightarrow{\text{row-flip}} \left[\begin{array}{cc|c} 1 & 4 & 3 \\ 2 & 7 & 5 \end{array}\right] \xrightarrow{\text{row } 2 - 2\cdot\text{row } 1} \left[\begin{array}{cc|c} 1 & 4 & 3 \\ 0 & -1 & -1 \end{array}\right] \tag{3.1}$$

At this point, we could write down the solution: From the second row, we see that $x_2 = 1$. Plugging this into the first equation we get $x_1 + 4 \cdot 1 = 3$, or $x_1 = -1$.

But Gauss and Jordan don't stop here. They proceed to further simplify the system so that the solution is even easier to read:

$$\left[\begin{array}{rr|r} 1 & 4 & 3 \\ 0 & -1 & -1 \end{array}\right] \xrightarrow{\text{sign-flip}} \left[\begin{array}{rr|r} 1 & 4 & 3 \\ 0 & 1 & 1 \end{array}\right] \xrightarrow{\text{row } 1 - 4\cdot\text{row } 2} \left[\begin{array}{rr|r} 1 & 0 & -1 \\ 0 & 1 & 1 \end{array}\right] \tag{3.2}$$

Now we have the identity matrix sitting inside the last matrix in 3.2, and the right-hand side is the *only* solution to the system: $\boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$.

Key Point Gaussian elimination uses 3 operations on the rows of the augmented matrix:

- swapping rows;
- multiplying rows by a nonzero constant;
- adding a multiple of one row, row i, to another row, row j, replacing row j.

These operations are called *elementary row operations*, and they *do not change the solutions* to a system of linear equations.

3.2 Homogeneous Systems, $Ax = 0$

To solve the system $A\boldsymbol{x} = \mathbf{0}$ is to determine the nullspace of A. That is, find all the $\boldsymbol{x}$'s that get crushed into the $\mathbf{0}$ vector when A is applied to them.

Example 11. The system $\begin{cases} x_1 + 3x_2 + 3x_3 + 2x_4 = 0 \\ 2x_1 + 6x_2 + 9x_3 + 7x_4 = 0 \end{cases}$ in its matrix form $A\boldsymbol{x} = \mathbf{0}$ is

$$\begin{bmatrix} 1 & 3 & 3 & 2 \\ 2 & 6 & 9 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

The very first row operation gives

$$\begin{bmatrix} 1 & 3 & 3 & 2 \\ 2 & 6 & 9 & 7 \end{bmatrix} \xrightarrow{\text{row } 2 - 2\times\text{row } 1} \begin{bmatrix} \boxed{1} & 3 & 3 & 2 \\ 0 & 0 & \boxed{3} & 3 \end{bmatrix}. \tag{3.3}$$

free and pivot variables

The columns with highlighted entries correspond to the *pivot* variables x_1 and x_3; the second and fourth columns correspond to the *free* variables x_2 and x_4.

special solutions

Because we have two free variables, we are going to have a *2-parameter space of solutions* (a 2-dimensional plane embedded in $\mathbb{R}^4$). To describe the plane of solutions we hand-pick two special vectors that lie on that plane. We call them the *special solutions*.

We hand-pick these special solutions and explain how we picked them in the following paragraph:

$$\begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \end{bmatrix} \begin{array}{l} \leftarrow \textit{find pivot from row 1} \\ \leftarrow \textit{set free } x_2 = \mathbf{1} \\ \leftarrow \textit{find pivot from row 2} \\ \leftarrow \textit{set free } x_4 = \mathbf{0} \end{array}, \quad \text{and} \quad \begin{bmatrix} 1 \\ 0 \\ -1 \\ 1 \end{bmatrix} \begin{array}{l} \leftarrow \textit{find pivot from row 1} \\ \leftarrow \textit{set free } x_2 = \mathbf{0} \\ \leftarrow \textit{find pivot from row 2} \\ \leftarrow \textit{set free } x_4 = \mathbf{1} \end{array}$$

Here's how we hand-picked these special solutions. We started by setting one of the free variables to 1 (say, $x_2 = 1$) and the other to 0 ($x_4 = 0$). Once the free variables were set, we figured out the values of the pivots, which were uniquely determined: x_3 worked out to 0 and x_1 to -3. That's one special solution.

We then repeated the same process to find the second special solution, this time setting $x_4 = 1$ and $x_2 = 0$, and again determining the values of the pivot variables.

These two vectors – the two special solutions found this way – generate *all* the solutions to $A\boldsymbol{x} = \mathbf{0}$:

$$\boldsymbol{x} = x_2 \begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 1 \\ 0 \\ -1 \\ 1 \end{bmatrix}$$

are all linear combinations of the special solutions. □

The last matrix in 3.3 Example 11 is said to be in steplike (or echelon) form and is denoted by U:

$$U = \begin{bmatrix} 1 & 3 & 3 & 2 \\ 0 & 0 & 3 & 3 \end{bmatrix} \qquad \text{(the steplike (or echelon) matrix } U\text{)}$$

U can be further simplified to a *row-reduced echelon form* we denote by R:

$$U = \begin{bmatrix} 1 & 3 & 3 & 2 \\ 0 & 0 & 3 & 3 \end{bmatrix} \xrightarrow{\text{row 1}-\text{row 2}} \begin{bmatrix} 1 & 3 & 0 & -1 \\ 0 & 0 & 3 & 3 \end{bmatrix} \xrightarrow{\frac{1}{3}\text{row 2}} \begin{bmatrix} 1 & 3 & 0 & -1 \\ 0 & 0 & 1 & 1 \end{bmatrix} = R$$

The advantage of R is that the identity matrix is sitting in the pivot columns and the solution can be just read off from the other entries (with signs flipped):

$$R = \begin{bmatrix} \boxed{1} & 3 & \boxed{0} & -1 \\ \boxed{0} & 0 & \boxed{1} & 1 \end{bmatrix}$$

In R, the identity matrix sits in its pivot columns.

☞ It's good to do Exercise 3.1 now. The solution is included at the end.

Example 12. Solve the system $A\boldsymbol{x} = \mathbf{0}$ where $A = \begin{bmatrix} 1 & 2 \\ 3 & 6 \\ 3 & 9 \\ 2 & 7 \end{bmatrix}$. This is a "tall" A, unlike the A in Example 11, which was "wide." Here we have more equations than unknowns — more constraints on the way the unknowns are related than unknowns. The $\mathbf{0}$ vector will always be a solution to $A\boldsymbol{x} = \mathbf{0}$, but are there any more solutions? Let's do the usual row operations:

$$\begin{bmatrix} 1 & 2 \\ 3 & 6 \\ 3 & 9 \\ 2 & 7 \end{bmatrix} \xrightarrow{\text{row ops} \ldots} \begin{bmatrix} 1 & 2 \\ 0 & 0 \\ 0 & 3 \\ 0 & 3 \end{bmatrix} \xrightarrow{\text{more row ops} \ldots} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} = R$$

(We subtracted the multiples of row 1 from the other rows to get the north-west pivot 1, combining these operations under "row ops," and then exchanged and scaled the rows to get to R, combining those operations under "more row ops.")

We can now see that the only solution to $R\boldsymbol{x} = \mathbf{0}$ is $\boldsymbol{x} = \mathbf{0}$: we have no free variables — both x_1 and x_2 are pivots, and the only way to satisfy the system is to set both of them to 0.

$$\text{The only solution to } \begin{bmatrix} 1 & 2 \\ 3 & 6 \\ 3 & 9 \\ 2 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \text{ is } \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

☞ It's good to do Exercise 3.2 now. The solution is included at the end.

Example 13. To solve $\begin{cases} 2x_1 + 4x_2 - 2x_3 = 0 \\ 4x_1 + 9x_2 - 3x_3 = 0 \\ -2x_1 - 3x_2 + 7x_3 = 0 \end{cases}$ is to find the nullspace of $\begin{bmatrix} 2 & 4 & -2 \\ 4 & 9 & -3 \\ -2 & -3 & 7 \end{bmatrix}$.

We find the nullspace by Gaussian elimination:

$$\begin{bmatrix} 2 & 4 & -2 \\ 4 & 9 & -3 \\ -2 & -3 & 7 \end{bmatrix} \xrightarrow{\text{row } 2 - 2\times\text{row } 1} \begin{bmatrix} 2 & 4 & -2 \\ 0 & 1 & 1 \\ -2 & -3 & 7 \end{bmatrix} \xrightarrow{\text{row } 3+\text{row } 1} \begin{bmatrix} 2 & 4 & -2 \\ 0 & 1 & 1 \\ 0 & 1 & 5 \end{bmatrix}$$

$$\xrightarrow{\text{row } 3-\text{row } 2} \begin{bmatrix} 2 & 4 & -2 \\ 0 & 1 & 1 \\ 0 & 0 & 4 \end{bmatrix} \xrightarrow{\frac{1}{4}\times\text{row } 3} \begin{bmatrix} 2 & 4 & -2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\xrightarrow{\text{row } 2-\text{row } 3} \begin{bmatrix} 2 & 4 & -2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{\text{row } 1 + 2\times\text{row } 3} \begin{bmatrix} 2 & 4 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\xrightarrow{\text{row } 1 - 4\times\text{row } 2} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{\frac{1}{2}\times\text{row } 1} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Pivots are highlighted as they appear during Gaussian elimination. We just brought A to its row-reduced echelon form R, which in this case happens to be the the identity matrix. Whenever you can do this (whenever you do *not* get a row of zeros as you carry out row-reductions), there will be a unique solution to $A\boldsymbol{x} = \mathbf{0}$, just the $\boldsymbol{x} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = \mathbf{0}$ solution.

3.3 The Rank of a Matrix

You are solving $A\boldsymbol{x} = \mathbf{0}$. You do Gaussian row reductions the usual way to transform A into its step-like (echelon) form U and then into its row-reduced echelon form R. Benefit? The pivot variables are readily seen in either U or R. For example, in Example 11, we had

$$A = \begin{bmatrix} 1 & 3 & 3 & 2 \\ 2 & 6 & 9 & 7 \end{bmatrix} \xrightarrow{\text{Gaussian steps } \ldots} \begin{bmatrix} \boxed{1} & 3 & 0 & -1 \\ 0 & 0 & \boxed{1} & 1 \end{bmatrix} = R.$$

The pivot variables are x_1 and x_3. The highlighted pivots are in columns 1 and 3. These columns are called the *pivot columns* and their number – two of them here – is known as the rank of A; we denote the rank by r:

r = number of pivot columns

$$r = \text{rank}(A) = 2 \qquad \text{(because } A \text{ has 2 pivot columns)}$$

Definition The rank of matrix A is the number of its pivot columns.

$A_{m \times n}$

$r = m$

$n - m$ = number of free variable

This 2-by-4 A has a pivot in each of its rows, so its rank is equal to the number of its rows. Such a matrix is said to have a *full row rank*.

If A has m rows and r pivots and $m = r$ then A is called a ***full row rank matrix***. If A has a full row rank,

- all $m = r$ rows have pivots, and R has no zero rows,
- the other $n - m$ variables are free and each generates a special solution.

But it can certainly happen that not every row of A contains a pivot. For example, the tall A of Example 12 has four rows but two pivots. Recall that Gaussian elimination in that example produced rows of zeros:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 6 \\ 3 & 9 \\ 2 & 7 \end{bmatrix} \xrightarrow{\text{Gaussian steps } \ldots} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} = R$$

Both of A's columns are pivot columns, so its rank, 2, equals $r = n$, the number of A's columns. Such A's are known as *full-column-rank* matrices.

If A has n columns and r pivots and $n = r$ then A is called a ***full column rank matrix***. When A has a full column rank,

- all $n = r$ columns are pivot columns,
- $A\boldsymbol{x} = \mathbf{0}$ has no special solutions,
- The nullspace $N(A)$ has only the zero vector $\boldsymbol{x} = 0$.

3.3.1 Transposing a Matrix Doesn't Change Its Rank

The transpose of A, denoted by A^T, is a matrix whose i-th row is the same as the i-th column of A. For example, these two matrices are transposes of each other:

$$\begin{bmatrix} 1 & 3 & 3 & 2 \\ 2 & 6 & 9 & 7 \end{bmatrix} \text{ and } \begin{bmatrix} 1 & 2 \\ 3 & 6 \\ 3 & 9 \\ 2 & 7 \end{bmatrix} \text{ are transposes of each other.}$$

These are the "wide" and "tall" matrices from Examples 11 and 12. The long rows in the wide matrix now stand tall as columns in the tall matrix, and vice versa.

According to the observations made in this section, the rank of each of these matrices is equal to 2. Generally, for any A, this fact is true:

$$\text{rank}(A) = \text{rank}(A^T)$$

This is a fundamental fact in linear algebra, and we'll prove it later[1].

[1] The proof is nice and short, but it relies on making a few observations on the reduced row echelon forms of A and A^T. There are also even simpler, more elegant proofs. We'll talk about them during or after the lectures with those interested.

3.4 Takeaways

An m by n matrix A comes in three shapes:

- $m < n$: A is *wide* — it has more columns than rows.

$$\text{Example: } A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \end{bmatrix}.$$

- $m > n$: A is *tall* — it has more rows than columns.

$$\text{Example: } A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \\ 7 & 8 \end{bmatrix}.$$

- $m = n$: A is *square*.

$$\text{Example: } A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}.$$

No matter what the shape, $\boldsymbol{x} = \mathbf{0}$ is always a solution to $A\boldsymbol{x} = \mathbf{0}$.

But if A has more columns than rows, you know for sure there is going to be some $\boldsymbol{x} \neq \mathbf{0}$ in A's nullspace.

- Gaussian row reductions take A to its echelon form U and ultimately to its reduced row echelon form R.
- Those row reductions do not change the solutions to $A\boldsymbol{x} = \mathbf{0}$, which is called the nullspace of A.
- A's pivot columns can be readily seen in U or in R. The columns that are not pivots are called free.

3.5 Exercises

Exercise 3.1. Solve $A\boldsymbol{x} = \mathbf{0}$ where $A = \begin{bmatrix} 1 & 3 & 3 & 2 \\ 2 & 6 & 9 & 7 \\ 4 & 12 & 15 & 11 \end{bmatrix}$.

Exercise 3.2. Change the entries in rows 3 and 4 of A from Example 12 so that $A\boldsymbol{x} = \mathbf{0}$ has a nonzero solution.

Exercise 3.3. Suppose you keep doing the usual Gaussian row reductions on an m by n matrix A until you find all the pivots. Call the number of pivots r. How does r relate to m and n? (Is r larger, smaller, or equal to m or n?)

Exercise 3.4. How many special solutions does $R\boldsymbol{x} = \mathbf{0}$ have if

$$R = \begin{bmatrix} 1 & 0 & * & * & 0 & * & * \\ 0 & 1 & * & * & 0 & * & * \\ 0 & 0 & 0 & 0 & 1 & * & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix},$$

where $*$ stands for any number?

If $r = m$, number of free variable : $n - m$
which means that $r \leq n$

If $r = n$ there are no free variables and $r \leq m$

3.6 Answers

3.1 Gaussian elimination gives the matrix $R = \begin{bmatrix} 1 & 3 & 0 & -1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$. The last row doesn't add any new constraints on the way the variables are related: A's third row is $2\times$ row 1 + row 2, and Gaussian elimination revealed this by giving the third row of zeros in R. The solution is the same as in Example 11.

3.2 If you make all the rows multiples of one another, you will end up with a free variable. For example,

$$\begin{bmatrix} 1 & 2 \\ 3 & 6 \\ 3 & 6 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \text{ has a nonzero solution } \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \end{bmatrix}.$$

All the solutions to $A = \mathbf{0}$ lie on the same line as $\begin{bmatrix} -2 \\ 1 \end{bmatrix}$. The solution**s** can be written down in this compact form: $c \begin{bmatrix} -2 \\ 1 \end{bmatrix}$, where c is any number whatsoever.

4 $A\mathrm{x} = \mathrm{b}$ Non-Homogeneous Systems

To solve the system $A\boldsymbol{x} = \boldsymbol{b}$ is to determine whether $\boldsymbol{b}$ is in the column space of A. Whenever $\boldsymbol{b} \neq \mathbf{0}$, the row operations on A must also act on the right-hand side $\boldsymbol{b}$.

4.1 A Is Wide: More Unknowns Than Equations

Example 14. $A\boldsymbol{x} = \boldsymbol{b}$ with $A = \begin{bmatrix} 1 & 3 & 3 & 2 \\ 2 & 6 & 9 & 7 \end{bmatrix}$ and $\boldsymbol{b} = \begin{bmatrix} 1 \\ 8 \end{bmatrix}$ corresponds to the system $\begin{cases} x_1 + 3x_2 + 3x_3 + 2x_4 = 1 \\ 2x_1 + 6x_2 + 9x_3 + 7x_4 = 8 \end{cases}$.

Observe that A's four columns all live in the $\mathbb{R}^2$ plane. Of those four column vectors only at most two will really matter in expressing $\boldsymbol{b}$ as a combination of the columns. OK, let's see what row ops say:

$$\left[\begin{array}{cccc|c} 1 & 3 & 3 & 2 & 1 \\ 2 & 6 & 9 & 7 & 8 \end{array}\right] \xrightarrow{\text{row } 2 - 2\times\text{row}1} \left[\begin{array}{cccc|c} 1 & 3 & 3 & 2 & 1 \\ 0 & 0 & 3 & 3 & 6 \end{array}\right] = U \qquad \text{(steplike form)}$$

Row-reduce it further to R:

$$U = \left[\begin{array}{cccc|c} 1 & 3 & 3 & 2 & 1 \\ 0 & 0 & 3 & 3 & 6 \end{array}\right] \xrightarrow{\text{row } 1-\text{row } 2\text{; then } \frac{1}{3}\text{row } 2} \left[\begin{array}{cccc|c} 1 & 3 & 0 & -1 & -5 \\ 0 & 0 & 1 & 1 & 2 \end{array}\right] = R$$

(row-reduced form)

Now let's pick a *particular* solution that solves the system by setting all the free variables to

0s:

$$\textbf{Particular solution:} \quad \boldsymbol{x}_p = \begin{bmatrix} -5 \\ 0 \\ 2 \\ 0 \end{bmatrix} \begin{array}{l} \leftarrow \textit{find pivot from row 1} \\ \leftarrow \textit{set free } x_2 = \mathbf{0} \\ \leftarrow \textit{find pivot from row 2} \\ \leftarrow \textit{set free } x_4 = \mathbf{0} \end{array}$$

This is just one solution. There are many particular solutions you could have picked, but setting the free variables to 0s and then figuring out the pivots makes life easier.

How about all the solutions$A\boldsymbol{x} = \boldsymbol{b}$? If you *add everything in the nullspace to the particular solution*, you will get all the solutions. The nullspace is generated by the *special* solutions, and we already found them back in Example 11. These were the special solutions that generated *all* the solutions to $A\boldsymbol{x} = \mathbf{0}$:

$$\textbf{The nullspace of } A\textbf{:} \quad \boldsymbol{x} = x_2 \begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 1 \\ 0 \\ -1 \\ 1 \end{bmatrix}$$

(all linear combinations of special solutions)

Now we can write down all the solutions to the original $A\boldsymbol{x} = \boldsymbol{b}$:

All solutions to $A\boldsymbol{x} = \boldsymbol{b}$ are:

the particular solution + the nullspace of A.

$$\boldsymbol{x}_{\text{complete}} = \begin{bmatrix} -5 \\ 0 \\ 2 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 1 \\ 0 \\ -1 \\ 1 \end{bmatrix}.$$

Key Point As Example 14 shows,

- the particular solution $\boldsymbol{x}_p$ solves $A\boldsymbol{x}_p = \boldsymbol{b}$ (find one by setting free variables to 0s),
- the special solution(s) $\boldsymbol{x}_n$ solve $A\boldsymbol{x}_n = \mathbf{0}$ (A's nullspace found by solving $A\boldsymbol{x} = \mathbf{0}$),
- The complete solution is $\boldsymbol{x} = \boldsymbol{x}_p + \boldsymbol{x}_n$ (just add them up).

It's good to do Exercise 4.1 after reading Example 14.

Remark Note that for the matrix $A = \begin{bmatrix} 1 & 3 & 3 & 2 \\ 2 & 6 & 9 & 7 \end{bmatrix}$ of Example 14, $2 \times$ column-3 $-$ $5 \times$ colum-1 $= 2 \times \begin{bmatrix} 3 \\ 9 \end{bmatrix} - 5 \times \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 8 \end{bmatrix} = \boldsymbol{b}$, the right-hand side.

So it's no surprise that $\begin{bmatrix} -5 \\ 0 \\ 2 \\ 0 \end{bmatrix}$ is a particular solution.

For small matrices you can sometimes guess a solution like this, but Gaussian eliminations will always show the solution.

The reduced row echelon form of matrix $A = \begin{bmatrix} 1 & 3 & 3 & 2 \\ 2 & 6 & 9 & 7 \end{bmatrix}$ from Example 14 is $R = \begin{bmatrix} 1 & 3 & 0 & -1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$ with two pivots (columns 1 and 3), so A's rank $r = 2$, which is also equal to A's number of rows: $m = 2 = r$. Thus A is a *full row rank matrix* because it has as many pivots as it has rows.

This implies that A's column space is 2-dimensional. Moreover, because $A\boldsymbol{x}$ gives the linear combinations of A's columns as you vary $\boldsymbol{x}$, any 2-dimensional vector $\boldsymbol{b}$ can be expressed as a linear combination of A's columns. Thus, we are certain to find the solution to $A\boldsymbol{x} = \boldsymbol{b}$ for *any* right-hand side $\boldsymbol{b}$ thanks to A having full row rank.

This justifies adding the following key points to the inventory of observations we made earlier on page 29 about full row rank matrices.

Key Point Whenever A is has a full row rank, that is, when $m = r$,

- The column space $C(A)$ is the whole $\mathbb{R}^m$ space, spanned by the m pivot columns.
- $A\boldsymbol{x} = \boldsymbol{b}$ has a solution for any $\boldsymbol{b}$.

Okay, so $A\boldsymbol{x} = \boldsymbol{b}$ always has a solution when A is a wide, full row rank matrix.

But how about tall matrices that show up in systems with more equations than unknowns? When A has more rows than columns, then some rows have no pivots (Gaussian elimination will put all zeros in those rows).

The next example is an excellent opportunity to make a few new observations about such systems.

4.2 A Is Tall: More Equations Than Unknowns

If A is tall and has as many pivots r as it has columns n — as in Example 15 below — then $A\boldsymbol{x} = \boldsymbol{b}$ has either 1 solution or no solutions. There is no way to get an infinite number of solutions as we could with wide systems. And you can only find that unique solution if $\boldsymbol{b}$ is just the right combination of A's columns. (Recall that when the number of pivots $r = n$, the number of columns, then A is said to have *full column rank*.)

Example 15. Solve $A\boldsymbol{x} = \boldsymbol{b}$ for $A = \begin{bmatrix} 1 & 2 \\ 1 & 1 \\ -3 & -2 \end{bmatrix}$ and $\boldsymbol{b} = \begin{bmatrix} 1 \\ 2 \\ -7 \end{bmatrix}$. The usual Gaussian row reductions give

$$\left[\begin{array}{cc|c} 1 & 2 & 1 \\ 1 & 1 & 2 \\ -3 & -2 & -7 \end{array}\right] \xrightarrow{\text{row reductions } \ldots} \left[\begin{array}{cc|c} 1 & 0 & 3 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{array}\right].$$

We were lucky to end up with all zeros in the third row! The unique solution here is $\boldsymbol{x} = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$. As a check, this works because 3 times column-1 minus column-2 gives exactly

the right-hand side $\boldsymbol{b}$:

$$\begin{bmatrix} 1 & 2 \\ 1 & 1 \\ -3 & -2 \end{bmatrix} \begin{bmatrix} 3 \\ -1 \end{bmatrix} = (3) \begin{bmatrix} 1 \\ 1 \\ -3 \end{bmatrix} + (-1) \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ -7 \end{bmatrix}.$$

☞ It's good to do Exercise 4.2 after reading Example 15.

Example 16. Find the conditions on the components of $\boldsymbol{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$ so that the system

$$\begin{bmatrix} 1 & 2 \\ 1 & 1 \\ -3 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

has a solution.

We do Gaussian row reductions to find out what the conditions are.

$$\left[\begin{array}{rr|l} 1 & 2 & b_1 \\ 1 & 1 & b_2 \\ -3 & -2 & b_3 \end{array}\right] \xrightarrow[\text{and row 3 } +3\times \text{ row 1}]{\text{row 2 } - \text{ row 1,}} \left[\begin{array}{rr|l} 1 & 2 & b_1 \\ 0 & -1 & b_2 - b_1 \\ 0 & 4 & b_3 + 3b_1 \end{array}\right]$$

$$\xrightarrow{\text{row 3 } +4\times\text{row 2}} \left[\begin{array}{rr|l} 1 & 2 & b_1 \\ 0 & -1 & b_2 - b_1 \\ 0 & 0 & b_3 + 4b_2 - b_1 \end{array}\right]$$

We see that unless $b_3 + 4b_2 - b_1 = 0$, this system has no solution. (As a check, in Example 15 this condition was satisfied: $b_3 + 4b_2 - b_1 = -7 + 4 \cdot 2 - 1 = 0$, and in Exercise 15 it was not.) Continuing with row deductions towards R we get

$$\left[\begin{array}{rr|l} 1 & 2 & b_1 \\ 0 & -1 & b_2 - b_1 \\ 0 & 0 & b_3 + 4b_2 - b_1 \end{array}\right] \xrightarrow[\text{then } (-1)\times\text{row 2}]{\text{row 1 } +2\times\text{row 2,}} \left[\begin{array}{rr|l} 1 & 0 & 2b_2 - b_1 \\ 0 & 1 & b_1 - b_2 \\ 0 & 0 & b_3 + 4b_2 - b_1 \end{array}\right]$$

This gives us the particular solution in the first two rows. Moreover, there is nothing other

than the $\boldsymbol{x}_n = \mathbf{0} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ vector in the nullspace because we have no free variables, so

$$\textbf{The only solution} \text{ to } A\boldsymbol{x} = \boldsymbol{b} \quad \text{is} \quad \boldsymbol{x} = \boldsymbol{x}_p + \boldsymbol{x}_n = \begin{bmatrix} 2b_2 - b_1 \\ b_1 - b_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

$$\textbf{and} \text{ if } b_3 + 4b_2 - b_1 \text{ is not zero, then there is no solution.}$$

☞ It's good to do Exercise 4.3 after reading Example 15.

The reduced row echelon form of matrix $A = \begin{bmatrix} 1 & 2 \\ 1 & 1 \\ -3 & -2 \end{bmatrix}$ from Examples 15 and 16 is $R = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$, whose both columns are pivot columns.

So A's rank $r = 2$, which is also equal to A's number of columns: $n = 2 = r$. This makes A a *full column rank matrix* because it has as many pivots as it has columns.

If $A\boldsymbol{x} = \boldsymbol{b}$, then $\boldsymbol{b}$ lives in A's 3-dimensional column space, and we can express $\boldsymbol{b}$ as a linear combination of A's columns. Geometrically, this means that $\boldsymbol{b}$ lies in the 2-dimensional plane spanned by A's 3-dimensional column vectors. When this happens, we have exactly one solution because there is exactly one way to express $\boldsymbol{b}$ as a linear combination of A's columns.

If, on the other hand, $\boldsymbol{b}$ is an arbitrary 3-dimensional vector that does not live on the plane spanned by A's columns, then $A\boldsymbol{x} = \boldsymbol{b}$ has no solution at all. Geometrically, this means that $\boldsymbol{b}$ "juts out" from the plane spanned by A's columns.

To summarize: if we get lucky, we have one and only one solution to $A\boldsymbol{x} = \boldsymbol{b}$ or we have no solution at all. There's no way to have an infinite number of solutions when A is tall.

This justifies adding the following key points to the inventory of observations we made earlier on page 30 about full row rank matrices.

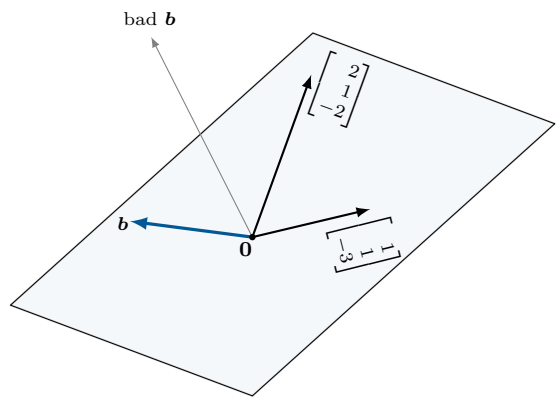


Figure 4.1: the plane in $\mathbb{R}^3$ spanned by A's columns. For $A\boldsymbol{x} = \boldsymbol{b}$ to have a solution, the vector $\boldsymbol{b}$ must lie in this plane. If $\boldsymbol{b}$ protrudes from the plane, then $A\boldsymbol{x} = \boldsymbol{b}$ has no solutions.

Key Point Whenever A has a full column rank, that is, when $n = r$,

- The nullspace $N(A)$ has only the zero vector $\boldsymbol{x} = 0$.
- $A\boldsymbol{x} = \boldsymbol{b}$ can only have one solution or no solutions at all.

4.3 A Is Square

Finally, let's look at square As. The nice case is when the matrix rank r is full, meaning that $r = m = n$ (where, as usual, m and n stand for the numbers of A's rows and columns, respectively). In the nice case, there will be exactly one solution to $A\boldsymbol{x} = \boldsymbol{b}$, and in the not-so-nice case, when $r < m = n$, the nullspace of A will have non-zero vectors in it.

4.3.1 A Is Nice: Full Rank Case

Example 17. To solve $\begin{cases} 2x_1 + 4x_2 - 2x_3 = 2 \\ 4x_1 + 9x_2 - 3x_3 = 8 \\ -2x_1 - 3x_2 + 7x_3 = 10 \end{cases}$ is to find a linear combination of A's columns that is equal to $\boldsymbol{b} = \begin{bmatrix} 2 \\ 8 \\ 10 \end{bmatrix}$. Let's verify that $\boldsymbol{x} = \begin{bmatrix} -1 \\ 2 \\ 2 \end{bmatrix}$ gives just the right linear combination of A's columns that lead to $\boldsymbol{b}$. Indeed,

$$A\boldsymbol{x} = (-1)\begin{bmatrix} 2 \\ 4 \\ -2 \end{bmatrix} + 2\begin{bmatrix} 4 \\ 9 \\ -3 \end{bmatrix} + 2\begin{bmatrix} -2 \\ -3 \\ 7 \end{bmatrix} = \begin{bmatrix} 2 \\ 8 \\ 10 \end{bmatrix} = \boldsymbol{b}.$$

This $\boldsymbol{x}$ is the only solution to the system.

Example 18. We know the solution to $A\boldsymbol{x} = \boldsymbol{b}$ from Example 17. To actually find it, we use Gaussian elimination on the augmented matrix $[\,A \mid \boldsymbol{b}\,]$:

$$\left[\begin{array}{rrr|r} 2 & 4 & -2 & 2 \\ 4 & 9 & -3 & 8 \\ -2 & -3 & 7 & 10 \end{array}\right] \xrightarrow{\text{row } 2 - 2\times\text{row } 1} \left[\begin{array}{rrr|r} 2 & 4 & -2 & 2 \\ 0 & 1 & 1 & 4 \\ -2 & -3 & 7 & 10 \end{array}\right] \xrightarrow{\text{row } 3+\text{row } 1}$$

$$\left[\begin{array}{rrr|r} 2 & 4 & -2 & 2 \\ 0 & 1 & 1 & 4 \\ 0 & 1 & 5 & 12 \end{array}\right] \xrightarrow{\text{row } 3-\text{row } 2} \left[\begin{array}{rrr|r} 2 & 4 & -2 & 2 \\ 0 & 1 & 1 & 4 \\ 0 & 0 & 4 & 8 \end{array}\right] \xrightarrow{\frac{1}{4}\times\text{row } 3}$$

$$\left[\begin{array}{rrr|r} 2 & 4 & -2 & 2 \\ 0 & 1 & 1 & 4 \\ 0 & 0 & 1 & 2 \end{array}\right] \xrightarrow{\text{row } 2-\text{row } 3} \left[\begin{array}{rrr|r} 2 & 4 & -2 & 2 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 2 \end{array}\right] \xrightarrow{\text{row } 1 + 2\times\text{row } 3}$$

$$\left[\begin{array}{rrr|r} 2 & 4 & 0 & 6 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 2 \end{array}\right] \xrightarrow{\text{row } 1 - 4\times\text{row } 2} \left[\begin{array}{rrr|r} 2 & 0 & 0 & -2 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 2 \end{array}\right] \xrightarrow{\frac{1}{2}\times\text{row } 1}$$

$$\left[\begin{array}{rrr|r} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 2 \end{array}\right] \qquad \text{(the simplest “}[\,I \mid \boldsymbol{b}_{\text{new}}\,]\text{ form”)}$$

We just brought the augmented matrix $[\,A \mid \boldsymbol{b}\,]$ to its simplest $[\,I \mid \boldsymbol{b}_{\text{new}}\,]$ form — the identity matrix is to the left of the vertical bar, and to the right of the vertical bar sits the solution $\boldsymbol{b}_{\text{new}}$.

Key Point If you can bring the augmented system $[A \mid \boldsymbol{b}]$ to the form $[\,I \mid \boldsymbol{b}_{\text{new}}\,]$ as in Example 18, then you've solved the system: $\boldsymbol{b}_{\text{new}}$ is the solution.

Gaussian elimination brings A to I whenever A has full rank.

Full-rank matrices are also called non-singular or invertible. We'll justify why it makes sense to call such matrices invertible when we talk about inverting square As next time.

If A doesn't have full rank, meaning that A has fewer than n pivots, then Gaussian elimination

will produce a row of zeros during its run. Such matrices are not nice enough to be invertible and are also called singular. Let's look at a singular A next.

4.3.2 A Is Not Nice: $r < n$

Here's a not-so-nice square $A = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}$ with $m = 2$ rows and $n = 2$ columns.

What is A's rank r? To find out, we do Gaussian elimination as for any $A\boldsymbol{x} = \mathbf{0}$ system.

The very first elimination step produces 0s in the second row. This means that A has only one pivot: column 1 is the pivot, corresponding to the variable x_1. So rank $r = 1$.

Because $1 = r < n = 2$, this A is *not* a full column rank matrix and therefore $A\boldsymbol{x} = \mathbf{0}$ will have infinite solutions: A's nullspace $N(A)$ has much more than just the $\mathbf{0}$ vector in it.

Can you write down all solutions to this $A\boldsymbol{x} = \mathbf{0}$?

This A will show up in Example 20 when we'll talk about matrix inverses and try to invert it.

As we'll see, whenever $r < n$, as for this A, then A is not invertible. Non-invertible matrices like A are also called *singular*. More on this: next up.

4.4 Takeaways

The takeaways are in the "key points" – the blue boxes – scattered throughout this chapter.

`//TODO:` further summarize the key points here if need be.

4.5 Exercises

Exercise 4.1. Set the x_2 and x_4 parameters in the complete solution to $A\boldsymbol{x} = \boldsymbol{b}$ of Example 14 to any numbers that make you happy and verify using a calculator that what you picked is indeed a solution.

Exercise 4.2. What is the solution to the system $\begin{bmatrix} 1 & 2 \\ 1 & 1 \\ -3 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$? Carry out the usual Gaussian row reductions as in the previous examples to find it.

Exercise 4.3. Verify that the right-hand side $\boldsymbol{b}$ in Example 15 agrees with the solution we found in Example 16.

Exercise 4.4. You are given three different vectors $\boldsymbol{b}_1, \boldsymbol{b}_2$ and $\boldsymbol{b}_3$. Your job is to construct a matrix A so that the equations $A\boldsymbol{x} = \boldsymbol{b}_1$ and $A\boldsymbol{x} = \boldsymbol{b}_2$ are solvable, but $A\boldsymbol{x} = \boldsymbol{b}_3$ is not solvable.

Is constructing such A always possible for any $\boldsymbol{b}_1, \boldsymbol{b}_2$ and $\boldsymbol{b}_3$? Give an example of $\boldsymbol{b}_1, \boldsymbol{b}_2$ and $\boldsymbol{b}_3$ when this is possible. If this is not always possible, give an example when it's not.

Exercise 4.5. Find the condition on b_1, b_2, and b_3 for $A\boldsymbol{x} = \boldsymbol{b}$ to be solvable if

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ -2 & -3 \end{bmatrix} \quad \text{and} \quad \boldsymbol{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}.$$

Exercise 4.6. Find the complete solution to $\begin{cases} x + y + z = 3 \\ x + 2y - z = 4 \end{cases}$.

Answers to Selected Problems

4.2 Row reductions get us to

$$\left[\begin{array}{rr|r} 1 & 2 & 1 \\ 1 & 1 & 2 \\ -3 & -2 & 3 \end{array}\right] \xrightarrow{\text{make pivot in column 1 ...}} \left[\begin{array}{rr|r} 1 & 2 & 1 \\ 0 & -1 & 1 \\ 0 & 4 & 6 \end{array}\right] \xrightarrow{\text{make pivot in column 2 ...}}$$

We can already see from the last two rows that there will be no solution. Continuing towards R, we add 4 times row-2 to row-3 to get:

$$\xrightarrow{\text{make pivot in column 2 ...}} \left[\begin{array}{rr|r} 1 & 2 & 1 \\ 0 & -1 & 1 \\ 0 & 0 & 10 \end{array}\right]$$

The last row makes it now very clear that there is now way to satisfy the third equation: there is no way $0x_1 + 0x_2 = 10$ no matter what x_1 and x_2 are. This system doesn't have a solution.

4.3 The solution we obtained in Example 15 was $\boldsymbol{x} = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$, which should agree with the findings in Example 16 which says that $\boldsymbol{x} = \begin{bmatrix} 2b_2 - b_1 \\ b_1 - b_2 \end{bmatrix}$. Indeed, in Example 15, $\boldsymbol{b} = \begin{bmatrix} 1 \\ 2 \\ -7 \end{bmatrix}$, and this gives $2b_2 - b_1 = 2 \cdot 2 - 1 = 3$ and $b_1 - b_2 = 1 - 2 = -1$.

4.6 Row reductions of the augmented matrix give

$$\left[\begin{array}{rrr|r} 1 & 1 & 1 & 3 \\ 1 & 2 & -1 & 4 \end{array}\right] \to \left[\begin{array}{rrr|r} 1 & 1 & 1 & 3 \\ 0 & 1 & -2 & 1 \end{array}\right] \to \left[\begin{array}{rrr|r} 1 & 0 & 3 & 2 \\ 0 & 1 & -2 & 1 \end{array}\right]$$

To find a particular solution, set the free variable x_3 to 0 and figure out what the pivots are. And to find the special solution, set x_3 to 1 and again figure out what the pivots should be. We get

$$\boldsymbol{x}_{\text{particular}} = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$$

$$\boldsymbol{x}_{\text{special}} = \begin{bmatrix} -3 \\ 2 \\ 1 \end{bmatrix} \qquad \text{(the nullspace consists of all multiples of this)}$$

So we write down the solution as

$$\textbf{Complete solution:} \qquad \boldsymbol{x} = \boldsymbol{x}_p + \boldsymbol{x}_n = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -3 \\ 2 \\ 1 \end{bmatrix}.$$

Figure 4.2 shows what the line of complete solution looks like (in this example it's a line in $\mathbb{R}^3$).

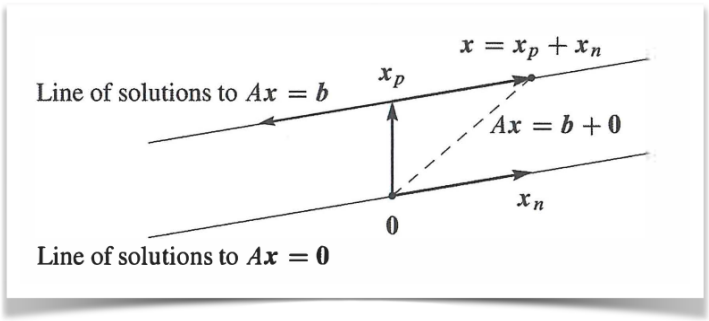


Figure 4.2: Complete solutions lie on the line shifted from the origin. The line that goes through the origin is the nullspace of A. The particular solution $\boldsymbol{x}_p$ can be any vector that juts out from the origin and lands on the line of complete solutions. The complete solution is given by choosing *one* particular $\boldsymbol{x}_p$ and adding *all* the nullspace vectors $\boldsymbol{x}_n$ to it.

5 The Inverse of A

You see $A\boldsymbol{x} = \boldsymbol{y}$ and think: "A acts on the vector $\boldsymbol{x}$ and out comes the vector $\boldsymbol{y}$." You ask: "How do I *undo* what A just did?" That is, what matrix B when applied to $\boldsymbol{y}$ will get the same $\boldsymbol{x}$ back?

Example 19. Apply this $A = \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix}$ to $\boldsymbol{x} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$ to get $\begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \boldsymbol{y}$.

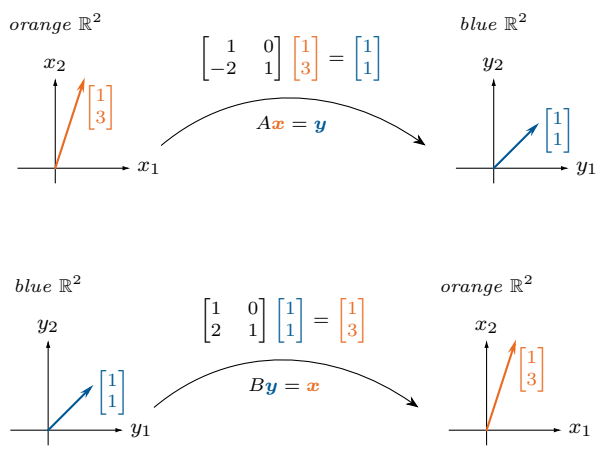


Figure 5.1: $A\boldsymbol{x} = \boldsymbol{y}$ and $B\boldsymbol{y} = \boldsymbol{x}$, so B is A's inverse: B udoes what A does: $B(A\boldsymbol{x}) = \boldsymbol{x}$.

Next, apply this matrix $B = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$ to the result $\boldsymbol{y}$. You'll get the original $\boldsymbol{x}$ back: $B\boldsymbol{y} = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$. So, B "undoes" what A does. □

Observe the following about A and B of Example 19:

$$AB = \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

$$BA = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

Whether you multiply A from the left or from the right by B, you get the same answer, I. We call I the *identity* matrix. It plays the role of unity in matrix algebra.

$$AB = I = BA.$$

An n-by-n identity matrix has 1s on the main diagonal and 0s elsewhere:

$$I = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \\ 0 & 0 & 0 & \cdots & 1 \end{bmatrix}$$

Suppose A and B are n-by-n matrices such that $AB = I$ and $BA = I$ Then, we call A and B *invertible*. We use this special notation for the inverse of A: we denote it by A^{-1}.

Definition The *inverse* of A, denoted by A^{-1}, is a matrix B such that $BA = I = AB$.

But not every matrix is invertible.

Example 20. This $A = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}$ has no inverse. There is no "undo matrix" that reverses its action. Why?

Non-invertible A Watch what happens when you apply A to $x = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$:

$$A\boldsymbol{x} = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} = 2 \cdot \begin{bmatrix} 1 \\ 3 \end{bmatrix} - 1 \cdot \begin{bmatrix} 2 \\ 6 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \mathbf{0}.$$

The non-zero vector $\begin{bmatrix} 2 \\ -1 \end{bmatrix}$ got squashed into the zero vector. No matrix can now bring that zero vector back to the original $\begin{bmatrix} 2 \\ -1 \end{bmatrix}$. Matrix A is not invertible. □

So not every matrix has an inverse. But if A-inverse exists, then it's unique.

Fact There's no way A has two different inverses.

To see why, argue by contradiction. Suppose you found two different matrices B and C such that $BA = I$ and $AC = I$. Then, because we can put the parentheses as we please in the product BAC, we have:

$$B(AC) = (BA)C \quad \text{gives} \quad BI = IC \quad \text{which is the same as} \quad B = C.$$

It's good to attempt Exercise 5.2 now.

Fact If A is invertible, the one and only solution to $A\boldsymbol{x} = \boldsymbol{b}$ is $\boldsymbol{x} = A^{-1}\boldsymbol{b}$. Here's why:

$$\text{Multiply} \quad A\boldsymbol{x} = \boldsymbol{b} \quad \text{by} \quad A^{-1}. \quad \text{Then} \quad \boldsymbol{x} = A^{-1}A\boldsymbol{x} = A^{-1}\boldsymbol{b}.$$

5.1 Inverses You Can "See" Without Computing Too Much

Sometimes you can "see" what A^{-1} is by thinking of what it means to undo the action of A.

Example 21. What is the inverse of $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$? The effect of left-multiplying P by some matrix A is to switch the rows of A:

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} c & d \\ a & b \end{bmatrix}.$$

What does it mean to "undo" the effect of P? Switching the rows once more will get us back to the original A :

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} c & d \\ a & b \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}.$$

This means that P is its own inverse! Let's check. Indeed:

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I.$$

So $P^{-1} = P$. □

☞ It's good to attempt Exercise 5.3 right after reading Example 21.

Example 22. What is the inverse of $E = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$? The effect of left-multiplying E by some matrix A is to subtract 2 times row 1 from row 2 of A. For instance,

$$\begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 9 & 9 & 9 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 9 & 9 & 9 \end{bmatrix}.$$

What does it mean to "undo" the effect of E? Adding 2 times row 1 back to row 2 will undo what E did. And this matrix

$$E^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

does exactly that. Let's check that this E^{-1} is really E-inverse:

$$EE^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I. \qquad \square$$

It's good to do a few parts of Exercise 5.4 right after reading Example 22.

Key Point The inverse of AB is $B^{-1}A^{-1}$:

$$(AB)^{-1} = B^{-1}A^{-1} \qquad \text{(notice the reversed order)}$$

To see why the order reversed, multiply AB by $B^{-1}A^{-1}$. Because you can move the parentheses around (by associativity of matrix multiplication), you'll get $BB^{-1} = I$ inside and then $AA^{-1} = I$ again:

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AIA^{-1} = AA^{-1} = I$$

Remark That inverses come in reverse order is consistent with everyday experience: If you put your socks on first and then shoes, you'd be better off reversing that procedure by taking off your shoes first and then socks.

5.2 Elementary Matrices and Row Operations

Matrices P and E from Examples 21 and 22 are known as *elementary matrices* because they correspond to the elementary row operations we discussed when we solved $Ax = 0$ and $Ax = b$ systems. For example, P will switch A's rows if you left-multiply A by P. And if you left-multiply by E, the effect is to add a multiple of one row to another.

Example 23. One of our in-class examples was to invert $A = \left[\begin{smallmatrix} 2 & 7 \\ 1 & 4 \end{smallmatrix}\right]$. The first step is to write down the augmented matrix, which is A with the identity matrix I tacked on the right of the vertical

bar: $[\,A\,|\,I\,] = \left[\begin{array}{cc|cc} 2 & 7 & 1 & 0 \\ 1 & 4 & 0 & 1 \end{array}\right]$. The goal is to row-reduce the augmented matrix so that the identity matrix sits in place of A, as shown in the last step. Then the matrix to the right of the vertical bar is the A^{-1} we seek. A^{-1} is shown in blue at the last step:

The row operation	*Changes made to the augmented matrix*	*The elementary matrix that did the job*
Augment A with the identity matrix on the right side:	$\left[\begin{array}{cc\|cc} 2 & 7 & 1 & 0 \\ 1 & 4 & 0 & 1 \end{array}\right]$	
Swap the rows:	$\left[\begin{array}{cc\|cc} 1 & 4 & 0 & 1 \\ 2 & 7 & 1 & 0 \end{array}\right]$	$P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
Subtract twice the first row from the second:	$\left[\begin{array}{cc\|cc} 1 & 4 & 0 & 1 \\ 0 & -1 & 1 & -2 \end{array}\right]$	$E = \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix}$
Multiply the second row by -1:	$\left[\begin{array}{cc\|cc} 1 & 4 & 0 & 1 \\ 0 & 1 & -1 & 2 \end{array}\right]$	$F = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$
Subtract 4 times second row from the first:	$\left[\begin{array}{cc\|cc} 1 & 0 & \mathbf{4} & \mathbf{-7} \\ 0 & 1 & \mathbf{-1} & \mathbf{2} \end{array}\right]$	$G = \begin{bmatrix} 1 & -4 \\ 0 & 1 \end{bmatrix}$

Table 5.1: Inverting $A = \begin{bmatrix} 2 & 7 \\ 1 & 4 \end{bmatrix}$ using Gauss-Jordan row-reductions gives $A^{-1} = \begin{bmatrix} 4 & -7 \\ -1 & 2 \end{bmatrix}$. Each row corresponds to an elimination step. The elementary matrix that appears in each row left-multiplies the augmented matrix that appears in the row immediately above it.

Key Point The product of the elementary matrices in Table 5.1, $GFEP$ — in that order — gives the inverse of A:

$$A = \begin{bmatrix} 2 & 7 \\ 1 & 4 \end{bmatrix},$$

$$A^{-1} = GFEP = \begin{bmatrix} 1 & -4 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 4 & -7 \\ -1 & 2 \end{bmatrix}.$$

Note that every elementary matrix is invertible. Why? Because each elementary matrix corresponds to a row operation, and any row operation can be undone by another row operation, which also corresponds to some elementary matrix. So

Key Point The inverse of an elementary matrix is an elementary matrix.

5.3 Calculating A^{-1} the Gauss-Jordan Way

Elimination – the row-reduction operations we used to solve linear systems – can also be used as a way to calculate A^{-1}. In this context, the row-reductions are known as Gauss-Jordan elimination.

The Gauss-Jordan idea is to $\boxed{\text{solve } AA^{-1} = I}$ to find each unknown column of A^{-1}.

Say you wanted to invert $A = \left[\begin{smallmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{smallmatrix}\right]$. You know that the unknown A^{-1} is such that

$$\underbrace{\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{bmatrix}}_{A} \cdot \underbrace{\begin{bmatrix} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \\ x_{31} & x_{32} & x_{33} \end{bmatrix}}_{\text{unknown } A^{-1}} = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{\text{Identity } I}. \tag{5.1}$$

When you find all the unknown x-entries in (5.1), you'll have your A^{-1}. Your first thought might be to solve for the unknown *columns* of A^{-1}, one at a time:

$$\underbrace{\begin{bmatrix}1&1&1\\1&2&2\\1&2&3\end{bmatrix}\cdot\begin{bmatrix}x_{11}\\x_{21}\\x_{31}\end{bmatrix}=\begin{bmatrix}1\\0\\0\end{bmatrix}}_{\textbf{Linear System 1}},\quad \underbrace{\begin{bmatrix}1&1&1\\1&2&2\\1&2&3\end{bmatrix}\cdot\begin{bmatrix}x_{12}\\x_{22}\\x_{32}\end{bmatrix}=\begin{bmatrix}0\\1\\0\end{bmatrix}}_{\textbf{Linear System 2}},\quad \underbrace{\begin{bmatrix}1&1&1\\1&2&2\\1&2&3\end{bmatrix}\cdot\begin{bmatrix}x_{13}\\x_{23}\\x_{33}\end{bmatrix}=\begin{bmatrix}0\\0\\1\end{bmatrix}}_{\textbf{Linear System 3}}.$$

But elimination is possible on all systems simultaneously. This is the Gauss-Jordan way of finding A^{-1}. Elimination is carried out in two phases. First, the usual Gaussian elimination leads to an upper triangular matrix. In the second phase, elimination continues by subtracting multiples of a row from the rows above. This produces zeros above the diagonal as well as below. When it reaches the identity matrix we have found A^{-1}.

This is exactly what we did in Example 23. Here's another detailed 3-by-3 example.

5.3.1 A Detailed Example

Example 24. *Finding A^{-1} the Gauss-Jordan way.* Find the inverse of $A = \begin{bmatrix}2&1&1\\4&-6&0\\-2&7&2\end{bmatrix}$.

1. Forward elimination:

$$\left[\begin{array}{rrr|rrr} \mathbf{2} & \mathbf{1} & \mathbf{1} & 1 & 0 & 0 \\ \mathbf{4} & \mathbf{-6} & \mathbf{0} & 0 & 1 & 0 \\ \mathbf{-2} & \mathbf{7} & \mathbf{2} & 0 & 0 & 1 \end{array}\right] \to \left[\begin{array}{rrr|rrr} \mathbf{2} & \mathbf{1} & \mathbf{1} & 1 & 0 & 0 \\ \mathbf{0} & \mathbf{-8} & \mathbf{-2} & -2 & 1 & 0 \\ \mathbf{-2} & \mathbf{7} & \mathbf{2} & 0 & 0 & 1 \end{array}\right]$$

row 2 - 2×row 1:

$$\text{Myltiply by } E = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\to \left[\begin{array}{rrr|rrr} \mathbf{2} & \mathbf{1} & \mathbf{1} & 1 & 0 & 0 \\ \mathbf{0} & \mathbf{-8} & \mathbf{-2} & -2 & 1 & 0 \\ \mathbf{0} & \mathbf{8} & \mathbf{3} & 1 & 0 & 1 \end{array}\right]$$

row 3 + row 1:

$$\text{Myltiply by } F = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

$$\to \left[\begin{array}{rrr|rrr} \boxed{\mathbf{2}} & 1 & 1 & 1 & 0 & 0 \\ \mathbf{0} & \boxed{\mathbf{-8}} & -2 & -2 & 1 & 0 \\ \mathbf{0} & \mathbf{0} & \boxed{\mathbf{1}} & -1 & 1 & 1 \end{array}\right]$$

row 3 + row 2:

$$\text{Myltiply by } G = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

This completes the first half – forward elimination. We have the pivots on the diagonal (boxed) and the upper triangular U appears in the first three columns. In the last three columns we have a lower triangular matrix.

Now the second half will take us from U to I in the first three columns. The inverse A^{-1} will then show up in the last three columns.

continued on the next page . . .

. . . continued from the previous page.

2. Creating zeros above the pivots:

$$\left[\begin{array}{rrr|rrr} 2 & 1 & 1 & 1 & 0 & 0 \\ 0 & -8 & -2 & -2 & 1 & 0 \\ 0 & 0 & 1 & -1 & 1 & 1 \end{array}\right] \to \left[\begin{array}{rrr|rrr} 2 & 1 & 1 & 1 & 0 & 0 \\ 0 & -8 & \mathbf{0} & -4 & 3 & 2 \\ 0 & 0 & 1 & -1 & 1 & 1 \end{array}\right]$$

row 2 + 2×row 3:

Myltiply by $K = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$

$$\to \left[\begin{array}{rrr|rrr} 2 & 1 & \mathbf{0} & 2 & -1 & -1 \\ 0 & -8 & \mathbf{0} & -4 & 3 & 2 \\ 0 & 0 & 1 & -1 & 1 & 1 \end{array}\right]$$

row 1 - row 3:

Myltiply by $M = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$$\to \left[\begin{array}{rrr|rrr} 2 & \mathbf{0} & \mathbf{0} & \frac{12}{8} & -\frac{5}{8} & -\frac{6}{8} \\ 0 & -8 & \mathbf{0} & -4 & 3 & 2 \\ 0 & 0 & 1 & -1 & 1 & 1 \end{array}\right]$$

row 1 + $\frac{1}{8}$×row 2:

Myltiply by $N = \begin{bmatrix} 1 & \frac{1}{8} & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Finally, scale each row: $$\to \left[\begin{array}{rrr|rrr} 1 & 0 & 0 & \frac{12}{16} & -\frac{5}{16} & -\frac{6}{16} \\ 0 & 1 & 0 & \frac{4}{8} & -\frac{3}{8} & -\frac{2}{8} \\ 0 & 0 & 1 & -1 & 1 & 1 \end{array}\right]$$

$\frac{\text{row1}}{2}$ and $-\frac{\text{row2}}{8}$:

Myltiply by $S = \begin{bmatrix} \frac{1}{2} & 0 & 0 \\ 0 & -\frac{1}{8} & 0 \\ 0 & 0 & 1 \end{bmatrix}$

The inverse of A is now sitting in the last 3 columns: $A^{-1} = \begin{bmatrix} \frac{12}{16} & -\frac{5}{16} & -\frac{6}{16} \\ \frac{4}{8} & -\frac{3}{8} & -\frac{2}{8} \\ -1 & 1 & 1 \end{bmatrix}$. □

☞ It's good to do Exercise 5.5 and some of Exercise 5.6 right after reading Example 24.

5.4 Exercises

Exercise 5.1. Is $(A+B)^{-1} = A^{-1} + B^{-1}$? Explain.

Exercise 5.2. We said that matrix A in Example 20 above had no inverse without really explaining why. Based on the definition of A^{-1} and the remarks made in Example 20, show the following:

If there is a nonzero vector x such that $Ax = 0$ then A cannot have an inverse.

Suggestion: argue by contradiction (as, for example, in the Fact box on page 49 about uniqueness of the inverse).

Exercise 5.3. What is the effect of right-multiplying a 2-by-2 matrix by $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$? Double-check your conjecture by calculating $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$.

Exercise 5.4. What are the inverses of the following matrices?

(a) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & 0 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ (c) $\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 3 & 1 \end{bmatrix}$

(d) $\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & 3 & 1 \end{bmatrix}$ (e) $\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ (f) $\begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Exercise 5.5. Find the inverse of $A = \begin{bmatrix} 1 & 3 & 0 \\ -2 & 3 & 1 \\ 0 & 1 & 1 \end{bmatrix}$ using Gauss-Jordan elimination.

Exercise 5.6. Compute the inverses of the following matrices using Gauss-Jordan elimination.

(a) $\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$. (b) $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{bmatrix}$. (c) $\begin{bmatrix} 2 & 3 \\ 4 & 7 \end{bmatrix}$.

Exercise 5.7. Let $E = \begin{bmatrix} 1 & 0 \\ 5 & 1 \end{bmatrix}$.

(a) Verify that E times any 2-by-2 matrix A performs the elementary row operation that adds 5 times the first row to the second row of A.

(b) Find the inverse of E and verify that $E^{-1}A$ is the matrix you get after subtracting 5 times the first row from the second row of A.

Exercise 5.8. Find the inverse of $A = \begin{bmatrix} 0 & 1 \\ 2 & 3 \end{bmatrix}$ without a matrix calculator. For an extra challenge, do it without writing anything down.

Exercise 5.9. Time to strike out on your own. Show that $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ has no inverse in a way that is different than what we've been doing in class: show that if you multiply A on the right by a generic 2-by-2 matrix, then the result must be a matrix with identical rows. Infer that the product cannot be an identity matrix.

Exercise 5.10. Find the inverse of $A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$.

Exercise 5.11. Find the conditions on a and b that make the matrix A invertible and find A^{-1} when it exists, where

$$A = \begin{bmatrix} a & b & b \\ a & a & b \\ a & a & a \end{bmatrix}.$$

Exercise 5.12. Give examples of matrices A and B such that

(a) $A + B$ is not invertible, although A and B are invertible.

(b) $A + B$ is invertible, although A and B are not invertible.

(c) all of A, B, and $A + B$ are invertible.

Exercise 5.13. Find the inverse of $A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -\frac{1}{2} & 1 & 0 & 0 \\ 0 & -\frac{2}{3} & 1 & 0 \\ 0 & 0 & -\frac{3}{4} & 1 \end{bmatrix}$.

Do it without a calculator, by thinking about what A does. How do you undo what A does? (A is a product of elementary matrices. Which ones? How do you reverse their effect?)

Then verify your answer using a matrix calculator. Did you get it right? Do you find the inverse has a nice structure related to A?

Exercise 5.14. Without a calculator: What is $FE = \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 11 & 1 \end{bmatrix}}_{F} \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 10 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}}_{E}$?

What is $(FE)^{-1}$? Keep in mind that the product of lower triangular matrices is always lower triangular. These elementary matrices are lower triangular.

Exercise 5.15. Find the numbers a and b such that $\begin{bmatrix} 3 & -1 & -1 \\ -1 & 3 & -1 \\ -1 & -1 & 3 \end{bmatrix}^{-1} = \begin{bmatrix} a & b & b \\ b & a & b \\ b & b & a \end{bmatrix}$.

Answers to Selected Problems

5.1 Of course not! This isn't even true when A and B are 1-by-1 matrices (i.e., numbers). $(2+3)^{-1} \neq 2^{-1} + 3^{-1}$.

5.2 Suppose, by way of contradiction, that A^{-1} exists and let x be a nonzero vector. Then

$$x = Ix = (A^{-1}A)x = A^{-1}(Ax) = A^{-1}0 = 0,$$

which is a contradiction because x was taken to be a nonzero vector.

5.3 Now P flips the columns, not the rows: $\begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} b & a \\ d & c \end{bmatrix}$.

5.4 We hope you didn't succumb to the temptation to plug these matrices into a matrix calculator before predicting the answers. Did you, for example, expect that the entries in positions a_{31} for parts (c) and (d) would be 6 and 5, respectively? Why did those numbers show up there? How about part (e)? Did you expect that matrix to be its own inverse?

(a) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ (c) $\begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 6 & -3 & 1 \end{bmatrix}$

(d) $\begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 5 & -3 & 1 \end{bmatrix}$ (e) $\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ (f) $\begin{bmatrix} 0 & 1 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

5.5 We carry out the usual elimination steps as in Example 24:

$$\left[\begin{array}{ccc|ccc} 1 & 3 & 0 & 1 & 0 & 0 \\ -2 & 3 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right] \to \left[\begin{array}{ccc|ccc} 1 & 3 & 0 & 1 & 0 & 0 \\ 0 & 9 & 1 & 2 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right] \to \left[\begin{array}{ccc|ccc} 1 & 3 & 0 & 1 & 0 & 0 \\ 0 & 9 & 1 & 2 & 1 & 0 \\ 0 & 0 & \frac{8}{9} & -\frac{2}{9} & -\frac{1}{9} & 1 \end{array}\right]$$

$$\to \left[\begin{array}{ccc|ccc} 1 & 3 & 0 & 1 & 0 & 0 \\ 0 & 9 & 1 & 2 & 1 & 0 \\ 0 & 0 & 1 & -\frac{1}{4} & -\frac{1}{8} & \frac{9}{8} \end{array}\right] \to \left[\begin{array}{ccc|ccc} 1 & 3 & 0 & 1 & 0 & 0 \\ 0 & 9 & 0 & \frac{9}{4} & \frac{9}{8} & -\frac{9}{8} \\ 0 & 0 & 1 & -\frac{1}{4} & -\frac{1}{8} & \frac{9}{8} \end{array}\right]$$

$$\to \left[\begin{array}{ccc|ccc} 1 & 3 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & \frac{1}{4} & \frac{1}{8} & -\frac{1}{8} \\ 0 & 0 & 1 & -\frac{1}{4} & -\frac{1}{8} & \frac{9}{8} \end{array}\right] \to \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{4} & -\frac{3}{8} & \frac{3}{8} \\ 0 & 1 & 0 & \frac{1}{4} & \frac{1}{8} & -\frac{1}{8} \\ 0 & 0 & 1 & -\frac{1}{4} & -\frac{1}{8} & \frac{9}{8} \end{array}\right]$$

The inverse of A can be read from the last 3 columns of the last matrix: $A^{-1} = \begin{bmatrix} \frac{1}{4} & -\frac{3}{8} & \frac{3}{8} \\ \frac{1}{4} & \frac{1}{8} & -\frac{1}{8} \\ -\frac{1}{4} & -\frac{1}{8} & \frac{5}{8} \end{bmatrix}$.

5.6 Following the Gauss-Jordan procedure, we get (a): $\begin{bmatrix} 3/4 & 1/2 & 1/4 \\ 1/2 & 1 & 1/2 \\ 1/4 & 1/2 & 3/4 \end{bmatrix}$, (b): $\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix}$, (c) $\begin{bmatrix} 7/2 & -3/2 \\ -2 & 1 \end{bmatrix}$.

5.10 The answer is $A^{-1} = \begin{bmatrix} 0 & -1 & 1 \\ -1 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$.

$$\left[\begin{array}{ccc|ccc} 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 \end{array}\right] \xrightarrow{\text{flip rows 1 \& 3}} \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \end{array}\right] \xrightarrow{\text{row 2 - row 3}} \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \end{array}\right]$$

$$\xrightarrow{\text{row 1 - row 3}} \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & -1 & 0 & 1 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \end{array}\right] \xrightarrow{\text{row 1 - row 2}} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & -1 & 1 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \end{array}\right]$$

5.15 $a = 1/2, b = 1/4$. For an extra challenge (optional), can you guess what $\begin{bmatrix} 4 & -1 & -1 & -1 \\ -1 & 4 & -1 & -1 \\ -1 & -1 & 4 & -1 \\ -1 & -1 & -1 & 4 \end{bmatrix}^{-1}$ is?

The answer follows the same pattern, only this time with $a = 2/5, b = 1/5$: $\begin{bmatrix} \frac{2}{5} & \frac{1}{5} & \frac{1}{5} & \frac{1}{5} \\ \frac{1}{5} & \frac{2}{5} & \frac{1}{5} & \frac{1}{5} \\ \frac{1}{5} & \frac{1}{5} & \frac{2}{5} & \frac{1}{5} \\ \frac{1}{5} & \frac{1}{5} & \frac{1}{5} & \frac{2}{5} \end{bmatrix}$.

Part II

Eigenvalues & Eigenvectors

6 Eigenstuff (Part 1 of 3) – Basics

A always sends $\mathbf{0}$ to $\mathbf{0}$ but *generally moves vectors* around: $\boldsymbol{x}$ and $A\boldsymbol{x}$ usually don't lie along the same line. For example, the blue vectors $\boldsymbol{x}$ and $A\boldsymbol{x}$ in Figure 6.1 point in different directions.

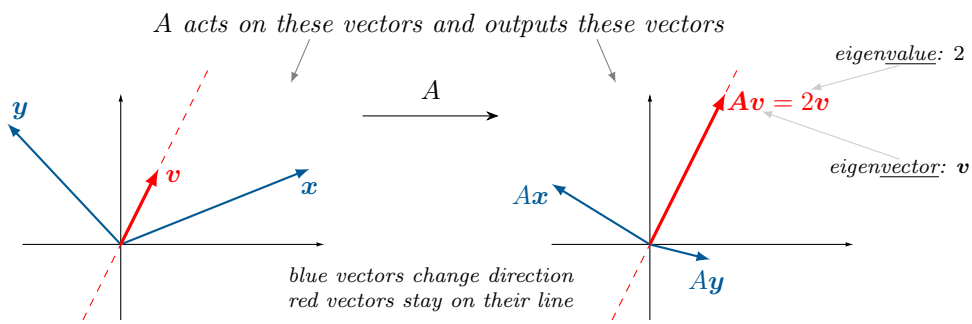


Figure 6.1: Any vector on the red dotted line is an eigenvector of A. For example, $\boldsymbol{v}$ is an eigenvector. Because $\boldsymbol{v}$ gets stretched twice, $\lambda = 2$ is the eigenvalue associated with $\boldsymbol{v}$.

> But what if some vector $\boldsymbol{v}$ had the *special* property that $A\boldsymbol{v}$ lies on the same line as $\boldsymbol{v}$? That would mean that $A\boldsymbol{v}$ is a scalar multiple of $\boldsymbol{v}$. For example, it might be that $A\boldsymbol{v} = 2\boldsymbol{v}$ or it might be that $A\boldsymbol{v} = -\frac{1}{5}\boldsymbol{v}$. That scalar multiple of $\boldsymbol{v}$ is notated by λ, which is just a number like 2, $-\frac{1}{5}$, or -2166.

Calling that scalar multiple λ is just a convention – you could call it anything that makes you happy if you are not a stickler for conventions – but we'll stick with λ.

Key Point A acts on $\boldsymbol{v}$ and out comes $A\boldsymbol{v}$, usually pointing in a different direction from $\boldsymbol{v}$.

But some exceptional vectors do not change direction. Those are the *eigenvectors* of A.

Definition We say that λ is an *eigenvalue* of A if $A\boldsymbol{v} = \lambda\boldsymbol{v}$ for some nonzero vector $\boldsymbol{v}$. If such a vector exists, we call it the *eigenvector* of A.

6.1 Making Sense of What $Av = \lambda v$ Means

The first time you see an equation like $A\boldsymbol{v} = \lambda\boldsymbol{v}$, you might wonder how on earth would one go about solving it, especially when you realize that you have to solve for $\boldsymbol{v}$ *and* you don't know what λ is!

One obstacle seems to be that on the left side in $A\boldsymbol{v} = \lambda\boldsymbol{v}$, you have a matrix times $\boldsymbol{v}$ and on the right side you have a number times $\boldsymbol{v}$. The key to dealing with this is to rewrite the equation so as to make both sides a matrix times $\boldsymbol{v}$ by simply introducing the identity matrix on the right side.

Example 25. $A\boldsymbol{v} = \lambda\boldsymbol{v}$ says that matrix times vector $\boldsymbol{v}$ equals the number λ times $\boldsymbol{v}$. Rewrite $A\boldsymbol{v} = \lambda\boldsymbol{v}$ in the form that says matrix times $\boldsymbol{v}$ equals matrix times $\boldsymbol{v}$.

Answer:

$$\boxed{A\boldsymbol{v} = \lambda\boldsymbol{v} \quad \textit{is the same as} \quad A\boldsymbol{v} = \lambda I\boldsymbol{v}.}$$

To see why, take $A\boldsymbol{v} = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \lambda \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \lambda\boldsymbol{v}$ as a concrete example.

$$\begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \lambda \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \quad \text{is the same as} \quad \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} \lambda v_1 \\ \lambda v_2 \end{bmatrix}$$

$$\text{is the same as} \quad \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} \lambda v_1 \\ \lambda v_2 \end{bmatrix}$$

$$\text{is the same as} \quad \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

$$\text{is the same as} \quad A\boldsymbol{v} = \lambda I \boldsymbol{v}.$$

Example 26. Continuing from Example 25, rewrite $A\boldsymbol{v} = \lambda\boldsymbol{v}$ as a homogeneous system of equations. That is, rewrite it as "matrix times $\boldsymbol{v}$ equals the $\mathbf{0}$ vector."

Answer:

$$A\boldsymbol{v} = \lambda\boldsymbol{v} \quad \textit{is is the same as} \quad (A - \lambda I)\boldsymbol{v} = \mathbf{0}.$$

$$\begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \lambda \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \quad \text{is the same as} \quad \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

$$\text{is the same as} \quad \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\text{is the same as} \quad \begin{bmatrix} 1-\lambda & 4 \\ 2 & 3-\lambda \end{bmatrix}\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\text{is the same as} \quad (A - \lambda I)\boldsymbol{v} = \mathbf{0}.$$

The point of Examples 25 and 26 was to see the following key points.

Key Point $A\boldsymbol{v} = \lambda\boldsymbol{v}$ is the same as the homogeneous system of equations $(A-\lambda I)\boldsymbol{v} = \mathbf{0}$.

Solving $(A - \lambda I)\boldsymbol{v} = \mathbf{0}$ for nonzero $\boldsymbol{v}$'s will give you the eigenvectors.

In other words, the question becomes: Is there anything other than the zero vector in the nullspace of $A - \lambda I$? For some λ's there may be, and those λ's are the eigenvalues of A.

Key Point Those λ's that make $A - \lambda I$ singular are the eigenvalues of A.

Now, remembering that a matrix is singular if and only if its determinant is zero, we have our final key point of this section: You find the eigenvalues by setting the determinant of $A - \lambda I$ to zero and solving for λ.

Key Point Solving $\det(A - \lambda I) = 0$ for λ will give you the eigenvalues.

When A is an n-by-n matrix, $\det(A - \lambda I)$ is a polynomial of degree n in the unknown λ. It is called the *characteristic polynomial*, and it will have n roots (this fact is the so-called fundamental theorem of algebra). Some of those roots you may not like because they may be complex numbers, and some of those roots may be repeated, but there will be n roots. If you get lucky, all roots will be real (this will happen on the quizzes). And if you get even luckier, all the roots will be distinct. We'll see soon why having n distinct real roots is the dream scenario.

6.2 How to Find Eigenvalues and Eigenvectors

To solve the eigenvalue problem for an n by n matrix, follow these steps:

- ***Compute the determinant of*** $A - \lambda I$. With λ subtracted along the diagonal, this determinant is a polynomial in λ of degree n and includes the term λ^n or $-\lambda^n$.

- ***Find the roots of this polynomial*** by solving $\det(A - \lambda I) = 0$. The n roots are the n eigenvalues of A. They make $A - \lambda I$ singular.

- For each eigenvalue λ, ***solve*** $(A - \lambda I)\boldsymbol{x} = 0$ ***to find an eigenvector*** $\boldsymbol{x}$.

Example 27. Find the eigenvalues and eigenvectors of $A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$.

The first step is to write down $A - \lambda I$ and compute its determinant.

$$\det \begin{bmatrix} 1-\lambda & 4 \\ 2 & 3-\lambda \end{bmatrix} \qquad \Rightarrow \qquad (1-\lambda)\cdot(3-\lambda) - 2\cdot 4 \qquad \Rightarrow \qquad \lambda^2 - 4\lambda - 5.$$

This is the the characteristic polynomial of A.

The next step is to find the roots of the characteristic polynomial $\lambda^2 - 4\lambda - 5$. This is done by setting it to zero and solving for λ.

$$\lambda^2 - 4\lambda - 5 = 0. \tag{6.1}$$

Equation (6.1) is of degree 2 and has two solutions:

$$\lambda^2 - 4\lambda - 5 = 0 \qquad \Rightarrow \qquad (\lambda - 5)\cdot(\lambda + 1) = 0 \qquad \Rightarrow \qquad \boxed{\lambda_1 = -1 \quad \text{and} \quad \lambda_2 = 5}.$$

The final step is to find an eigenvector associated with each λ.

For each eigenvalue λ solve $(A - \lambda I)\boldsymbol{x} = \mathbf{0}$ to find an eigenvector $\boldsymbol{x}$.

- $\lambda_1 = -1$: $\begin{bmatrix} 1-\lambda_1 & 4 \\ 2 & 3-\lambda_1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow 2x_1 - 4x_2 = 0$

  yields an eigenvector $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$.

- $\lambda_2 = 5$: $\begin{bmatrix} 1-\lambda_2 & 4 \\ 2 & 3-\lambda_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -4 & 4 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow x_1 - x_2 = 0$

  yields an eigenvector $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

☞ It's good to do Exercises 6.1, 6.2, 6.3, and 6.4 right after reading Example 27.

The matrix you worked with in Exercises 6.1 and 6.2, $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, is called a reflection matrix — you give it any vector $\boldsymbol{x}$ and it gives back $A\boldsymbol{x}$ which is a reflection of $\boldsymbol{x}$ across the $45°$ line. The one in Example 28 rotates $\mathbb{R}^2$ by $90°$.

Example 28. What are the eigenvalues and eigenvectors of $A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$?

This A is called the rotation matrix – you give it a vector like $\boldsymbol{v} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and out comes the vector $A\boldsymbol{v} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$. Those two vectors, $\boldsymbol{v}$ and $A\boldsymbol{v}$, are perpendicular to one another. (Draw them if you are unsure.) A rotates every vector in the plane counterclockwise by $90°$.

It's not shocking, then, that this matrix doesn't have any eigenvectors in the real $\mathbb{R}^2$ plane: If all vectors are rotated, no vector can come out parallel to itself. But let's check what our standard procedure has to say. Setting the characteristic polynomial of A to zero, we get this:

$$\det(A - \lambda I) = 0 \quad \Rightarrow \quad \det\left(\begin{bmatrix} -\lambda & 1 \\ -1 & -\lambda \end{bmatrix}\right) = 0 \quad \Rightarrow \quad \lambda^2 + 1 = 0.$$

And $\lambda^2 + 1 = 0$ doesn't have any real solutions.

Aside (optional): If you are familiar with the complex numbers, you'll spot the complex roots of $\lambda^2 + 1 = 0$. Those are $-i$ and i, the eigenvalues of A. And i shows up in the eigenvectors as well:

$$\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}\begin{bmatrix} 1 \\ i \end{bmatrix} = i\begin{bmatrix} 1 \\ i \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}\begin{bmatrix} i \\ 1 \end{bmatrix} = -i\begin{bmatrix} i \\ 1 \end{bmatrix}.$$

Somehow the complex vectors $\begin{bmatrix} 1 \\ i \end{bmatrix}$ and $\begin{bmatrix} i \\ 1 \end{bmatrix}$ keep their direction as they are rotated.

Moral of the story: real matrices can easily have complex eigenvalues and eigenvectors.

6.3 Eigenspaces

Key Point Suppose $\boldsymbol{x}$ is an eigenvector of A associated with the eigenvalue λ. Then any vector that lies on the same line as $\boldsymbol{x}$ is also an eigenvector of A.

In other words, the vector $c\,\boldsymbol{x}$, where c is any number, is an eigenvector or A whenever $\boldsymbol{x}$ is an eigenvector of A.

We denote the space of eigenvectors associated with λ by E_λ:

<u>*Notation*</u>*:* E_λ stands for the nullspace of $A - \lambda I$.

The following examples will make the notation introduced in the above Key Point clear.

Example 29. Find the eigenvalues of $A = \begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix}$ and the eigenspaces associated with those eigenvalues.

Step 1. Compute the determinant of $A - \lambda I$.

$$\det\left(\begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}\right) = \det\begin{bmatrix} 1-\lambda & 2 \\ 5 & 4-\lambda \end{bmatrix} \quad \Rightarrow \quad (1-\lambda)\cdot(4-\lambda) - 10$$

$$\Rightarrow \quad \lambda^2 - 5\lambda - 6.$$

This is the the characteristic polynomial of A.

Step 2. Find the roots of the characteristic polynomial $\lambda^2 - 5\lambda - 6$ by setting it to zero and solving for λ.

$$\lambda^2 - 5\lambda - 6 = 0 \qquad \text{(factor the polynomial)}$$

$$(\lambda+1)(\lambda-6) = 0$$

$$\boxed{\lambda_1 = -1 \text{ and } \lambda_2 = 6} \text{ are the eigenvalues of } A.$$

Step 3. For each eigenvalue λ, find the nullspace of $(A - \lambda I)$. That is, for each λ, solve $(A - \lambda I)\boldsymbol{x} = \mathbf{0}$. The solutions give the eigenspaces associated with the λs.

- $\lambda_1 = -1$: Solve $\begin{bmatrix} 1-(-1) & 2 \\ 5 & 4-(-1) \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ 5 & 5 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$.

As expected, only one of the equations matters[1]. The solution is any multiple of $\begin{bmatrix} -1 \\ 1 \end{bmatrix}$:

$$\begin{bmatrix} 2 & 2 \\ 5 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow 2x_1 + 2x_2 = 0 \text{ has the special solution}$$

$$\boldsymbol{x}_1 = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}, \text{ which is an eigenvector for } \lambda_1 = -1.$$

Observation: Every multiple of an eigenvector is also an eigenvector.

For example, $20\,\boldsymbol{x}_1 = 20 \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} -20 \\ 20 \end{bmatrix}$ is also an eigenvector. Similarly, $-2166\,\boldsymbol{x}_1 = \begin{bmatrix} 2166 \\ -2166 \end{bmatrix}$ is also an eigenvector.

We call the set of all eigenvectors associated with λ the *eigenspace* of A corresponding to λ and denote it by E_λ.

Here, $E_{-1} = \text{span}\left(\begin{bmatrix} -1 \\ 1 \end{bmatrix}\right) = c \begin{bmatrix} -1 \\ 1 \end{bmatrix}$, where c is an arbitrary number, is the eigenspace associated with $\lambda = -1$.

[1] We reiterate that one of the equations here is redundant. Gaussian elimination shows this clearly: the first elimination step applied to the matrix $A - (-1)I = \begin{bmatrix} 2 & 2 \\ 5 & 5 \end{bmatrix}$ produces

$$\begin{bmatrix} 2 & 2 \\ 5 & 5 \end{bmatrix} \xrightarrow{\text{row } 2 - \frac{5}{2}\cdot\text{row } 1} \begin{bmatrix} 2 & 2 \\ 0 & 0 \end{bmatrix}, \text{ making the second equation redundant.}$$

As usual, setting the free variable $x_2 = 1$ uniquely determines the pivot: $x_1 = -1$, producing a special solution $\boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$. Thus, the entire nullspace of $\begin{bmatrix} 2 & 2 \\ 5 & 5 \end{bmatrix}$ consists of all multiples of $\begin{bmatrix} -1 \\ 1 \end{bmatrix}$: the nullspace is $c \begin{bmatrix} -1 \\ 1 \end{bmatrix}$.

- $\lambda_2 = 6$: Solve $\begin{bmatrix} 1-6 & 2 \\ 5 & 4-6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -5 & 2 \\ 5 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$. Again, as expected, only one equation matters. The solution is any multiple of $\begin{bmatrix} 2 \\ 5 \end{bmatrix}$:

$$\begin{bmatrix} -5 & 2 \\ 5 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow -5x_1 + 2x_2 = 0 \text{ has the special solution}$$

$$\boldsymbol{x}_1 = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2/5 \\ 1 \end{bmatrix}, \text{ which is an eigenvector for } \lambda_2 = 6.$$

The *eigenspace* of $A = \begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix}$ corresponding to $\lambda = 6$ is

$$E_6 = \text{span}\left(\begin{bmatrix} 2/5 \\ 1 \end{bmatrix}\right) = c \begin{bmatrix} 2/5 \\ 1 \end{bmatrix}, \text{ where } c \text{ is an arbitrary number.}$$

$A = \begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix}$ has two linearly independent eigenvectors. One eigenvector corresponds to the eigenvalue $\lambda = -1$ and the other to $\lambda = 6$. Each eigenvector generates its own eigenspace.

Any vector on the plane can be expressed as a linear combination of A's eigenvectors because the eigenvectors are linearly independent.

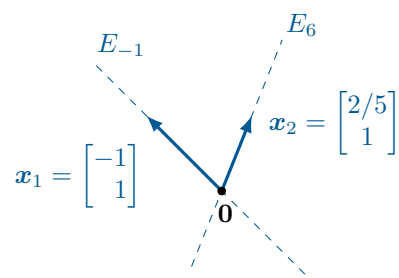


Figure 6.2 shows the vector $\boldsymbol{v} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$ expressed as a linear combination of A's eigenvectors:

$$\boldsymbol{v} = \begin{bmatrix} 4 \\ 3 \end{bmatrix} = (-2) \begin{bmatrix} -1 \\ 1 \end{bmatrix} + 5 \begin{bmatrix} 2/5 \\ 1 \end{bmatrix} = (-2)\,\boldsymbol{x}_1 + 5\,\boldsymbol{x}_2$$

In fact, any vector $\boldsymbol{v}$ in $\mathbb{R}^2$ can be written as $\boldsymbol{v} = a\,\boldsymbol{x}_1 + b\,\boldsymbol{x}_2$, where a and b are some numbers.

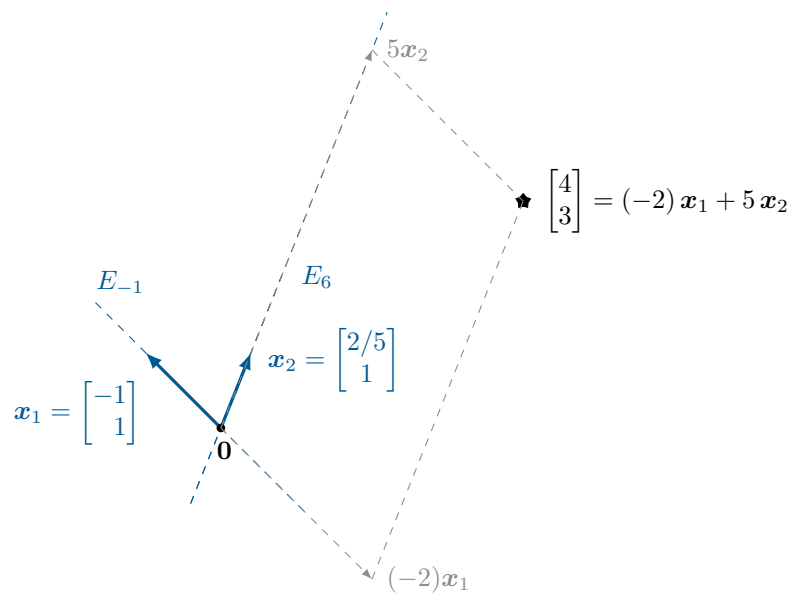


Figure 6.2: *The nice case:* The n eigenvectors of an n-by-n matrix A form a basis in $\mathbb{R}^n$, so any vector $\mathbb{R}^n$ is a linear combination of the eigenvectors. In Example 29, $n = 2$, so any vector in the plane is a combination of A's two eigenvectors.

As we'll see in the following sections, when an n-by-n matrix has n linearly independent eigenvectors as in Figure 6.2, then calculating A^n becomes particularly easy.

Unfortunately, not every n-by-n matrix is nice enough to have n linearly independent eigenvectors. When that happens, the eigenvectors do not form a basis in $\mathbb{R}^n$, and we can not express arbitrary vectors as linear combinations of A's eigenvectors. We'll see examples of such matrices in the coming sections.

6.3.1 Repeatedly Applying A To Eigenvectors

Consider $A = \begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix}$ of Example 29 again. Suppose you wanted to find A^{42} times the vector $\begin{bmatrix} 2 \\ 5 \end{bmatrix}$. Because $\begin{bmatrix} 2 \\ 5 \end{bmatrix}$ is an eigenvector with eigenvalue $\lambda = 6$, the matrix A acts like the multiplication by the number 6 :

$$\begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix}^{42} \begin{bmatrix} 2 \\ 5 \end{bmatrix} = 6^{42} \begin{bmatrix} 2 \\ 5 \end{bmatrix}.$$

So instead of calculating A^{42}, which can be time-consuming, we just calculate 6^{42}, which is not at all time-consuming, and multiply the result by the components of the vector $\begin{bmatrix} 2 \\ 5 \end{bmatrix}$ to get the answer.

6.3.2 Repeatedly Applying A To Any Vector

Now suppose you want to calculate A^{42} times the vector $\begin{bmatrix} 0 \\ 7 \end{bmatrix}$. Unfortunately, the vector $\begin{bmatrix} 0 \\ 7 \end{bmatrix}$ is not an eigenvector of A, so at first a shortcut for computation isn't obvious. But notice that $\begin{bmatrix} 0 \\ 7 \end{bmatrix}$ is a linear combination of A's eigenvectors:

$$\begin{bmatrix} 0 \\ 7 \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \end{bmatrix} + \begin{bmatrix} -2 \\ 2 \end{bmatrix}.$$

So A^{42} can apply easily to both terms of the sum:

$$\begin{aligned} \begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix}^{42} \begin{bmatrix} 0 \\ 7 \end{bmatrix} &= \begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix}^{42} \left(\begin{bmatrix} 2 \\ 5 \end{bmatrix} + \begin{bmatrix} -2 \\ 2 \end{bmatrix} \right) \\ &= 6^{42} \begin{bmatrix} 2 \\ 5 \end{bmatrix} + (-1)^{42} \begin{bmatrix} -2 \\ 2 \end{bmatrix} \\ &= \begin{bmatrix} 2 \times 6^{42} - 2 \\ 5 \times 6^{42} - 2 \end{bmatrix} \end{aligned}$$

So again, we don't need to calculate A^{42} to know what A^{42} times $\begin{bmatrix} 0 \\ 7 \end{bmatrix}$ is. Notice that the same kind of calculation can be done with any vector other than $\begin{bmatrix} 0 \\ 7 \end{bmatrix}$ because the eigenvectors $\begin{bmatrix} 2 \\ 5 \end{bmatrix}$ and $\begin{bmatrix} -2 \\ 2 \end{bmatrix}$ form a basis in $\mathbb{R}^2$ (as shown in Figure 6.2).

Thus, we made calculating A^{42} times $\begin{bmatrix} 0 \\ 7 \end{bmatrix}$ easier by expressing $\begin{bmatrix} 0 \\ 7 \end{bmatrix}$ as a linear combination of A's eigenvectors.

The basis of eigenvectors is a more convenient basis than any other when you try to understand the structure of the matrix A or try to calculate its powers A^n.

It is not always possible to find n eigenvectors for an n-by-n matrix that form a basis in $\mathbb{R}^n$. But when it is, as in the examples so far, calculating A^n becomes very easy.

6.4 Some Shortcuts

The *trace* of a A is the sum of its diagonal entries. For example, $\text{trace}\left(\begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix}\right) = 1+4 = 5$.

> **Key Point** $\text{trace}(A)$ is equal to the sum of A's eigenvalues.

For example, for $A = \begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix}$, we have $5 = 1 + 4 = \text{trace}(A) = \lambda_1 + \lambda_2 = (-1) + 6 = 5$.

> **Key Point** $\det(A)$ is equal to the product of A's eigenvalues.

For example, $\det\left(\begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix}\right) = 1 \cdot 4 - 5 \cdot 2 = -6$. On the other hand, $\lambda_1 \cdot \lambda_2 = (-1) \cdot 6 = -6$.

Example 30. The characteristic polynomial of $A = \begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix}$ is

$$\det\left(\begin{bmatrix} 1-\lambda & 2 \\ 5 & 4-\lambda \end{bmatrix}\right) = \lambda^2 - 5\lambda - 6.$$

Do you see how the $\text{trace}(A)$ shows up as the coefficient by λ? And do you see how the determinant makes an appearance as the last term in the characteristic polynomial? □

The last two Key Points (page 77) we made about the determinant and trace work for any square matrices. But for 2-by-2 matrices, you can sometimes use these facts to quickly compute the eigenvalues.

Example 31. What are the eigenvalues of $A = \begin{bmatrix} 1 & 4 \\ 2 & 8 \end{bmatrix}$?

Instead of chugging through the usual process of finding λ_1 and λ_2, we take a shortcut.

- First we note that $\text{trace}(A) = 1 + 8 = 9 = \lambda_1 + \lambda_2$.

- Then we note that because A's rows (or columns) are multiples of one another, A's determinant must be $0 = \lambda_1 \cdot \lambda_2$.
- Finally, we conclude that $\lambda_1 = 0$ and $\lambda_2 = 9$ because these are the only two numbers that add up to 9 whose product is 0.

Answer: $\lambda_1 = 0$ and $\lambda_2 = 9$ are the eigenvalues of $\begin{bmatrix} 1 & 4 \\ 2 & 8 \end{bmatrix}$. □

☞ It's good to do Exercise 6.5 right after reading Examples 30 and 31.

6.4.1 What Happens to the Eigenstuff If You Shift, Square, or Invert A?

What about the eigenvalues and eigenvectors of $A + 5I, A^2$, or A^{-1}?

The matrix $A = \begin{bmatrix} 1 & 1 \\ -2 & 4 \end{bmatrix}$

- has an eigenvalue $\lambda_1 = 2$ that goes with the eigenvector $\boldsymbol{x}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$,
- and an eigenvalue $\lambda_2 = 3$ that goes with the eigenvector $\boldsymbol{x}_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$.

☞ It's good to do Exercise 6.6 now, before reading on.

Key Point $A + 9I, A^2$, or A^{-1} all keep the same eigenvectors as A.

Their eigenvalues are $\lambda + 9, \lambda^2$, and $\dfrac{1}{\lambda}$, respectively, whenever λ is A's eigenvalue.

Why? Suppose $\boldsymbol{x}$ is A's eigenvector. Then $A\boldsymbol{x} = \lambda\boldsymbol{x}$. Now see what happens if you multiply through by A:

$$A^2\boldsymbol{x} = A(A\boldsymbol{x}) = A(\lambda\boldsymbol{x}) = \lambda(A\boldsymbol{x}) = \lambda(\lambda\boldsymbol{x}) = \lambda^2\boldsymbol{x} \tag{6.2}$$

So λ^2 must be an eigenvalue for A^2 and $\boldsymbol{x}$ its eigenvector. You should check if a similar argument works on $A + 2022\,I$ and A^{-1}.

☞ If you didn't do exercise 6.6, do it now and explain why the above Key Point makes sense.

6.5 Examples: Column-Stochastic Matrices

Example 32. Consider a matrix like $A = \begin{bmatrix} .8 & .3 \\ .2 & .7 \end{bmatrix}$ whose entries are positive numbers and whose columns sum up to 1. Such matrices are called column-stochastic[2] or Markov matrices and come up in applications often.

Going through the usual process of finding eigenvalues and eigenvectors for $A = \begin{bmatrix} .8 & .3 \\ .2 & .7 \end{bmatrix}$, we find that the eigenvalues are $\lambda_1 = 1$ and $\lambda_2 = 1/2$. Check:

- $\begin{bmatrix} .8 & .3 \\ .2 & .7 \end{bmatrix} \begin{bmatrix} .6 \\ .4 \end{bmatrix} = \begin{bmatrix} .6 \\ .4 \end{bmatrix}$, so $\boldsymbol{x}_1 = \begin{bmatrix} .6 \\ .4 \end{bmatrix}$ is an eigenvector for $\lambda_1 = 1$.
- $\begin{bmatrix} .8 & .3 \\ .2 & .7 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} .5 \\ -.5 \end{bmatrix} = \frac{1}{2}\begin{bmatrix} 1 \\ -1 \end{bmatrix}$, so $\boldsymbol{x}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ is an eigenvector for $\lambda_2 = \frac{1}{2}$.

Note the peculiar property of the eigenvector $\boldsymbol{x}_1 = \begin{bmatrix} .6 \\ .4 \end{bmatrix}$ associated with the eigenvalue $\lambda_1 = 1$: when $\boldsymbol{x}_1$ is multiplied by A again, we still get $\boldsymbol{x}_1$. In fact, every power of A will give $A^n\boldsymbol{x}_1 = \boldsymbol{x}_1$:

$$A\boldsymbol{x}_1 = \boldsymbol{x}_1 \qquad \text{(because } \lambda_1 = 1 \text{ is an eigenvalue)}$$
$$A^2\boldsymbol{x}_1 = A(A\boldsymbol{x}_1) = A(\lambda_1\boldsymbol{x}_1) = A\boldsymbol{x}_1 = \boldsymbol{x}_1 \qquad \text{(applying } A \text{ again still gives } \boldsymbol{x}_1\text{)}$$

The blue vector in Figure 6.3 is the eigenvector that corresponds to the eigenvalue $\lambda_1 = 1$. No matter how many times A is applied to it, this eigenvector stays put.

On the other hand, the red vector in Figure 6.3 is the eigenvector that corresponds to the eigenvalue $\lambda_2 = 1/2$. Each successive application of A to this vector shrinks it twice. As n increases, $A^n\boldsymbol{x}_2$ approaches the $\mathbf{0}$ vector:

$$A\boldsymbol{x}_2 = \frac{1}{2}\boldsymbol{x}_2 \qquad \text{(because } \lambda_2 = \frac{1}{2} \text{ is an eigenvalue)}$$
$$A^2\boldsymbol{x}_2 = A(A\boldsymbol{x}_2) = A(\lambda_2\boldsymbol{x}_2) = \frac{1}{2}A\boldsymbol{x}_2 = \frac{1}{2^2}\boldsymbol{x}_2 \qquad \text{(applying } A \text{ again shrinks } \boldsymbol{x}_2 \text{ twice)}$$

[2] In probabilistic models – weather prediction is an example of such a model – the matrix entries are interpreted as probabilities. A given matrix entry is viewed as a transition probability from, say, a sunny day to a rainy day. The meteorologist may want to examine matrix powers A^n to estimate the probability of observing a rainy day n days from now.

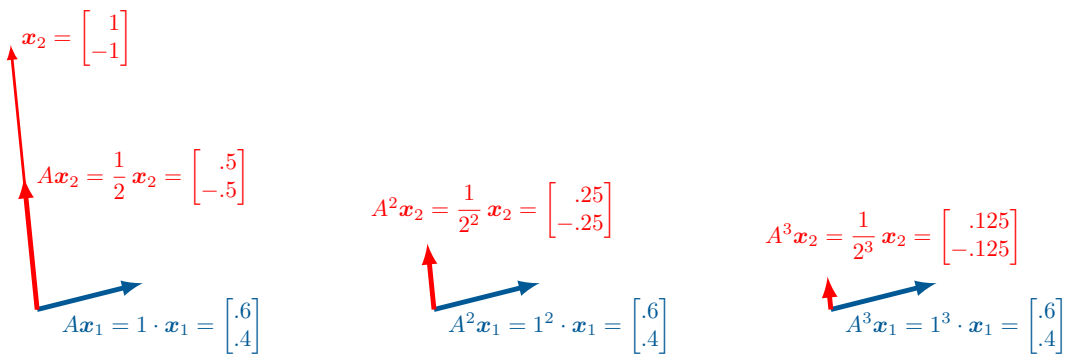


Figure 6.3: Repeated applications of A to its eigenvectors. The blue eigenvector corresponds to $\lambda_1 = 1$ and stays put no matter how many times A is applied to it. The red eigenvector corresponds to $\lambda_2 = \frac{1}{2}$ and shrinks twice every time A is applied to it. Of course, each eigenvector stays on its own line.

The last line echoes the point we made earlier in Equation 6.2: if A has eigenvalue λ, then λ^2 is an eigenvalue of A^2.

Both of $A = \begin{bmatrix} .8 & .3 \\ .2 & .7 \end{bmatrix}$'s columns add up to 1 and A has an eigenvalue $\lambda_1 = 1$. This is true generally: any column-stochastic matrix has an eigenvalue $\lambda = 1$.

The eigenvalues of A^2 are 1^2 and $\left(\frac{1}{2}\right)^2$ — the squares of A's eigenvalues.

When A is squared, the eigenvectors stay the same. The eigenvalues are squared.

Generally, $A^n\boldsymbol{x} = \lambda^n\boldsymbol{x}$ whenever $\boldsymbol{x}$ is an eigenvector for λ.

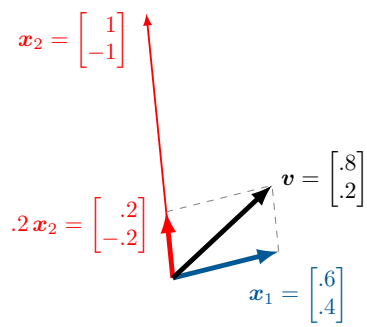
It's good to do Exercise 6.7 right after reading Example 32.

Repeated Applications of A To Any Vector

Example 33. The eigenvectors of A from Example 32 do not change direction when A is applied to them. All other vectors *do* change direction. But all other vectors are also combinations of the two eigenvectors. For example, consider the vector $\boldsymbol{v} = \begin{bmatrix} .8 \\ .2 \end{bmatrix}$, which is expressed as this combination $\boldsymbol{x}_1 + (.2)\boldsymbol{x}_2$.

Separate $\boldsymbol{v}$ into eigenvectors:

$$\begin{aligned} \boldsymbol{v} = \begin{bmatrix} .8 \\ .2 \end{bmatrix} &= \boldsymbol{x}_1 + (.2)\,\boldsymbol{x}_2 \\ &= \begin{bmatrix} .6 \\ .4 \end{bmatrix} + (.2) \begin{bmatrix} 1 \\ -1 \end{bmatrix} \end{aligned}$$



Because $\boldsymbol{v}$ is separated into eigenvectors (that is, $\boldsymbol{v}$ is expressed as a linear combination of the eigenvectors), we can easily calculate repeated applications of A on $\boldsymbol{v}$. That is, it is easy to calculate $A^n\boldsymbol{v}$. Here's why:

$$\begin{aligned} A\boldsymbol{v} = A \begin{bmatrix} .8 \\ .2 \end{bmatrix} &= A\left(\boldsymbol{x}_1 + (.2)\,\boldsymbol{x}_2\right) = A\boldsymbol{x}_1 + (.2)\,A\boldsymbol{x}_2 = \lambda_1\boldsymbol{x}_1 + (.2)\,\lambda_2\boldsymbol{x}_2 \\ &= 1 \cdot \begin{bmatrix} .6 \\ .4 \end{bmatrix} + (.2)\,\frac{1}{2} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} .7 \\ .3 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} A^2\boldsymbol{v} = A^2 \begin{bmatrix} .8 \\ .2 \end{bmatrix} &= A^2\left(\boldsymbol{x}_1 + (.2)\,\boldsymbol{x}_2\right) = A^2\boldsymbol{x}_1 + (.2)\,A^2\boldsymbol{x}_2 = \lambda_1^2\boldsymbol{x}_1 + (.2)\,\lambda_2^2\boldsymbol{x}_2 \\ &= 1^2 \cdot \begin{bmatrix} .6 \\ .4 \end{bmatrix} + (.2)\,\frac{1}{2^2} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} .65 \\ .35 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} A^n\boldsymbol{v} = A^n \begin{bmatrix} .8 \\ .2 \end{bmatrix} &= A^n\left(\boldsymbol{x}_1 + (.2)\,\boldsymbol{x}_2\right) = A^n\boldsymbol{x}_1 + (.2)\,A^n\boldsymbol{x}_2 = \lambda_1^n\boldsymbol{x}_1 + (.2)\,\lambda_2^n\boldsymbol{x}_2 \\ &= 1^n \cdot \begin{bmatrix} .6 \\ .4 \end{bmatrix} + (.2)\,\frac{1}{2^n} \begin{bmatrix} 1 \\ -1 \end{bmatrix} \approx \begin{bmatrix} .6 \\ .4 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix} \end{aligned}$$

Notice that as n increases, the second term becomes arbitrarily close to $\mathbf{0}$ because of the $\frac{1}{2^n}$

factor. So

$$A^n \begin{bmatrix} .8 \\ .2 \end{bmatrix} \approx \begin{bmatrix} .6 \\ .4 \end{bmatrix} \qquad (A^n \boldsymbol{v} \text{ "settles" to } A\text{'s eigenvector } \boldsymbol{x}_1)$$

Takeaways from Examples 32 and 33.

The matrix $A = \begin{bmatrix} .8 & .3 \\ .2 & .7 \end{bmatrix}$ has eigenvalues $\lambda_1 = 1$ and $\lambda_2 < 1$.

- The eigenvectors $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$ corresponding to the distinct eigenvalues λ_1 and λ_2 are linearly independent.
- Because the eigenvectors are linearly independent, *any* vector $\boldsymbol{v}$ can be expressed as a linear combination of the eigenvectors. That is, we can find numbers a and b for which

$$\boldsymbol{v} = a\,\boldsymbol{x}_1 + b\,\boldsymbol{x}_2.$$

- Because any vector $\boldsymbol{v}$ can be expressed as a linear combination of the eigenvectors, it is easy to calculate $A^n\boldsymbol{v}$:

$$A^n\boldsymbol{v} = A^n(a\,\boldsymbol{x}_1 + b\,\boldsymbol{x}_2) = a\,A^n\boldsymbol{x}_1 + b\,A^n\boldsymbol{x}_2 = a\,\lambda_1^n\boldsymbol{x}_1 + b\,\lambda_2^n\boldsymbol{x}_2$$

  So, to know the effect of applying A^n to any vector $\boldsymbol{v}$, we do not need to calculate A^n, which can be expensive. Instead, we only calculate λ_i^ns, which is inexpensive.
- Moreover, because $\lambda_1 = 1$ and $\lambda_2 < 1$, the first term on the right-hand side of the above expression approaches $a\,\boldsymbol{x}_1$ and the second term approaches $\mathbf{0}$ as n grows, so

$$A^n\boldsymbol{v} \approx a\boldsymbol{x}_1$$

We haven't proved this yet, but the claim of the first bullet point above always holds:

Key Point Eigenvectors from different eigenspaces are always linearly independent.

If $\boldsymbol{x}_1$ through $\boldsymbol{x}_k$ are A's eigenvectors corresponding to *distinct* eigenvalues λ_1 through λ_k, then the eigenvectors are linearly independent.

6.6 A First Look at Matrix Powers

One of the many uses of eigenvalues is in computing matrix powers.

We just saw what A does to the eigenvectors. How about computing higher powers of A? Consider the same matrix $A = \begin{bmatrix} .8 & .3 \\ .2 & .7 \end{bmatrix}$ as in Examples 32 and 33. These are A's powers:

$$\underset{A}{\begin{bmatrix} .8 & .3 \\ .2 & .7 \end{bmatrix}} \quad \underset{A^2}{\begin{bmatrix} .7 & .45 \\ .3 & .55 \end{bmatrix}} \quad \underset{A^3}{\begin{bmatrix} .65 & .525 \\ .35 & .475 \end{bmatrix}} \quad \underset{A^4}{\begin{bmatrix} .625 & .5625 \\ .375 & .4375 \end{bmatrix}} \quad \cdots \quad \underset{A^{99}}{\begin{bmatrix} .6 & .6 \\ .4 & .4 \end{bmatrix}} \quad \cdots \quad \underset{A^{2166}}{\begin{bmatrix} .6 & .6 \\ .4 & .4 \end{bmatrix}} \quad \cdots$$

These A^n were calculated on a computer. Note what happens as n gets larger: both columns of A^n eventually "settle" to $\begin{bmatrix} .6 \\ .4 \end{bmatrix}$, the eigenvector that corresponds to the eigenvalue $\lambda = 1$.

Why does this happen?

If you worked through Example 33, where we applied $A = \begin{bmatrix} .8 & .3 \\ .2 & .7 \end{bmatrix}$ to the vector $v = \begin{bmatrix} .8 \\ .2 \end{bmatrix}$, which happens to be the first column of A, you might guess the answer.

The reason is that every vector in the column space of A, which of course includes its first column, $v = \begin{bmatrix} .8 \\ .2 \end{bmatrix}$, can be expressed as a linear combination of A's eigenvectors[3]. A's eigenvectors are $\boldsymbol{x}_1 = \begin{bmatrix} .6 \\ .4 \end{bmatrix}$ (for $\lambda_1 = 1$) and $\boldsymbol{x}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ (for $\lambda_2 = 1/2$). So a thousand applications of A to its first column gives this:

$$A^{1000} \begin{bmatrix} .8 \\ .2 \end{bmatrix} \quad \text{is really} \quad \boldsymbol{x}_1 + (.2)\left(\frac{1}{2}\right)^{1000} \boldsymbol{x}_2 = \begin{bmatrix} .6 \\ .4 \end{bmatrix} + \begin{bmatrix} \text{very, very, very} \\ \text{small vector} \end{bmatrix} \approx \begin{bmatrix} .6 \\ .4 \end{bmatrix}.$$

Thus, at every step, $\boldsymbol{x}_1$ is unchanged and $\boldsymbol{x}_2$ is multiplied by $\frac{1}{2}$. Therefore, $\boldsymbol{x}_1$ is the first column of A^{1000}. When you ask the computer to calculate A^{1000}, it returns $\begin{bmatrix} .6 \\ .4 \end{bmatrix}$ as the first column of A^{1000}. But this number is not exact — it leaves out the $(.2)\left(\frac{1}{2}\right)^{1000}$ part.

[3] which is why we called A "nice": – A's eigenvectors form a basis in $\mathbb{R}^2$, the column space of A.

The eigenvector $\boldsymbol{x}_1$ is a "steady state" that doesn't change (because $\lambda_1 = 1$). The eigenvector $\boldsymbol{x}_2$ is a "decaying mode" that disappears (because $\lambda_2 = .5 < 1$). The higher the power of A, the closer its columns approach the steady state.

6.7 Exercises

Exercise 6.1. What are the eigenvalues and eigenvectors of $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$?

Exercise 6.2. In Exercise 6.1, you followed the procedure to find the eigenvalues and eigenvectors of $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. Can you argue "geometrically" and guess A's eigenvectors and eigenvalues? (*Hint:* what does this matrix do when it's applied to 2-dimensional vectors?)

Exercise 6.3. What are the eigenvalues and eigenvectors of $A = \begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix}$?

Exercise 6.4. What are the eigenvalues and eigenvectors of $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$?

Exercise 6.5. Find the eigenvalues and eigenvectors of $A = \begin{bmatrix} 1 & -1 \\ 2 & 4 \end{bmatrix}$.

Exercise 6.6. Try to guess what the eigenvectors of $A + 5I, A^2$, or A^{-1} are. Verify your guess on $A = \begin{bmatrix} 1 & 1 \\ -2 & 4 \end{bmatrix}$. Can you also guess the eigenvalues of these matrices?

Exercise 6.7. Show that if A's *rows* add up to 1, then A must have an eigenvalue equal to 1.

Use the fact that the eigenvalues of A and A^T are the same to conclude that if A's *columns* add up to 1, then A must have an eigenvalue equal to 1.

Exercise 6.8. Find all eigenvalues of the following matrices, as well as the corresponding eigenspaces.

$$\text{(a)}\ A = \begin{bmatrix} 1 & -1 \\ -6 & 0 \end{bmatrix} \qquad \text{(b)}\ A = \begin{bmatrix} 3 & -1 \\ 0 & 3 \end{bmatrix} \qquad \text{(c)}\ A = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(*Note:* Later, in Exercise 8.5, we'll ask whether these matrices are diagonalizable.)

Answers to Selected Problems

6.1 *Finding the eigenvalues:*

$$\det\left(\begin{bmatrix} -\lambda & 1 \\ 1 & -\lambda \end{bmatrix}\right) = 0 \quad \Rightarrow \quad \lambda^2 - 1 = 0 \quad \Rightarrow \quad (\lambda - 1)(\lambda + 1) = 0 \quad \Rightarrow \quad \boxed{\lambda_1 = -1} \text{ and } \boxed{\lambda_2 = 1}.$$

Finding an eigenvector associated with each eigenvalue:

- $\lambda_1 = -1$

$$(A - \lambda_1 I)\boldsymbol{x} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \Rightarrow \quad x_1 + x_2 = 0 \quad \Rightarrow \quad \boxed{\boldsymbol{v}_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}}.$$

- $\lambda_2 = 1$

$$(A - \lambda_2 I)\boldsymbol{x} = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \Rightarrow \quad -x_1 + x_2 = 0 \quad \Rightarrow \quad \boxed{\boldsymbol{v}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}}.$$

The final answer: The eigenvalues of $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ are $\lambda_1 = -1$ and $\lambda_2 = 1$.

- Egenvalue $\lambda_1 = -1$ has an associated egenvector $\boldsymbol{v}_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$,
- Egenvalue $\lambda_2 = 1$ has an associated egenvector $\boldsymbol{v}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

6.2 When A is applied to a 2-by-2 vector, A switches the coordinates of the vector:

$$A\boldsymbol{x} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_2 \\ x_1 \end{bmatrix}.$$

Another way of saying this is that the vectors on the 45° line (whose coordinates are equal) stay put under A; other vectors get reflected across that line and change direction. For example, $\boldsymbol{v} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$ stays put under A: $A\boldsymbol{v} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 3 \\ 3 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$. But $\boldsymbol{u} = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$ points in a different direction: $A\boldsymbol{u} = \begin{bmatrix} 5 \\ 2 \end{bmatrix}$.

So the vectors on the 45° line must be the eigenvectors.

Are there any other eigenvectors? Yes: $\boldsymbol{v} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ is taken to $A\boldsymbol{v} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$. We see that $\boldsymbol{v}$ stays on the same line, although it points in the opposite direction.

6.3 This is one of the first examples we did in class. The procedure is exactly the same as in Example 27. The characteristic polynomial is $\det(\begin{bmatrix} 1-\lambda & 2 \\ 5 & 4-\lambda \end{bmatrix}) = (\lambda+1)(\lambda-6)$, giving the eigenvalues $\lambda_1 = -1$ and $\lambda = 6$. The final answer:

- Egenvalue $\lambda_1 = -1$ has an associated egenvector $\boldsymbol{v}_1 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$,
- Egenvalue $\lambda_2 = 6$ has an associated egenvector $\boldsymbol{v}_2 = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$.

6.4 The identity matrix leaves every vector it acts upon in place. Every vector in $\mathbb{R}^2$ is an eigenvector for $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$. Its characteristic polynomial is

$$\det(I - \lambda I) \quad \Rightarrow \quad \det\left(\begin{bmatrix} 1-\lambda & 0 \\ 0 & 1-\lambda \end{bmatrix} \right) \quad \Rightarrow \quad (\lambda - 1)^2$$

Whose roots are $\boxed{\lambda_1 = 1}$ and $\boxed{\lambda_2 = 1}$ (a repeated root).

6.5 $\lambda_1 = 2$ and $\lambda_2 = 3$.

6.6 The eigenvalues of $A = \begin{bmatrix} 1 & 1 \\ -2 & 4 \end{bmatrix}$ are $\lambda_1 = 2$ and $\lambda_2 = 3$, whose corresponding eigenvectors are $\boldsymbol{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\boldsymbol{v}_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, respectively.

When you shift A by a multiple of I or raise A to some power, the eigenvalues change as described in Section 6.4.1, but the eigenvectors don't. So the answers are

- $A + 5I$ has eigenvalues $\lambda_1 = 7$, $\lambda_2 = 8$ and the same eigenvectors as A.
- A^2 has eigenvalues $\lambda_1 = 4$, $\lambda_2 = 9$ and the same eigenvectors as A.
- A^{-1} has eigenvalues $\lambda_1 = 1/2$, $\lambda_2 = 1/3$ and the same eigenvectors as A.

6.7 This follows from the fact that A and A^T have the same eigenvalues. We know that when A's columns add up to 1, then A has an eigenvalue equal to 1. The rows of A^T are the same as the columns of A. Because the rows of A^T add up to 1 and and because A^T has the same eigenvalues as A, we conclude that A^T must have an eigenvalue equal to 1.

6.8 (a) The characteristic polynomial is $\lambda^2 - \lambda - 6 = (\lambda - 3)(\lambda + 2)$, so $\lambda_1 = -2$ and $\lambda_2 = 3$.

- The eigenspace for $\lambda_1 = -2$:

$$\begin{bmatrix} 3 & -1 \\ -6 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \text{ so } \begin{bmatrix} 1 \\ 3 \end{bmatrix} \text{ is an eigenvector and}$$

$$E_{-2} = \text{span}\left(\begin{bmatrix} 1 \\ 3 \end{bmatrix}\right) = c \begin{bmatrix} 1 \\ 3 \end{bmatrix}.$$

- The eigenspace for $\lambda_1 = 3$:

$$\begin{bmatrix} -2 & -1 \\ -6 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \text{ so } \begin{bmatrix} -1 \\ 2 \end{bmatrix} \text{ is an eigenvector and}$$

$$E_3 = \text{span}\left(\begin{bmatrix} -1 \\ 2 \end{bmatrix}\right) = c \begin{bmatrix} -1 \\ 2 \end{bmatrix}.$$

(b) The characteristic polynomial is $\lambda^2 - 6\lambda + 9 = (\lambda - 3)^2$ so $\lambda_1 = \lambda_2 = 3$ is the repeated eigenvalue.

- The eigenspace for $\lambda = 3$:

$$\begin{bmatrix} 0 & -1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \text{ so } \begin{bmatrix} 1 \\ 0 \end{bmatrix} \text{ is an eigenvector and}$$

$$E_3 = \text{span}\left(\begin{bmatrix} 1 \\ 3 \end{bmatrix}\right) = c \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Notice that, unlike the result in part (a) where we found two linearly eigenvectors, here we only have a one-dimensional eigenspace associated with the repeated eigenvalue $\lambda = 3$.

(c) The characteristic equation is

$$\det(A - \lambda I) = 0 \quad \Rightarrow \quad \det\left(\begin{bmatrix} 1-\lambda & 0 & 1 \\ 1 & 1-\lambda & 0 \\ 0 & 0 & 1-\lambda \end{bmatrix}\right) = 0 \quad \Rightarrow \quad (1-\lambda)^3 = 0$$

with roots $\lambda = 1, 1, 1$. So we have one eigenvalue of multiplicity three. The associated eigenspace is found, as usual, by writing out $(A - \lambda I)\boldsymbol{x} = \mathbf{0}$:

$$\begin{bmatrix} 1-1 & 0 & 1 \\ 1 & 1-1 & 0 \\ 0 & 0 & 1-1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \text{ so } \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$\text{is an eigenvector and } E_1 = \text{span}\left(\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}\right) = c \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}.$$

Note that in this case, an eigenvalue of multiplicity three gave rise to an eigenspace of dimension one.

7 Eigenstuff (Part 2 of 3) – Root Multiplicity

We discussed the standard bases in Section B.1 of the appendix and noted that A takes the standard basis vectors to its own columns, as shown in Figure B.1 on page 141.

But the standard basis is not always the most "natural" basis. Instead, the *basis of eigenvectors* – if one exists – is often more convenient when examining linear transformations because then the grid based on these eigenvectors wouldn't change much under the transformation; the grid may change scale along these directions, but its orientation would stay the same.

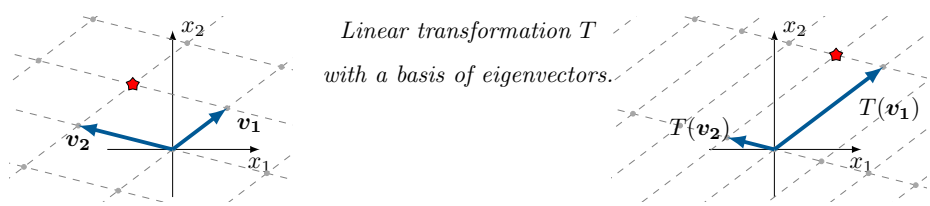


Figure 7.1: Eigenvector $\boldsymbol{v_1}$ goes to $T(\boldsymbol{v_1})$, a multiple of itself. Eigenvector $\boldsymbol{v_2}$ goes to $T(\boldsymbol{v_2})$, again, a multiple of itself. The scaling factors are the corresponding eigenvalues. Because $\boldsymbol{v_1}$ and $\boldsymbol{v_2}$ form a basis in $\mathbb{R}^2$, any vector can be written in terms of $\boldsymbol{v_1}$ and $\boldsymbol{v_2}$. And once you know where T sends the basis vectors, you know where T sends every other vector, just by looking at the image grid.

The nice thing about the basis if eigenvectors as in Figure 7.1 is that you can see what the transformation is doing to *all* vectors simply by looking at their components in the eigenvector directions. So the red star on the left, which is the sum $\boldsymbol{v_1} + \boldsymbol{v_2}$, gets sent to the red star on the right, which is the sum $T(\boldsymbol{v_1}) + T(\boldsymbol{v_2})$. You can easily follow any point by looking at its location on the grid and looking for the corresponding point on the image grid.

Thus, eigenvectors and eigenvalues can give you a more natural structure by which to describe

the action of a linear transformation.

In this lecture, we examine the conditions under which a linear transformation has a basis of eigenvectors. As we'll see later, having the basis of eigenvectors simplifies various calculations; for example, it reduces the computational effort involved in raising A to the n-th power.

7.1 The Shortcuts

Take $A = \begin{bmatrix} 1 & 4 \\ 2 & 8 \end{bmatrix}$ whose eigenvalues are $\lambda_1 = 0$ and $\lambda_2 = 9$ and observe the following:

- The determinant of A is $1 \cdot 8 - 2 \cdot 4 = 0$, which is also equal to $\lambda_1 \cdot \lambda_2 = 0 \cdot 9 = 0$.
- The trace[1] of A us $1 + 8 = 9$, which is also equal to $\lambda_1 + \lambda_2 = 0 + 9 = 9$.

It is nothing short of magical[2] that these two properties should hold for any n-by-n matrix A:

- The product of A's n eigenvalues equals A's determinant.
- The sum of A's n eigenvalues equals A's trace.

7.1.1 $\det(A)$ Is the Product of A's Eigenvalues

Recall the first step in the process of finding the eigenvalues of A: We set $\det(A - \lambda I)$ to zero to compute the roots of the characteristic polynomial; those roots are the eigenvalues. For an n-by-n matrix A, that polynomial has degree n (meaning that the highest degree of λ that appears in it is λ^n). The characteristic polynomial always has n roots (some may be complex, and some may be repeated, but if you count them all, you'll get n of them).

The roots of the characteristic polynomial appear in its factorization:

$$\det(A - \lambda I) = (-1)^n (\lambda - \lambda_1)(\lambda - \lambda_2) \ldots (\lambda - \lambda_n) \tag{7.1}$$

[1] Recall that the trace of A is defined to be the sum of A's main diagonal entries, $1 + 8$ for this A.

[2] Why "magical"? The procedure for finding A's eigenvalues is generally quite elaborate. Moreover, a tiny change in a single matrix entry can significantly alter its eigenvalues. In light of this complexity, it is not at all intuitive why these two properties should hold for *all* square matrices.

where the $(-1)^n$ in the front accounts for the sign of the largest power of λ, and the λ_i are the eigenvalues of A. If you set λ to zero in Equation 7.1, you'll get:

Key Point The determinant of A is the product of its eigenvalues.

$$\det(A) = \lambda_1 \lambda_2 \ldots \lambda_n, \tag{7.2}$$

This **Key Point** makes intuitive sense if you recall that the eigenvalues are the expansion factors (under multiplication by A) in certain special directions, but the determinant is a kind of overall expansion factor — it is how much the volume expands under multiplication by A.

From **Key Point** (7.2) we see yet another criterion for invertibility. Since a number is zero if and only if one of its factors is zero, and the matrix having the determinant equal to zero is equivalent to the matrix being non-invertible, we get

Key Point A has an eigenvalue 0 if and only if A is not invertible.

7.1.2 trace(A) Is the Sum of A's Eigenvalues

Why? This footnote[3] explains why it is true for any n-by-n A. But the following 2-by-2 example captures some of the essence of that truth:

[3] A's characteristic polynomial is $\det(A - \lambda I) = \det\left(\begin{bmatrix} a_{11}-\lambda & * & \cdots & * \\ * & a_{22}-\lambda & \cdots & * \\ \vdots & \vdots & \vdots & \ddots \\ * & * & \cdots & a_{nn}-\lambda \end{bmatrix}\right)$. The $*$s stand for the off-diagonal entries of A. If you expand this determinant, you'll find that the coefficient by λ^{n-1} is the sum $a_{11} + a_{22} + \cdots + a_{nn}$, which is the trace of A. On the other hand, if you expand the product on the right-hand side of the expression 7.1 for the determinant of $A - \lambda I$, you'll see that the coefficient by λ^{n-1} is the sum $\lambda_1 + \lambda_2 + \cdots + \lambda_n$. So, for the two polynomials $\det(A - \lambda I)$ to equal one another, the coefficients by λ^{n-1} must be equal, making $a_{11} + a_{22} + \cdots + a_{nn} = \lambda_1 + \lambda_2 + \cdots + \lambda_n$.

Example 34. The characteristic polynomial of $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is

$$\det(A - \lambda I) = \det\left(\begin{bmatrix} a-\lambda & b \\ c & d-\lambda \end{bmatrix}\right) = \lambda^2 - (a+d)\lambda + ad - bc = 0. \tag{7.3}$$

The quadratic formula gives the eigenvalues from Equation 7.3: $\lambda_1 = (a + d - \sqrt{})/2$ and $\lambda_2 = (a + d + \sqrt{})/2$. Adding the two lambdas gives $a + d$, which is equal to trace(A). □

Key Point The trace of A is the sum of its eigenvalues.

$$\text{trace}(A) = \lambda_1 + \lambda_2 + \ldots + \lambda_n, \tag{7.4}$$

7.2 The Case of Repeated Eigenvalues

Both of the matrices in Examples 35 and 36 below have the same characteristic polynomial $\lambda^2 - 6\lambda + 9 = (\lambda - 3)^2$, so both have repeated eigenvalues $\lambda_1 = \lambda_2 = 3$. But their eigenspaces are very different.

Example 35. The trace of $A = \begin{bmatrix} 4 & -1 \\ 1 & 2 \end{bmatrix}$ is $4 + 2 = 6$ and its determinant is $4 \cdot 2 - (1) \cdot (-1) = 9$.

So A's eigenvalues must be $\lambda_1 = \lambda_2 = 3$ because these are the only lambdas that make $\lambda_1 \cdot \lambda_2 = 9$ and $\lambda_1 + \lambda_2 = 6$. We now find the eigenspace E_3:

- $\lambda = 3$: Solve $\begin{bmatrix} 4-3 & -1 \\ 1 & 2-3 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$. The solution,

$$\boldsymbol{v} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \text{ is an eigenvector for } \lambda = 3.$$

- The *eigenspace* of $A = \begin{bmatrix} 4 & -1/3 \\ 3 & 2 \end{bmatrix}$ corresponding to the repeated $\lambda = 3$

  is <u>*one-dimensional*</u>: $E_3 = \text{span}\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) = c \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, where c is an arbitrary number.

> We denote the eigenspace associated with λ by E_λ and its dimension by $\dim(E_\lambda)$.

Example 36. The eigenvalues of $A = 3 \cdot I = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$ are sitting on its diagonal. As in Example 35, A's eigenvalues are $\lambda_1 = \lambda_2 = 3$. We look for the eigenspace E_3:

- $\lambda = 3$: Solve $\begin{bmatrix} 3-3 & 0 \\ 0 & 3-3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$.

  No pivots – both x_1 and x_2 are free. So the solution is the entire $\mathbb{R}^2$ plane:

  $$\boldsymbol{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \text{ and } \boldsymbol{v}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \text{ are eigenvectors for } \lambda = 3.$$

- The *eigenspace* of $A = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$ corresponding to the repeated $\lambda = 3$

  is <u>*two-dimensional*</u>: $E_3 = \text{span}\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = c \begin{bmatrix} 1 \\ 0 \end{bmatrix} + d \begin{bmatrix} 0 \\ 1 \end{bmatrix}$,

  where c and d are arbitrary numbers.

> **Takeaways** from Examples 35 and 36:
>
> The 2-by-2 matrix A had a repeated eigenvalue λ. That λ appeared twice in the characteristic polynomial $\det(A - \lambda I)$.
>
> But the eigenspace associated with the twice-repeated eigenvalue can be either one-dimensional or two-dimensional.

As we'll see in a bit, the dimension of the eigenspace associated with an eigenvalue that repeats n times can never be larger than n.

7.3 The Algebraic and Geometric Multiplicities of λ

The characteristic polynomial may have repeated eigenvalues. When an eigenvalue repeats n times, n is called the *algebraic multiplicity* of that eigenvalue. The number of linearly independent eigenvectors associated with a repeated eigenvalue is called the eigenvalue's *geometric multiplicity*. The geometric multiplicity can be equal to n, or it can be smaller than n, but it will never be bigger. We'll let the examples make this clear.

Definition The *algebraic multiplicity* of an eigenvalue is the number of times it appears as a root of the characteristic polynomial.

Example 37. The eigenvalues of $A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 6 \end{bmatrix}$ are the roots of this characteristic polynomial:

$$\det \begin{bmatrix} 2-\lambda & 1 & 0 \\ 0 & 6-\lambda & 0 \\ 0 & 0 & 6-\lambda \end{bmatrix} = (2-\lambda)(6-\lambda)^2,$$

giving three eigenvalues, one of which is repeated: $\lambda = 6$ is repeated. So the eigenvalue 6 has algebraic multiplicity 2 because it appears twice in the characteristic polynomial. The eigenvalue $\lambda = 2$ appears only once in the polynomial, so its algebraic multiplicity is 1.

We now look for the eigenspace associated with $\lambda = 6$.

- $\lambda = 6$: As usual, solving $(A - 6\,I)\boldsymbol{x} = \mathbf{0}$ gets us the eigenspace E_6. So we solve

$$(A - 6\,I)\boldsymbol{x} = \begin{bmatrix} 2-6 & 1 & 0 \\ 0 & 6-6 & 0 \\ 0 & 0 & 6-6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -4 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = \mathbf{0}.$$

  We have one pivot, x_1, and two free variables, x_2 and x_3, so the nullspace is 2-dimensional.

The *eigenspace* of $A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 6 \end{bmatrix}$ corresponding to the repeated $\lambda = 6$

$$E_6 \text{ is } \underline{\textit{two-dimensional}}\text{: } E_6 = \operatorname{span}\left(\begin{bmatrix} 1/4 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right) = c \begin{bmatrix} 1 \\ 4 \\ 0 \end{bmatrix} + d \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix},$$

where c and d are arbitrary numbers.

Example 37 takeaway: The algebraic multiplicity of the eigenvalue $\lambda = 6$ for

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 6 \end{bmatrix}$$

is equal to 2 and the associated eigenspace E_6 is *two-dimensional*.

☞ Compare Example 37 with the next one, Example 38, whose A is slightly different:

Example 38. The eigenvalues of $A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 6 & 1 \\ 0 & 0 & 6 \end{bmatrix}$ are the roots of this characteristic polynomial:

$$\det \begin{bmatrix} 2-\lambda & 0 & 0 \\ 0 & 6-\lambda & 1 \\ 0 & 0 & 6-\lambda \end{bmatrix} = (2-\lambda)(6-\lambda)^2,$$

again giving three eigenvalues, one of which is repeated: $\lambda = 6$ is repeated, making the algebraic multiplicity of the eigenvalue 6 equal to 2.

- $\lambda = 6$: As usual, solving $(A - 6\,I)\boldsymbol{x} = \mathbf{0}$ gets us the eigenspace E_6. So we solve

$$(A - 6\,I)\boldsymbol{x} = \begin{bmatrix} 2-6 & 0 & 0 \\ 0 & 6-6 & 1 \\ 0 & 0 & 6-6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -4 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = \mathbf{0}.$$

This time, we have two pivots, x_1 and x_3, and one free variable, x_2, so the nullspace is 1-dimensional.

The *eigenspace* of $A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 6 & 1 \\ 0 & 0 & 6 \end{bmatrix}$ corresponding to the repeated $\lambda = 6$

$$E_6 \text{ is } \underline{\textit{one-dimensional}}\text{: } E_6 = \text{span}\left(\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}\right) = c\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \text{ where } c \text{ is an arbitrary number.}$$

Example 38 takeaway: The algebraic multiplicity of the eigenvalue $\lambda = 6$ for

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 6 & 1 \\ 0 & 0 & 6 \end{bmatrix}$$

is equal to 2 but the associated eigenspace E_6 is *one-dimensional*.

Definition The *geometric multiplicity* is the dimension of the eigenspace of λ.

$$\text{The geometric multiplicity of } \lambda: \quad \dim(E_\lambda).$$

Remark When you see an upper triangular matrix, like the ones in Examples 37 and 38, you know that its determinant is easy to compute: the determinant is simply the product of the diagonal entries.

For upper (or lower) triangular matrices, the eigenvalues are the diagonal entries.

Takeaways from Examples 37 and 38:

The 3-by-3 matrix A had a repeated eigenvalue λ. That λ appeared twice in the characteristic polynomial $\det(A - \lambda I)$.

But the eigenspace associated with the twice-repeated eigenvalue can be either one-dimensional or two-dimensional.

Based on the results of Examples 37 and 38, we can claim that for any upper-triangular matrix of the form $\begin{bmatrix} 2 & * & * \\ 0 & 6 & * \\ 0 & 0 & 6 \end{bmatrix}$, whose eigenvalues are $2, 6, 6$, the nullspace of $A - 6I$ is only ever going to be 1-dimensional or 2-dimensional — it *can never be larger*. You know it's at least 1-dimensional because $\det(A - 6I) = 0$, but its never going to be more than 2-dimensional because you have two 6's on the diagonal and so $A - 6I$ is only going to have *at most* two columns with no leading entries in them.

This argument can be generalized to any matrix, not just upper triangular, giving us:

Fact The geometric multiplicity of λ is always $\leq$ the algebraic multiplicity of λ.

Key Point If all eigenvalues are real, then the sum of algebraic multiplicities is n.

If the geometric multiplicity of λ equals the algebraic multiplicity for <u>*every*</u> eigenvalue, then, since the sum of algebraic multiplicities is n, the sum of geometric multiplicities has to be n.

The situation described in the last **Key Point** — when the sum of geometric multiplicities is n — is the dream scenario because in this situation we have a full basis of eigenvectors (that is, the n eigenvectors span the entire $\mathbb{R}^n$ space). This is the case when computing A^n becomes very straightforward and inexpensive.

But if the geometric multiplicity is strictly less than the algebraic multiplicity for some

eigenvalue, then we are in the undesirable situation when we don't have a full basis of eigenvectors because there will not be enough dimensions in the eigenspaces to span all of $\mathbb{R}^n$. Figure 7.2 shows this.

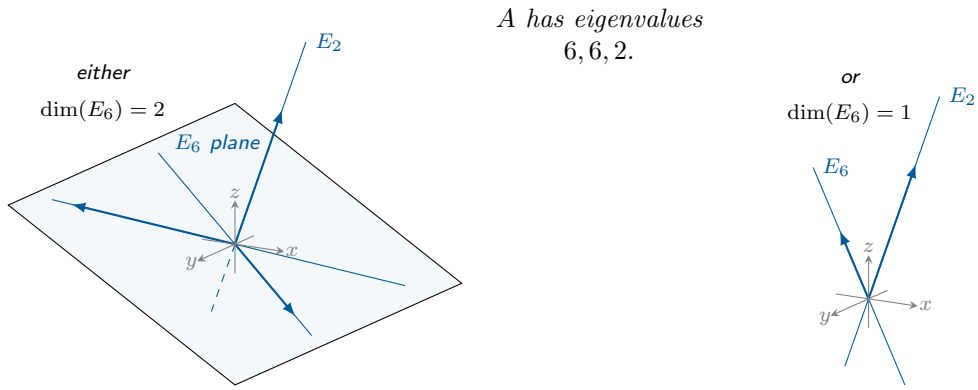


Figure 7.2: The left panel corresponds to Example 37, where $\lambda = 6$ has geometric multiplicity 2 ($\dim(E_6) = 2$). In this case, there is a basis of eigenvectors in $\mathbb{R}^3$; the three blue vectors in the left panel form the basis. The right panel corresponds to Example 38, where $\lambda = 6$ has geometric multiplicity 1 ($\dim(E_6) = 1$). In this case, the two blue eigenvectors in the right panel aren't enough to form a basis in $\mathbb{R}^3$.

7.4 Exercises

Exercise 7.1. You found the eigenvalues of $A = \begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix}$ in Exercise 6.3 of Chapter 6. They were $\lambda_1 = -1$ and $\lambda_2 = 6$. Check that these eigenvalues agree with A's trace and determinant.

Exercise 7.2. What are the algebraic multiplicities of the eigenvalues of $A = \begin{bmatrix} 9 & 2 \\ 0 & 0 \end{bmatrix}$?

Exercise 7.3. Find the eigenspace for each of the eigenvalues of A from Exercise 7.2.

Exercise 7.4. Find the eigenvalues of $A = \begin{bmatrix} 4 & -1/3 \\ 3 & 2 \end{bmatrix}$ by examining its trace and determinant. Find also the eigenspace of A and the dimension of the eigenspace.

Exercise 7.5. What is the nullspace associated with the eigenvalue $\lambda = 2$ for the matrix in Example 37?

What is the nullspace associated with the eigenvalue $\lambda = 2$ for the matrix in Example 38?

Exercise 7.6. The eigenvalues of this upper-triangular $A = \begin{bmatrix} 3 & 8 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 7 \end{bmatrix}$ are sitting on its diagonal.

(a) What is the algebraic multiplicity of $\lambda = 3$?

(b) What is the geometric multiplicity of $\lambda = 7$?

(c) What is the nullspace of $A - 3\,I$?

(d) What is $\dim(E_3)$?

Exercise 7.7. Suppose the middle entry in the top row of A from Exercise 7.6 is 0, not 8. What is the dimension of the nullspace of $A - 3I$ then?

Answers

7.1 Yes:

$$\det\left(\begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix}\right) = 1 \cdot 4 - 2 \cdot 5 = -6 \qquad \text{(which equals } \lambda_1 \cdot \lambda_2 = (-1)(6) = -6\text{)}$$

$$\operatorname{trace}\left(\begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix}\right) = 1 + 4 = 5 \qquad \text{(which equals } \lambda_1 + \lambda_2 = (-1) + 6 = 5\text{)}$$

7.2 The eigenvalues of $A = \begin{bmatrix} 9 & 2 \\ 0 & 0 \end{bmatrix}$ are the roots of this characteristic polynomial:

$$\det\begin{bmatrix} 9-\lambda & 2 \\ 0 & 0-\lambda \end{bmatrix} = (9-\lambda)(-\lambda).$$

Solving $(9-\lambda)(-\lambda) = 0$ gives the eigenvalues $\lambda_1 = 0$ and $\lambda_2 = 9$. Since the eigenvalues are distinct (not repeated), each eigenvalue has algebraic multiplicity 1.

7.3 As usual, we compute the nullspace of $A - 0I$ and of $A - 9I$. With a bit of work, you get:

- The eigenvalue $\lambda_1 = 0$ has an associated eigenvector $\boldsymbol{v}_1 = \begin{bmatrix} -2 \\ 9 \end{bmatrix}$.
- The eigenvalue $\lambda_2 = 9$ has an associated eigenvector $\boldsymbol{v}_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$.

Don't let the word "eigenspace" throw you off. Every multiple of an eigenvector is also an eigenvector. In this case, we have a 1-dimensional (a line) eigenspace associated with each of the eigenvectors. So the eigenspace is just "all the multiples of the eigenvector."

- The eigenspace for $\boldsymbol{v}_1 = \begin{bmatrix} -2 \\ 9 \end{bmatrix}$ consists of all the vectors in the $\operatorname{Span}\left(\begin{bmatrix} -2 \\ 9 \end{bmatrix}\right)$.
- The eigenspace for $\boldsymbol{v}_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ consists of all the vectors in the $\operatorname{Span}\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right)$.

Here by $\operatorname{Span}(\boldsymbol{v})$ we mean all the vectors that lie on the same line as $\boldsymbol{v}$. In other words, all the multiples $c\boldsymbol{v}$, where c is any real number.

7.4 The trace of $A = \begin{bmatrix} 4 & -1/3 \\ 3 & 2 \end{bmatrix}$ is $4 + 2 = 6$ and its determinant is $4 \cdot 2 - (3) \cdot (-1/3) = 9$.

So A's eigenvalues must be $\lambda_1 = \lambda_2 = 3$ because these are the only lambdas that make $\lambda_1 \cdot \lambda_2 = 9$ and $\lambda_1 + \lambda_2 = 6$. We now solve for the eigenspace E_3:

- $\lambda = 3$: Solve $\begin{bmatrix} 4-3 & -1/3 \\ 3 & 2-3 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 & -1/3 \\ 3 & -1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$.

The two rows are multiples of each other. So the solution,

$$\boldsymbol{x}_1 = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1/3 \\ 1 \end{bmatrix}, \text{ is an eigenvector for } \lambda_1 = 3.$$

The *eigenspace* of $A = \begin{bmatrix} 4 & -1/3 \\ 3 & 2 \end{bmatrix}$ corresponding to the repeated $\lambda = 3$

is *one-dimensional*: $E_3 = \text{span}\left(\begin{bmatrix} 1 \\ 3 \end{bmatrix}\right) = c \begin{bmatrix} 1 \\ 3 \end{bmatrix}$, where c is an arbitrary number.

7.5 It's $c \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ for both of them.

7.6 The eigenvalues of $A = \begin{bmatrix} 3 & 8 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 7 \end{bmatrix}$ can be read off its diagonal and are the roots of this characteristic polynomial:

$$\det \begin{bmatrix} 3-\lambda & 8 & 0 \\ 0 & 3-\lambda & 0 \\ 0 & 0 & 7-\lambda \end{bmatrix} = (3-\lambda)^2(7-\lambda).$$

This gives us three eigenvalues, one of which is repeated: $\lambda_{1,2} = 3, \lambda_3 = 7$.

(a) The eigenvalue 3 has algebraic multiplicity 2 (and the eigenvalue 7 has algebraic multiplicity 1).

(b) The geometric multiplicity of $\lambda = 7$ is 1 because the nullspace of $A - 7\,I$ is one-dimensional (it can't possibly be higher). We shorthand this by writing $\dim(E_7) = 1$.

(c) The matrix $A - 3I = \begin{bmatrix} 0 & 8 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 4 \end{bmatrix}$ is already almost in row echelon form — we just have to switch the last two rows. When we do this, the last two columns will be the pivots, and the first will be free.

$$\begin{bmatrix} 0 & 8 & 0 \\ 0 & 0 & 4 \\ 0 & 0 & 0 \end{bmatrix} \qquad \text{(the first column is free, last two are the pivots)}$$

This means that the nullspace is 1-dimensional. The dimension of the nullspace of $A-3I$ is E_3, the geometric multiplicity of $\lambda = 3$. So $\dim(E_3) = 1$.

(d) As explained in part (c), $\dim(E_3) = 1$.

7.7 In that case, $A-3I$ reduces to $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 4 \\ 0 & 0 & 0 \end{bmatrix}$, which has 1 pivot (the last column) and two free columns.

$$\boldsymbol{x}_{\text{complete}} = x_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}. \qquad \text{(the nullspace of } A - 3I\text{)}$$

This means the the nullspace is 2-dimensional (spanned by these two special solutions). So $\dim(E_3) = 2$.

8 Eigenstuff (Part 3 of 3) – Diagonalization

Diagonalization Suppose the n by n matrix A has n linearly independent eigenvectors $\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_n$. Put them into the columns of a matrix, and call it S, the *eigenvector matrix*:

$$\textbf{Eigenvector matrix } S \qquad S = \begin{bmatrix} | & & | \\ \boldsymbol{x}_1 & \cdots & \boldsymbol{x}_n \\ | & & | \end{bmatrix}.$$

Then $S^{-1}AS$ will be a diagonal matrix and A's eigenvalues will show up along its diagonal. We call it the *eigenvalue matrix* Λ :

$$\textbf{Eigenvalue matrix } \Lambda \qquad S^{-1}AS = \Lambda = \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix}.$$

Λ's off-diagonal entries are 0s, and whenever we can get from A to Λ this way, we say that A *is diagonalized*, or that the diagonal form of A is Λ.

Remark Recall from our last discussion that if A has n distinct real eigenvalues, then their corresponding n eigenvectors will point in different directions in $\mathbb{R}^n$ (technical term for "they will be linearly independent").

Remark Recall also that even if the eigenvalues are not distinct, you can still find n linearly independent eigenvectors if the geometric and algebraic multiplicities of all the λs are equal.

Example 39. To diagonalize $A = \begin{bmatrix} 1 & 5 \\ 0 & 6 \end{bmatrix}$, we first find its eigenvalues and eigenvectors. A is triangular, so the eigenvalues show up on the diagonal: $\lambda = 1$ and $\lambda = 6$. The corresponding eigenvectors are $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$. We put them into the columns of S :

$$\textbf{Eigenvectors of } A: \begin{bmatrix} 1 \\ 0 \end{bmatrix} \text{ and } \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \text{so} \quad S = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}.$$

$$\textbf{The inverse of } S: \qquad S^{-1} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}.$$

$$\textbf{The eigenvalue matrix } \Lambda: \qquad \underbrace{\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}}_{S^{-1}} \underbrace{\begin{bmatrix} 1 & 5 \\ 0 & 6 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}}_{S} = \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 6 \end{bmatrix}}_{\Lambda}.$$

In other words, $A = S\Lambda S^{-1}$:

$$A = \begin{bmatrix} 1 & 5 \\ 0 & 6 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 6 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} = S\Lambda S^{-1}.$$

8.1 Making Sense of $S^{-1}AS$

To make a better sense of the expression $S^{-1}AS$, we first look at the columns of the matrix AS. Multiply A by S; each column of S is an eigenvector of A, so we get

A times S

$$AS = A\underbrace{\begin{bmatrix} | & & | \\ \boldsymbol{x}_1 & \cdots & \boldsymbol{x}_n \\ | & & | \end{bmatrix}}_{S} = \underbrace{\begin{bmatrix} | & & | \\ A\boldsymbol{x}_1 & \cdots & A\boldsymbol{x}_n \\ | & & | \end{bmatrix}}_{AS} = \underbrace{\begin{bmatrix} | & & | \\ \lambda_1\boldsymbol{x}_1 & \cdots & \lambda_n\boldsymbol{x}_n \\ | & & | \end{bmatrix}}_{AS}.$$

(The vertical lines above and below the entries are there simply as a reminder that those entries are (n-tall) columns of the (n by n) matrices.)

The trick is to split the matrix AS into the product S times Λ :

S times Λ

$$\begin{bmatrix} | & & | \\ \lambda_1\boldsymbol{x}_1 & \cdots & \lambda_n\boldsymbol{x}_n \\ | & & | \end{bmatrix} = \begin{bmatrix} | & & | \\ \boldsymbol{x}_1 & \cdots & \boldsymbol{x}_n \\ | & & | \end{bmatrix} \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix} = S\Lambda.$$

So λ_1 multiplies the first column of S, which is $\boldsymbol{x}_1$, λ_2 multiplies the second column, $\boldsymbol{x}_2$, and so on, λ_n multiplies the n-th column of S, $\boldsymbol{x}_n$.

We see that $S\Lambda$ and AS are the same matrix! $AS = S\Lambda$, which can be rewritten in two useful ways:

$$AS = S\Lambda \quad \text{is} \quad S^{-1}AS = \Lambda \quad \text{or} \quad A = S\Lambda S^{-1}.$$

S has an inverse because its columns are the eigenvectors of A, which were assumed to be linearly independent. In particular, if all eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$ are distinct, we'll have linearly independent eigenvectors. *If the eigenvectors are not independent, then we cannot diagonalize* because then S^{-1} doesn't exist.

8.2 Matrix Powers

We already looked at matrix powers back in Section 6.6 and in a few examples that preceded and followed that section.

We now want to address the question: Why bother with diagonalizing a matrix? Watch what happens if you compute A^2 when A is diagonalized:

$$A^2 = AA = (S\Lambda S^{-1})(S\Lambda S^{-1}) = S\Lambda \underbrace{(S^{-1}S)}_{I} \Lambda S^{-1} = S\Lambda^2 S^{-1} \tag{8.1}$$

The right hand side of Equation (8.1) has the same S – the eigenvector matrix of A – and Λ^2, which is a diagonal matrix with squared eigenvalues of A on the diagonal. So if we wanted to compute A^2, we'd get

$$\begin{bmatrix} 1 & 5 \\ 0 & 6 \end{bmatrix}^2 = A^2 = S\Lambda^2 S^{-1} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1^2 & 0 \\ 0 & 6^2 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 6^2 - 1 \\ 0 & 6^2 \end{bmatrix} = \begin{bmatrix} 1 & 35 \\ 0 & 36 \end{bmatrix}$$

This may not sound like a big deal – after all, computing A^2 for this particular A would be faster if you did it directly. But notice what happens with A^k: the pattern shown in Equation (8.1) continues and we get

$$\boxed{A^k = S\Lambda^k S^{-1}}$$

This means that S and Λ have to be computed just once. Then any power of A is just a product of 3 matrices!

Example 40. Compute A^5 where $A = \begin{bmatrix} 1 & 5 \\ 0 & 6 \end{bmatrix}$.

Using diagonalization, A^5 is a cinch to compute. We already computed the eigenvectors and eigenvalues of A in Example 39, so using $A^k = S\Lambda^k S^{-1}$ we get:

$$A^k = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1^k & 0 \\ 0 & 6^k \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 6^k - 1 \\ 0 & 6^k \end{bmatrix},$$

and for $k = 5$ the answer is

$$A^5 = \begin{bmatrix} 1 & 6^5 - 1 \\ 0 & 6^5 \end{bmatrix} = \begin{bmatrix} 1 & 7775 \\ 0 & 7776 \end{bmatrix}.$$

Example 41. Compute A^{99} where $A = \begin{bmatrix} .8 & .3 \\ .2 & .7 \end{bmatrix}$ without using a computer or a calculator.

Answer We have seen this matrix before, in Section 6.5. Notice that both columns of A sum to 1. (Matrices with nonnegative entries whose columns sum to 1 are called *column-stochasitc* or *Markov* matrices.) The eigenvalues and eigenvectors of A are:

- $\boldsymbol{x}_1 = \begin{bmatrix} .6 \\ .4 \end{bmatrix}$ is an eigenvector that goes with the eigenvalue $\lambda_1 = 1$.
- $\boldsymbol{x}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ is an eigenvector that goes with the eigenvalue $\lambda_2 = \frac{1}{2}$.
- Then $S = \begin{bmatrix} .6 & 1 \\ .4 & -1 \end{bmatrix}$ and $S^{-1} = \begin{bmatrix} 1 & 1 \\ .4 & -.6 \end{bmatrix}$.

Using $A^k = S\Lambda^k S^{-1}$ we get:

$$A^k = \begin{bmatrix} .6 & 1 \\ .4 & -1 \end{bmatrix} \begin{bmatrix} 1^k & 0 \\ 0 & \left(\frac{1}{2}\right)^k \end{bmatrix} \begin{bmatrix} 1 & 1 \\ .4 & -.6 \end{bmatrix}.$$

and for $k = 99$ the answer (calculated "by hand") is

$$A^{99} = \begin{bmatrix} .6 & 1 \\ .4 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 2^{-99} \end{bmatrix} \begin{bmatrix} 1 & 1 \\ .4 & -.6 \end{bmatrix} = \begin{bmatrix} .6 & 1 \\ .4 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ .4 \times 2^{-99} & -.6 \times 2^{-99} \end{bmatrix}$$

$$= \begin{bmatrix} .6 + .4 \times 2^{-99} & .6 - .6 \times 2^{-99} \\ .4 - .4 \times 2^{-99} & .4 - .6 \times 2^{-99} \end{bmatrix} = \begin{bmatrix} .6 + \text{small value} & .6 - \text{small value} \\ .4 - \text{small value} & .4 - \text{small value} \end{bmatrix} \approx \begin{bmatrix} .6 & .6 \\ .4 & .4 \end{bmatrix}$$

It's worth reflecting on one point. Both matrices in Examples 40 and 41 had eigenvalues equal to 1. But A from Example 41 also had an eigenvalue $\lambda < 1$, while A from Example 40 had an eigenvalue $\lambda > 1$. In Example 40, some entries of A^k *grew larger and larger* as k increased, while in Example 41 the entries of A^k *settled to stable values*.

8.3 Exercises

Exercise 8.1. Diagonalize the matrix $A = \begin{bmatrix} 2 & 0 \\ -1 & 3 \end{bmatrix}$.

Exercise 8.2. Diagonalize the matrix $A = \begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$.

Exercise 8.3. Based on your results in Exercise 8.1, compute $A^{11} = \begin{bmatrix} 2 & 0 \\ -1 & 3 \end{bmatrix}^{11}$.

Exercise 8.4. Based on your results in Exercise 8.2, compute $A^{11} = \begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}^{11}$.

Exercise 8.5. Are the following two matrices diagonalizable?

(a) $A = \begin{bmatrix} 1 & -1 \\ -6 & 0 \end{bmatrix}$ (b) $A = \begin{bmatrix} 3 & -1 \\ 0 & 3 \end{bmatrix}$

(*Note:* These are the same matrices as in Exercise 6.8.)

Exercise 8.6. Are the following matrices diagonalizable?

(a) $A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 6 \end{bmatrix}$ (b) $A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 6 & 1 \\ 0 & 0 & 6 \end{bmatrix}$

(c) $A = \begin{bmatrix} 3 & 8 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 7 \end{bmatrix}$ (d) $A = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 8 \\ 0 & 0 & 7 \end{bmatrix}$

(*Note:* The matrices in parts (a) and (b) are from Examples 37 and 38 and the matrix in part (c) is from Exercise 7.6)

Exercise 8.7. Find the eigenvalues and eigenvectors of these two column-stochastic matrices:

$$A = \begin{bmatrix} .6 & .2 \\ .4 & .8 \end{bmatrix}, \text{ and } B = \begin{bmatrix} 1/3 & 1/3 \\ 2/3 & 2/3 \end{bmatrix}.$$

What is A^{100}? What is A^{∞}? Why are A^{100} and A^{∞} so close to one another?

Answers to Selected Problems

8.1 The eigenvalues are sitting on the diagonal of this lower triangular matrix: $\lambda_1 = 2$ and $\lambda_2 = 3$, whose eigenvectors are $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$, respectively. So $S = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$. Then $S^{-1} = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$ and we get

$$S^{-1}AS = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} = \Lambda.$$

8.2 Computing the eigenvalues is a bit more of a challenge this time, but not too bad. The characteristic polynomial is $\det(A - \lambda I) = \det \begin{bmatrix} 2-\lambda & 2 & 1 \\ 1 & 3-\lambda & 1 \\ 1 & 2 & 2-\lambda \end{bmatrix} = (\lambda-1)^2(\lambda-5)$, from which we get the repeated root $\lambda_{1,2} = 1$ and $\lambda_3 = 5$. We next compute the eigenvectors for each of the λs. The danger is that $\lambda_{1,2} = 1$ might have geometric multiplicity 1, in which case A wouldn't be diagonalizable. But here we get lucky be because $\lambda_{1,2}$'s geometric multiplicity is 2. We chug along to find the eigenvectors:

- $\lambda = 1$ has a 2-dimensional space of eigenvectors spanned by $\boldsymbol{x}_1 = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$.
- $\lambda = 5$ has an eigenvector $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$.

We put the eigenvectors into S to get $S = \begin{bmatrix} 1 & -2 & -1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$ Then $S^{-1} = \begin{bmatrix} \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ -\frac{1}{4} & \frac{1}{2} & -\frac{1}{4} \\ -\frac{1}{4} & -\frac{1}{2} & \frac{3}{4} \end{bmatrix}$ and we get

$$S^{-1}AS = \begin{bmatrix} \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ -\frac{1}{4} & \frac{1}{2} & -\frac{1}{4} \\ -\frac{1}{4} & -\frac{1}{2} & \frac{3}{4} \end{bmatrix} \begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix} \begin{bmatrix} 1 & -2 & -1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \Lambda.$$

Notice that the order in which we put the eigenvectors into the columns of S matters. Here we put $\lambda_3 = 5$'s eigenvector as the first column in S so the north-west entry in Λ ended up being 5. If you switch the order of the columns, the process still works, but the order of the diagonal entries in Λ will have to be switched accordingly.

8.3 Based on your results in Exercise 8.1, A^{11} can now be computed by hand:

$$A^{11} = S\Lambda^{11}S^{-1} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 2^{11} & 0 \\ 0 & 3^{11} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 2^{11} & 0 \\ -3^{11} & 3^{11} \end{bmatrix} = \begin{bmatrix} 2^{11} & 0 \\ -3^{11} + 2^{11} & 3^{11} \end{bmatrix}.$$

8.4 Based on your results in Exercise 8.2, A^{11} can now be computed with a little help using a calculator:

$$\begin{aligned} A^{11} = S\Lambda^{11}S^{-1} &= \begin{bmatrix} 1 & -2 & -1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 5^{11} & 0 & 0 \\ 0 & 1^{11} & 0 \\ 0 & 0 & 1^{11} \end{bmatrix} \begin{bmatrix} \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ -\frac{1}{4} & \frac{1}{2} & -\frac{1}{4} \\ -\frac{1}{4} & -\frac{1}{2} & \frac{3}{4} \end{bmatrix} \\ &= \begin{bmatrix} 5^{11} & -2 & -1 \\ 5^{11} & 1 & 0 \\ 5^{11} & 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ -\frac{1}{4} & \frac{1}{2} & -\frac{1}{4} \\ -\frac{1}{4} & -\frac{1}{2} & \frac{3}{4} \end{bmatrix} \\ &= \begin{bmatrix} 12\,207\,032 & 24\,414\,062 & 12\,207\,031 \\ 12\,207\,031 & 24\,414\,063 & 12\,207\,031 \\ 12\,207\,031 & 24\,414\,062 & 12\,207\,032 \end{bmatrix}. \end{aligned}$$

8.6

8.7 A's characteristic polynomial is $\det(A - \lambda I) = \lambda^2 - \frac{7}{5}\lambda + \frac{2}{5} = \Big(\lambda - 1\Big)\Big(\lambda - \frac{2}{5}\Big)$.

- $\lambda_1 = \frac{2}{5}$ has an eigenvector $\boldsymbol{x}_1 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$.
- $\lambda_2 = 1$ has an eigenvector $\begin{bmatrix} 1/2 \\ 1 \end{bmatrix}$.

Put the eigenvectors into the columns of S and compute A^{100} :

$$\begin{aligned} A^{100} = S\Lambda^{100}S^{-1} &= \begin{bmatrix} -1 & 1/2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} (2/5)^{100} & 0 \\ 0 & 1^{100} \end{bmatrix} \begin{bmatrix} -2/3 & 1/2 \\ 2/3 & 2/3 \end{bmatrix} \\ &= \begin{bmatrix} -1 & 1/2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \approx 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} -2/3 & 1/2 \\ 2/3 & 2/3 \end{bmatrix} \approx \begin{bmatrix} 1/3 & 1/3 \\ 2/3 & 2/3 \end{bmatrix}. \end{aligned}$$

This is equal to B. Squaring B now, or computing its higher powers doesn't change B any more: $B^2 = \begin{bmatrix} 1/3 & 1/3 \\ 2/3 & 2/3 \end{bmatrix}^2 = \begin{bmatrix} 1/3 & 1/3 \\ 2/3 & 2/3 \end{bmatrix}$. This matrix has "settled." The reason for this is that none of the eigenvalues were bigger than 1: $\lambda = 1$ or $|\lambda| < 1$.

Part III

Exam Practice Problems

The following tests are much longer than the actual final, but they cover all you need to know to do very well on the exam.

Some of the questions may be challenging. They will prompt you to review your lecture notes when you get stuck, allowing you to reflect on what we've covered once more. It's important not to get discouraged when you get stuck and to look at the solutions only after doing your best to tackle the problem. You will find all the clues you need in this booklet.

If you understand these problems thoroughly, you will ace the final!

Best of luck!

9 Sample Test Questions With Solutions

Problem 1. True or false? Let A and B be square matrices.

(a) $(A+B)^2 = (B+A)^2$

(b) If columns 1 and 3 of B are the same, then so are the columns 1 and 3 of BA.

(c) If rows 1 and 3 of A are the same, so are rows 1 and 3 of AB.

Problem 2. Find the products EFG and GFE if (upper triangular entries are all zeros)

$$E = \begin{bmatrix} 1 & & & \\ 2 & 1 & & \\ 0 & 0 & 1 & \\ 0 & 0 & 0 & 1 \end{bmatrix} \qquad F = \begin{bmatrix} 1 & & & \\ 0 & 1 & & \\ 0 & 2 & 1 & \\ 0 & 0 & 0 & 1 \end{bmatrix} \qquad G = \begin{bmatrix} 1 & & & \\ 0 & 1 & & \\ 0 & 0 & 1 & \\ 0 & 0 & 2 & 1 \end{bmatrix}.$$

Problem 3. There are 16 2-by-2 matrices $\begin{bmatrix} * & * \\ * & * \end{bmatrix}$ whose $*$ entries are 0s or 1s. How many of them are invertible?

Problem 4. Describe the column spaces of the following matrices:

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 0 & 0 \end{bmatrix} \quad \text{and} \quad C = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \end{bmatrix}.$$

Clearly indicate whether the column space is a line or the $\mathbb{R}^2$ plane.

Problem 5. Write the following problem in matrix form and solve it: "Two points with coordinates (x, y) are $(2, 5)$ and $(3, 7)$. The two points lie on the line $y = mx + c$. Find m and c."

Problem 6. Compute A^3 and A^{2166} where $A = \begin{bmatrix} 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$.

Problem 7. Suppose you solve $A\boldsymbol{x} = \boldsymbol{b}$ for the three right-hand sides shown below

$$A\boldsymbol{x}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad \text{and} \quad A\boldsymbol{x}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad \text{and} \quad A\boldsymbol{x}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

If the solutions $\boldsymbol{x}_1, \boldsymbol{x}_2, \boldsymbol{x}_3$ are the columns of a matrix X, what is A times X? (*Hint:* Recall how the Gauss-Jordan way of inverting a matrix works.)

Problem 8. Carry out Gauss-Jordan elimination to find the inverse of $A = \begin{bmatrix} a & b & b \\ a & a & b \\ a & a & a \end{bmatrix}$. Then prove that A is invertible if $a \neq 0$ and $a \neq b$.

Problem 9. Consider the $A\boldsymbol{x} = \boldsymbol{b}$ system $\begin{cases} x_1 + 2x_2 + 3x_3 + \ \ 5x_4 = b_1 \\ 2x_1 + 4x_2 + 8x_3 + 12x_4 = b_2 \\ 3x_1 + 6x_2 + 7x_3 + 13x_4 = b_3 \end{cases}$.

(a) Reduce $[\,A\ \boldsymbol{b}\,]$ to $[\,U\ \boldsymbol{c}\,]$ so that $A\boldsymbol{x} = \boldsymbol{b}$ becomes a triangular system $U\boldsymbol{x} = \boldsymbol{c}$.

(b) Find the conditions on b_1, b_2, b_3 for $A\boldsymbol{x} = \boldsymbol{b}$ to have a solution.

(c) Describe the column space of A. Is it a line or is it a plane. Is it in $\mathbb{R}^3$?

(d) Describe the nullspace of A by writing down the special solutions. Is the nullspace in $\mathbb{R}^4$?

(e) Reduce $[\,U\ \boldsymbol{c}\,]$ to $[\,R\ \boldsymbol{d}\,]$ and determine the special solutions and the particular solution.

(f) Find a particular solution to $A\boldsymbol{x} = \begin{bmatrix} 0 \\ 6 \\ -6 \end{bmatrix}$ and then the complete solution.

Problem 10. Suppose you know that the 3 by 4 matrix A has all the multiples of $\begin{bmatrix} 2 \\ 3 \\ 1 \\ 0 \end{bmatrix}$ as the only vectors in its nullspace.

(a) How many pivot variables does A have?

(b) Write down the complete solution to $A\boldsymbol{x} = \boldsymbol{0}$.

(c) What is the exact row reduced echelon form of A?

Problem 11. Recall that the allowed row operations of the Gaussian elimination algorithm correspond to the so-called elementary matrices. Elementary matrices E, F, G, and H left-multiply matrix

A in the order shown below. The result is the 4×5 matrix R on the right of the $=$ sign.

$$\underbrace{\begin{bmatrix} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}}_{H} \underbrace{\begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}}_{G} \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 11 & 1 \end{bmatrix}}_{F} \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 10 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}}_{E} A = \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 2 \\ 0 & 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 1 & 4 \end{bmatrix}}_{R}$$

(a) Which row operations do the four elimination matrices in the product correspond to? Please write down in words the operations corresponding to the matrices E, F, G, and H.

(b) What must be the upper left hand corner of A (the entry in the first row and the first column of A)? Why must it be zero?

(c) The equation implies that A factors as $A = LPUR$. Here R is the matrix on the right of the $=$ sign. The matrices U, P, and L are invertible 4×4 matrices: U is upper triangular, P is a permutation matrix, and L is lower triangular. Find U, P, and L without a calculator and explain how you got them.

Problem 12. Compute the eigenvalues and eigenvectors of A and represent it as a product $S\Lambda S^{-1}$ where Λ is a diagonal eigenvalue matrix and S the corresponding eigenvector matrix. Using that representation compute A^8, showing your steps.

$$A = \begin{bmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 0 \end{bmatrix}$$

Problem 13. Suppose A and B have the same eigenvalues $\lambda_1, \ldots, \lambda_n$ and the same independent eigenvectors $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n$. Is it true that then A must equal B? If it is true, prove it; if it is not, provide a counterexample with 2 by 2 matrices.

Problem 14. Let $\begin{bmatrix} .4 & 1-c \\ .6 & c \end{bmatrix}$ where c is some real number.

(a) Find the eigenvalues and eigenvectors of in terms of c.

(b) Show that A has just one line of eigenvectors when $c = 1.6$.

Problem 15. $B = \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix}$ has eigenvalues 1 and 9. Find matrix A such that $A^2 = B$.

Problem 16. Which of these matrices approaches the zero matrix as $n \to \infty$?

$$A_1 = \begin{bmatrix} .6 & .9 \\ .4 & .1 \end{bmatrix} \quad \text{and} \quad A_2 = \begin{bmatrix} .6 & .9 \\ .1 & .6 \end{bmatrix}$$

Hint: Try the $A = S\Lambda S^{-1}$ decomposition and look at the eigenvalues λ_1 and λ_2 on the diagonal of Λ.

Problem 17. Suppose A is a 4 by 4 upper triangular matrix with $1, 2, 3, 4$ on its main diagonal. (You can put all 1s above the diagonal.)

(a) For $A - 3I$, which columns have pivots? Which components of the eigenvector $\boldsymbol{x}_3$ (the special solution in the nullspace) are definitely zero?

(b) Using part (a), show that the eigenvector matrix is also upper triangular.

Problem 18. Choose the last row of A so that its eigenvalues are $1, 2, 3$:

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ * & * & * \end{bmatrix}.$$

9.1 Solutions

1 (a) True (because $A + B = B + A$).
(b) False.
(c) True.

2 $EFG = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 \\ 0 & 0 & 2 & 1 \end{bmatrix}$ and $GFE = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ 4 & 2 & 1 & 0 \\ 8 & 4 & 2 & 1 \end{bmatrix}$.

3 All such matrices other than $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$, and $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ have columns or rows that are multiples (-1 times or 1 times) of each other. So these are the only invertible ones.

4 A's column space is a *line* — all vectors of the form $\begin{bmatrix} x \\ 0 \end{bmatrix}$ (the x-axis). The column space of B is also a *line* generated by the $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ vector; that is all vectors of the form $\begin{bmatrix} x \\ 2x \end{bmatrix}$. For C, the column space is the entire $\mathbb{R}^2$ *plane*.

5 Since $(x, y) = (2, 5)$ lies on the $y = mx + c$ line, we have $5 = m \cdot 2 + c$. Similarly, for $(x, y) = (3, 7)$ we have $7 = m \cdot 3 + c$. This corresponds to the system $\begin{cases} 2m + c = 5 \\ 3m + c = 7 \end{cases}$, whose matrix form is $\begin{bmatrix} 2 & 1 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} m \\ c \end{bmatrix} = \begin{bmatrix} 5 \\ 7 \end{bmatrix}$ and whose solution is $m = 2$ and $c = 1$ (a line with slope 2).

6 Note how with successive powers of A the 2s turn to $2^2, 2^3$, etc. and move up north-east:

$$A = \begin{bmatrix} 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad A^2 = \begin{bmatrix} 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad A^3 = \begin{bmatrix} 0 & 0 & 0 & 8 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad A^4 = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \cdots$$

All higher powers of A give a 4 by 4 matrix of zeros.

7 AX will be the identity matrix $I = [A\boldsymbol{x}_1\ A\boldsymbol{x}_2\ A\boldsymbol{x}_3]$ when $X = [\boldsymbol{x}_1\ \boldsymbol{x}_2\ \boldsymbol{x}_3]$.

8 $A^{-1} = \dfrac{1}{a(a-b)} \begin{bmatrix} a & 0 & -b \\ -a & a & 0 \\ 0 & -a & a \end{bmatrix}$. As long as $a \neq 0$ *and* $a - b \neq 0$, A^{-1} exists.

9 (a)

$$\left[\begin{array}{cccc|c} 1 & 2 & 3 & 5 & b_1 \\ 2 & 4 & 8 & 12 & b_2 \\ 3 & 6 & 7 & 13 & b_3 \end{array}\right] \to \left[\begin{array}{cccc|c} 1 & 2 & 3 & 5 & b_1 \\ 0 & 0 & 2 & 2 & b_2 - 2b_1 \\ 0 & 0 & -2 & -2 & b_3 - 3b_1 \end{array}\right] \to \left[\begin{array}{cccc|c} 1 & 2 & 3 & 5 & b_1 \\ 0 & 0 & 2 & 2 & b_2 - 2b_1 \\ 0 & 0 & 0 & 0 & b_3 + b_2 - 5b_1 \end{array}\right] \tag{9.1}$$

(b) From (a) we get the solvability condition $b_3 + b_2 - 5b_1 = 0$, which makes the last row $0 = 0$.

(c) The column space is the plane in $\mathbb{R}^3$ containing all combinations of the pivot columns $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ and $\begin{bmatrix} 3 \\ 8 \\ 7 \end{bmatrix}$.

The pivots are in columns 1 and 3.

(d) The special solutions come from $\mathbb{R}^4$ and have free variables $x_2 = 1, x_4 = 0$ and then $x_2 = 0, x_4 = 1$:

$$\text{All combinations of the two special solutions:} \quad \boldsymbol{x}_n = x_2 \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -2 \\ 0 \\ -1 \\ 1 \end{bmatrix}$$

(the nullspace $N(A)$ in $\mathbb{R}^4$)

(e) Continue row reductions upward to find the reduced form $[\,R\ \boldsymbol{d}\,]$. Starting from from 9.1 above, the right-hand side is $\begin{bmatrix} b_1 \\ b_2 - b_1 \\ b_3 + b_2 - 5b_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 6 - 0 \\ (-6) + 6 - 5 \cdot 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 6 \\ 0 \end{bmatrix}$ (because the original vector $\boldsymbol{b}$ was equal to $\begin{bmatrix} 0 \\ 6 \\ -6 \end{bmatrix}$):

$$[\,U\ \boldsymbol{c}\,] = \left[\begin{array}{cccc|c} 1 & 2 & 3 & 5 & 0 \\ 0 & 0 & 2 & 2 & 6 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] \to \left[\begin{array}{cccc|c} 1 & 2 & 0 & 2 & -9 \\ 0 & 0 & 1 & 1 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] = [\,R\ \boldsymbol{d}\,].$$

So the right-hand side becomes $\boldsymbol{d} = \begin{bmatrix} -9 \\ 3 \\ 0 \end{bmatrix}$, showing -9 and 3 in the particular solution $\boldsymbol{x}_p$ (shown in part (f)).

(f) The particular solution is obtained by setting all free variables to zero and back substituting in $[\,R\ \boldsymbol{d}\,]$ to find the pivots is thus $\boldsymbol{x}_p = \begin{bmatrix} -9 \\ 0 \\ 3 \\ 0 \end{bmatrix}$ The complete solution to $A\boldsymbol{x} = \begin{bmatrix} 0 \\ 6 \\ -6 \end{bmatrix}$ is then $\boldsymbol{x}_p + \boldsymbol{x}_n$:

$$\text{The complete solution:} \quad \boldsymbol{x} = \begin{bmatrix} -9 \\ 0 \\ 3 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -2 \\ 0 \\ -1 \\ 1 \end{bmatrix}.$$

10 (a) The dimension of the nullspace is 1, so A must have $4 - 1 = 3$ pivots.

(b) The complete solution to $A\boldsymbol{x} = \mathbf{0}$ is $\boldsymbol{x} = x_3 \begin{bmatrix} 2 \\ 3 \\ 1 \\ 0 \end{bmatrix}$ (for any real number x_3).

(c) The row reduced echelon form has 3 pivots and the special solution $\boldsymbol{x} = \begin{bmatrix} 2 \\ 3 \\ 1 \\ 0 \end{bmatrix}$, so

$$R = \begin{bmatrix} 1 & 0 & -2 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

11 (a) The row operations are:

- E adds 10 times row 2 to row 3
- F adds 11 times row 3 to row 4
- G swaps rows 1 and 2
- H adds 4 times row 4 to row 1

(b) The upper right corner of A is zero because if it were anything other than zero we would never have to swap rows 1 and 2 (which is what G does).

(c) The upper triangular U corresponds to the upward elimination, so we have

$$U = \begin{bmatrix} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & 0 & 0 & -4 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

The permutation matrix P corresponds to the permutation of the first two rows, so we have

$$P = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

The lower triangular L corresponds to the first two elimination steps combined (the product FE):

$$L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 10 & 1 & 0 \\ 0 & 110 & 11 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -10 & 1 & 0 \\ 0 & 0 & -11 & 1 \end{bmatrix}$$

12 A's characteristic equation: $\det \begin{bmatrix} -\lambda & 2 & 2 \\ 2 & -\lambda & 2 \\ 2 & 2 & -\lambda \end{bmatrix} = 0 \quad \Rightarrow \quad (\lambda+2)^2(\lambda-4) = 0$ gives $\lambda_{1,2} = -2$ (a repeated eigenvalue of algebraic multiplicity 2) and $\lambda_3 = 4$.

- *Eigenvectors for* $\lambda_3 = 4$*:* $\begin{bmatrix} -4 & 2 & 2 \\ 2 & -4 & 2 \\ 2 & 2 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$.

  Gaussian elimination gives $\begin{bmatrix} -4 & 2 & 2 \\ 0 & -3 & 3 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$, and we pick the eigenvector $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$.

- *Eigenvectors for* $\lambda_{1,2} = -2$*:* $\begin{bmatrix} 0-(-2) & 2 & 2 \\ 2 & 0-(-2) & 2 \\ 2 & 2 & 0-(-2) \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$. Gaussian elimination of which gives $\begin{bmatrix} 2 & 2 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$. Here we have 2 free variables (x_3 and x_2) and one pivot (x_1). Setting the two free variables as usual (setting one free variable to 1, the other to 0, and determining the pivot), we see that the space of eigenvectors corresponding to $\lambda_2 = -2$ *is two-dimensional*: $x_2 \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$.

So the matrix S is $\begin{bmatrix} 1 & -1 & -1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$ and $S^{-1} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}$. Thus A can be diagonalized as

$$A = S\Lambda S^{-1} = \frac{1}{3} \cdot \begin{bmatrix} 1 & -1 & -1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 4 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}$$

And A^8 can be computed as

$$A^8 = S\Lambda^8 S^{-1} = \frac{1}{3} \cdot \begin{bmatrix} 1 & -1 & -1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 4^8 & 0 & 0 \\ 0 & (-2)^8 & 0 \\ 0 & 0 & (-2)^8 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix},$$

which works out to $A^8 = \begin{bmatrix} 22016 & 21760 & 21760 \\ 21760 & 22016 & 21760 \\ 21760 & 21760 & 22016 \end{bmatrix}$.

13 There's more than one way to argue why $A = B$, but one way is to note that the decompositions of $A = S\Lambda S^{-1}$ and $B = S\Lambda S^{-1}$ are identical because their S matrices are made up of the eigenvectors that are the same and their Λ matrices have the same entries because their eigenvalues are the same.

14 (a) Because both columns add up to 1, A has an eigenvalue $\lambda_1 = 1$. To find the other eigenvalue, recall that A's eigenvalues always add up to A's trace (which is the sum of A's diagonal entries). Checking the trace, $\texttt{trace}(A) = .4 + c$, we see that the other eigenvalue must be $\lambda_2 = .4 + c - \lambda_1 = .4 + c - 1 = c - .6$.

(b) If $c = 1.6$, then both eigenvalues of A are 1. The eigenvectors of $A = \begin{bmatrix} .4 & -.6 \\ .6 & 1.6 \end{bmatrix}$ are all multiples of $\boldsymbol{x} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ (that is, $\lambda = 1$ has geometric multiplicity equal to 1).

15 You want to make $A = S\Lambda^{1/2}S^{-1}$, where S is the eigenvector matrix shared by both A and B (recall that the eigenvalues of A^2 are the squares of the eigenvalues of A, and the eigenvectors of A^2 are the same as the eigenvectors of A). So, find the eigenvectors and put them in S to get

$$A = S\Lambda^{1/2}S^{-1} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \cdot \begin{bmatrix} 3 & \\ & 1 \end{bmatrix} \cdot \left(\frac{1}{2}\right) \cdot \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \quad \text{has } A^2 = B.$$

16 $A = S\Lambda S^{-1}$ approaches zero if and only if every $|\lambda| < 1$. A_1 is column stochastic and has an eigenvalue $= 1$, so it will *not* approach the zero matrix. But A_2 will because it has $\lambda = .9 < 1$ and $\lambda = .3 < 1$.

17 (a)

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & 1 & 1 \\ 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 4 \end{bmatrix}, \quad A - 3I = \begin{bmatrix} -2 & 1 & 1 & 1 \\ 0 & -1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Columns $1, 2$ and 4 have pivots. Because of the nonzero bottom-right element in $A - 3I$, the fourth component of $\boldsymbol{x}_3$ is definitely zero.

(b) In the same way as above, the special solutions for the matrices $A-1I, A-2I, A-3I$, and $A-4I$ must have $3, 2, 1$, and 0 zeros as the last components. The eigenvector matrix S is then upper triangular.

18 $A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 6 & -11 & 6 \end{bmatrix}$. Notice that $6 = 1 + 2 + 3$ (the sum of eigenvalues) and determinant is $6 = 1 \cdot 2 \cdot 3$ (the product of eigenvalues).

10 Sample Test Questions Without Solutions

Problem 19. True or false? (With a counterexample when false.) Let A and B be square matrices.

(a) $(AB)^2 = (BA)^2$

(b) If columns 1 and 3 of B are the same, then so are the columns 1 and 3 of AB.

(c) If rows 1 and 3 of B are the same, so are rows 1 and 3 of AB.

Problem 20. Find the product LU if (missing entries are all zeros)

$$L = \begin{bmatrix} 1 & & & \\ -1 & 1 & & \\ & -1 & 1 & \\ & & -1 & 1 \end{bmatrix} \qquad U = \begin{bmatrix} 1 & -1 & & \\ & 1 & -1 & \\ & & 1 & -1 \\ & & & 1 \end{bmatrix}.$$

Is $LU = UL$?

Problem 21. There are 16 2-by-2 matrices $\begin{bmatrix} * & * \\ * & * \end{bmatrix}$ whose $*$ entries are -1s or 1s. How many of them are invertible?

Problem 22. Describe the column spaces of the following matrices:

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 0 & 0 \end{bmatrix} \quad \text{and} \quad C = \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 0 & 0 \end{bmatrix}.$$

Clearly indicate whether the column space is a line, a plane, or the $\mathbb{R}^3$ space.

Problem 23. Write the following problem in matrix form and solve it: "X is twice as old as Y and their ages add to 39."

Problem 24. By trying out $n = 2$ and $n = 3$ predict A^n for these matrices:

$$A_1 = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \quad A_2 = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix} \quad A_3 = \begin{bmatrix} a & b \\ 0 & 0 \end{bmatrix}$$

Problem 25. If the three solutions in Problem 7 are $\boldsymbol{x}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, $\boldsymbol{x}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$, and $\boldsymbol{x}_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$, solve $A\boldsymbol{x} = \boldsymbol{b}$ when $\boldsymbol{b} = \begin{bmatrix} 3 \\ 5 \\ 8 \end{bmatrix}$.

Problem 26. Find three numbers a such that $A = \begin{bmatrix} 2 & a & a \\ a & a & a \\ 8 & 7 & a \end{bmatrix}$ is not invertible.

Problem 27. Consider the $A\boldsymbol{x} = \boldsymbol{b}$ system $\begin{cases} 2x_1 + 4x_2 + 6x_3 + 4x_4 = b_1 \\ 2x_1 + 5x_2 + 7x_3 + 6x_4 = b_2 \\ 2x_1 + 3x_2 + 5x_3 + 2x_4 = b_3 \end{cases}$.

(a) Reduce $[\,A\ \boldsymbol{b}\,]$ to $[\,U\ \boldsymbol{c}\,]$ so that $A\boldsymbol{x} = \boldsymbol{b}$ becomes a triangular system $U\boldsymbol{x} = \boldsymbol{c}$.

(b) Find the conditions on b_1, b_2, b_3 for $A\boldsymbol{x} = \boldsymbol{b}$ to have a solution.

(c) Describe the column space of A. Is it a line or is it a plane. Is it in $\mathbb{R}^3$?

(d) Describe the nullspace of A by writing down the special solutions. Is the nullspace in $\mathbb{R}^4$?

(e) Reduce $[\,U\ \boldsymbol{c}\,]$ to $[\,R\ \boldsymbol{d}\,]$ and determine the special solutions and the particular solution.

(f) Find a particular solution to $A\boldsymbol{x} = \begin{bmatrix} 4 \\ 3 \\ 5 \end{bmatrix}$ and then the complete solution.

Problem 28. Compute the eigenvalues and eigenvectors of A and represent it as a product $S\Lambda S^{-1}$ where Λ is a diagonal eigenvalue matrix and S the corresponding eigenvector matrix. Using that representation compute A^{2166}, showing your steps.

$$\text{(a)}\quad A = \begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \qquad \text{(b)}\quad A = \begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix} \qquad \text{(c)}\quad A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$$

Problem 29. When $a + b = c + d$ show that $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ is an eigenvector of $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and find both eigenvalues.

Problem 30. Let $\begin{bmatrix} .4 & .2 \\ .6 & .8 \end{bmatrix}$.

(a) Find the eigenvalues and eigenvectors of A.

(b) What is A^{2166} approximately?

Problem 31. Find the conditions on a and b that make the matrix A invertible. Then find A^{-1} when it exists.

$$A = \begin{bmatrix} a & b & b \\ a & a & b \\ a & a & a \end{bmatrix}.$$

Problem 32. What are the eigenvalues[1] of this 3 by 3 rank 1 matrix A?

$$A = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}$$

Hint 1: Used the shortcuts discussed in class (you shouldn't need a calculator here or do too many calculations).

Hint 2: What must be one of the eigenvalues?

Hint 3: What must be the geometric multiplicity of that eigenvalue? How about its algebraic multiplicity?

Hint 4: We discussed this problem in class on Friday, December 8. Please see your notes.

[1] If you do not see the answer just by looking at the matrix and using the shortcuts, there is no need to worry: you can always get the correct answer by doing the usual calculations.

Part IV

Appendices

A Vectors & Their Span

An m-by-1 matrix is an m-tall column of numbers. Example: $\boldsymbol{u} = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$ is a 2-tall column. We call such columns vectors. When columns are 2-tall or 3-tall, you can picture them as arrows on a plane (in $\mathbb{R}^2$) or in a 3-D space (in $\mathbb{R}^3$).

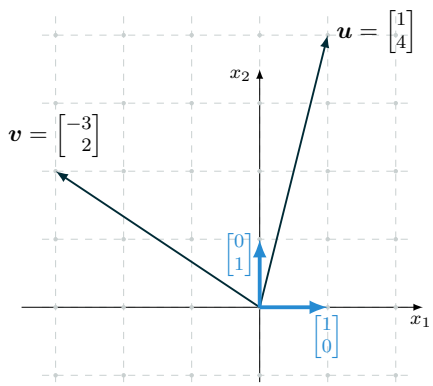


Figure A.1: Vectors that live in the $\mathbb{R}^2$ plane have two components.

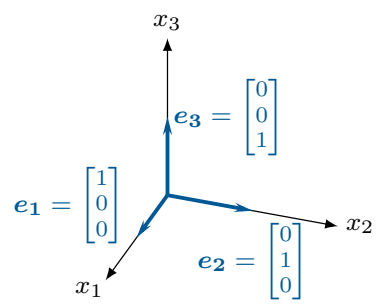


Figure A.2: Vectors that live in the $\mathbb{R}^3$ space have three components.

A.1 Basic Algebra on Vectors

Of all the ways you can manipulate vectors, one stands out: the *linear combination* of vectors.

Definition The *linear combination* of vectors $\boldsymbol{u}$ and $\boldsymbol{v}$ is any vector of the form

$$c \cdot \boldsymbol{u} + d \cdot \boldsymbol{v}$$

where c and d are arbitrary numbers.

The linear combination $c \cdot \boldsymbol{u} + d \cdot \boldsymbol{v}$ is also called the *span* of vectors $\boldsymbol{u}$ and $\boldsymbol{v}$.

To understand what this definition says, we need to know about two operations:

(1) multiplying a vector by a number (also known as "scaling" a vector), and

(2) adding two vectors to one another.

- ◇ An example of (1): you have the vector $\boldsymbol{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and the number $c = 3$.

  What is the vector $c \cdot \boldsymbol{u} = 2 \cdot \begin{bmatrix} 1 \\ 2 \end{bmatrix}$? Can you draw it?

- ◇ An example of (2): you have the vector $\boldsymbol{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and the vector $\boldsymbol{v} = \begin{bmatrix} 3 \\ -4 \end{bmatrix}$.

  What is the vector $\boldsymbol{u} + \boldsymbol{v} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} 3 \\ -4 \end{bmatrix}$? Can you draw it?

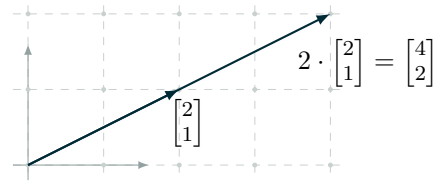
The good news is that when your vectors are 2-tall (plane $\mathbb{R}^2$ vectors) or 3-tall (space $\mathbb{R}^3$ vectors), then you can usually picture the results of these operations in your mind's eye and draw them.

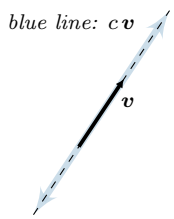
Scaling Vectors

To multiply a vector by a number c is to multiply each component of the vector by c. This amounts to stretching the vector c times. Example:

$$\text{if } \boldsymbol{u} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \text{ and } c = 2, \text{ then}$$

$$c \cdot \boldsymbol{u} = 2 \cdot \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \cdot 2 \\ 2 \cdot 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$$





If c is negative, then the vector reverses direction when multiplied by c. If $|c| < 1$, the the vector actually shrinks instead of stretching c times. And if $c = 0$, then the vector gets crushed into the zero vector: $0 \cdot \boldsymbol{u} = \mathbf{0}$.

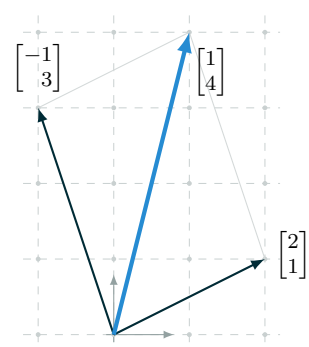
In short, all multiples $c\boldsymbol{v}$ of the vector $\boldsymbol{v}$ lie on the same line as the vector $\boldsymbol{v}$. So every vector on the blue line in this picture is some multiple c of the vector $\boldsymbol{v}$.

Adding Vectors

To add one vector to another, add them component-wise like this:

$$\text{if } \boldsymbol{u} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \ \boldsymbol{v} = \begin{bmatrix} -1 \\ 3 \end{bmatrix}, \text{ then}$$

$$\boldsymbol{u} + \boldsymbol{v} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} + \begin{bmatrix} -1 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 - 1 \\ 1 + 3 \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$$



One way to visualize this add-$\boldsymbol{u}$-and-$\boldsymbol{v}$ operation is to draw $\boldsymbol{u}$ and $\boldsymbol{v}$ and a parallelogram whose two sides are $\boldsymbol{u}$ and $\boldsymbol{v}$. Three corners of this parallelogram are the origin and the two tips of $\boldsymbol{u}$ and $\boldsymbol{v}$. Now draw a vector from the origin to the fourth corner of the parallelogram – this vector is the sum of $\boldsymbol{u}$ and $\boldsymbol{v}$.

A.2 The Span Vectors

If you have just one vector in $\mathbb{R}^n$, its span is all the multiples of that vector – a line, as in Figure A.3. If you have two vectors $\boldsymbol{v}_1$ and $\boldsymbol{v}_2$ somewhere in the 3-dimensional space, as in Figure A.4, any linear combination of them will produce a point on the plane containing $\boldsymbol{v}_1$ and $\boldsymbol{v}_2$. Those are the only points you can reach: you won't be able to reach any point that is above or below that plane.

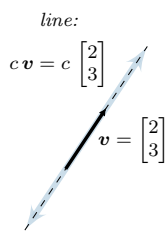


Figure A.3: The span of $\boldsymbol{v}$ consists of all the multiples of $\boldsymbol{v}$; those multiples lie on the blue line.

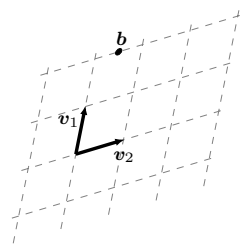


Figure A.4: The span of $\boldsymbol{v}_1$ and $\boldsymbol{v}_2$ forms a plane; for example, $\boldsymbol{b} = 2\boldsymbol{v}_1 + \frac{1}{2}\boldsymbol{v}_2$.

Similarly, a span of 3 vectors can be something that is at most 3-dimensional. But if all three vectors lie on a plane, any linear combination of them will lie on the plane.

Thus, the vectors $\boldsymbol{e}_1 = \begin{bmatrix}1\\0\end{bmatrix}$ and $\boldsymbol{e}_2 = \begin{bmatrix}0\\1\end{bmatrix}$ that live in $\mathbb{R}^2$ can at most span the $\mathbb{R}^2$ plane, and they do: you can get *any* vector $\boldsymbol{v} = \begin{bmatrix}a\\b\end{bmatrix}$ as a linear combination of $\boldsymbol{e}_1$ and $\boldsymbol{e}_2$ just by adding up their scalar multiples $a\boldsymbol{e}_1 + b\boldsymbol{e}_2$. For example, $\boldsymbol{v} = \begin{bmatrix}21\\66\end{bmatrix} = 21\begin{bmatrix}1\\0\end{bmatrix} + 66\begin{bmatrix}0\\1\end{bmatrix} = 21\boldsymbol{e}_1 + 66\boldsymbol{e}_2$. So

$$\text{Span}\left\{\begin{bmatrix}1\\0\end{bmatrix}, \begin{bmatrix}0\\1\end{bmatrix}\right\} \text{ is all of } \mathbb{R}^2. \tag{A.1}$$

Example 42. Can you get any vector in $\mathbb{R}^2$ by considering all linear combinations of

$$\boldsymbol{e} = \begin{bmatrix}1\\0\end{bmatrix} \text{ and } \boldsymbol{f} = \begin{bmatrix}2\\0\end{bmatrix}?$$

No, this would not be possible because $\boldsymbol{e}$ and $\boldsymbol{f}$ lie on the same line in $\mathbb{R}^2$ and their linear combinations also live on the same line. Taking linear combinations of $\boldsymbol{e}$ and $\boldsymbol{f}$ will only produce vectors whose second coordinate is 0 — there is no way to get the vector $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$, for example.

$\boldsymbol{e}$ and $\boldsymbol{f}$ span a 1-dimensional sub-space — a line — inside $\mathbb{R}^2$.

A.3 Exercises

Exercise A.1. It may be easiest to draw this on graph paper.

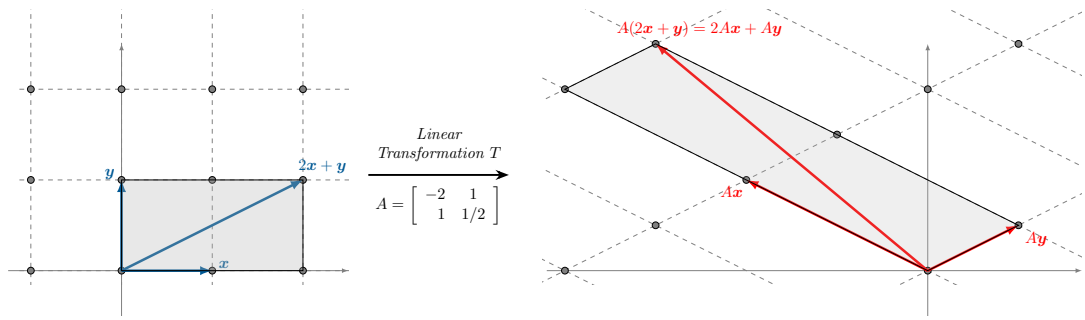
(a) draw the set of all scalar multiples of the vector $\boldsymbol{u} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ as points in $\mathbb{R}^2$ and indicate points that are integer multiples of the vector $\boldsymbol{u}$.

(b) draw the set of all scalar multiples of the vector $\boldsymbol{v} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ as points in $\mathbb{R}^2$ and indicate points that are integer multiples of the vector $\boldsymbol{v}$.

Exercise A.2. Consider the graph you drew in Exercise A.1. Now mark points on your graph that are linear combinations $c\boldsymbol{u} + d\boldsymbol{v}$, where c and d are integers. From your picture, can you estimate what linear combination of $\boldsymbol{u}$ and $\boldsymbol{v}$ would produce the point $(4, 0)$? Does the set of all linear combinations of the vectors $\boldsymbol{u}$ and $\boldsymbol{v}$ cover the entire plane?

B Linear Transformations

The word transformation conveys the idea that a whole set of inputs is simultaneously changed to a bunch of outputs. Here's an example of a *linear transformation* called T:



This transformation is represented by the matrix $A = \begin{bmatrix} -2 & 1 \\ 1 & 1/2 \end{bmatrix}$ that sends the blue vectors in $\mathbb{R}^2$ to the red vectors, also in $\mathbb{R}^2$. For example,

$$A\boldsymbol{x} = \begin{bmatrix} -2 & 1 \\ 1 & .5 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \end{bmatrix} \quad \text{and} \quad A\boldsymbol{y} = \begin{bmatrix} -2 & 1 \\ 1 & .5 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ .5 \end{bmatrix}.$$

Notice that

$$A(2\boldsymbol{x} + \boldsymbol{y}) = \begin{bmatrix} -2 & 1 \\ 1 & .5 \end{bmatrix} \left(2 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right) = \begin{bmatrix} -2 & 1 \\ 1 & .5 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} -3 \\ 2.5 \end{bmatrix},$$

while at the same time,

$$2A\boldsymbol{x} + A\boldsymbol{y} = 2 \cdot \begin{bmatrix} -2 & 1 \\ 1 & .5 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} -2 & 1 \\ 1 & .5 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 2 \begin{bmatrix} -2 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ .5 \end{bmatrix} = \begin{bmatrix} -3 \\ 2.5 \end{bmatrix}.$$

That is, $A(2\boldsymbol{x}+\boldsymbol{y}) = 2A\boldsymbol{x}+A\boldsymbol{y}$. Generally, any matrix A has these properties:

- When A is applied to the vector $\boldsymbol{x}+\boldsymbol{y}$, out comes the vector $A\boldsymbol{x}+A\boldsymbol{y}$. Thus, it doesn't matter whether you sum the vectors first and then apply A to them, or apply A first to each vector individually and then sum up the results. Either way, $A(\boldsymbol{x}+\boldsymbol{y}) = A\boldsymbol{x}+A\boldsymbol{y}$.
- When A is applied to the vector $2\boldsymbol{x}$, out comes the vector $2A\boldsymbol{x}$ Thus, it doesn't matter whether you scale the vector first and then apply A to it, or apply A first and then scale the result. Either way, $A(2\boldsymbol{x}) = 2A\boldsymbol{x}$.

Definition A mapping T from $\mathbb{R}^n$ to $\mathbb{R}^m$ is called a linear transformation if for any vectors $\boldsymbol{x}$ and $\boldsymbol{y}$ in $\mathbb{R}^n$ and any number a the following holds:

- $T(\boldsymbol{x}+\boldsymbol{y}) = T(\boldsymbol{x})+T(\boldsymbol{y})$,
- $T(a\boldsymbol{x}) = aT(\boldsymbol{x})$.

These two properties together can be combined into one: writing $T(a\boldsymbol{x}+b\boldsymbol{y}) = aT(\boldsymbol{x})+bT(\boldsymbol{y})$ is equivalent to writing the two properties separately[1].

Clearly, matrix multiplication is an example of a linear transformation: every m-by-n matrix A meets this definition because $A(a\boldsymbol{x}+b\boldsymbol{y}) = aA\boldsymbol{x}+bA\boldsymbol{y}$ whenever $\boldsymbol{x}$ and $\boldsymbol{y}$ are n-tall vectors and a and b are any numbers.

The amazing fact is that there are really no other examples of linear transformations – matrix multiplication is the only kind there is: every linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$ can be represented by an m-by-n matrix.

Key Point Every matrix is a linear transformation, and every linear transformation can be represented by some matrix.

It's good to do Exercise B.1 now.

[1] Just let $a = b = 1$ in $T(a\boldsymbol{x}+b\boldsymbol{y}) = aT(\boldsymbol{x})+bT(\boldsymbol{y})$ to get the first property; let $b = 0$ to get the second.

Key Facts If T is a linear transformation, then

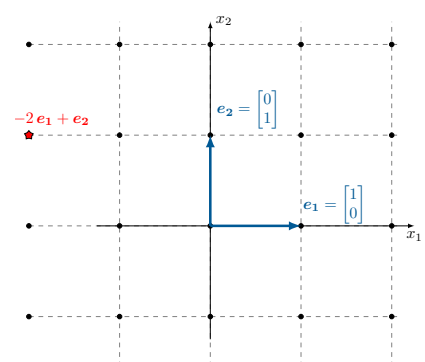
T takes the zero vector to the zero vector

- T takes lines to lines
- T takes parallel lines to parallel lines
- T takes triangles to triangles

B.1 The Standard Basis

Recall that a basis for $\mathbb{R}^n$ is a set of linearly independent vectors that span all of $\mathbb{R}^n$. Any vector in $\mathbb{R}^n$ can be written in terms of basis vectors in exactly one way.

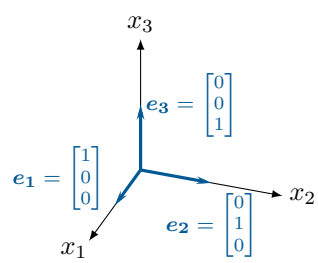
The *standard* basis vectors in $\mathbb{R}^n$ are usually denoted by $\boldsymbol{e_1}, \boldsymbol{e_2}, \ldots \boldsymbol{e_n}$. The standard basis vectors for $\mathbb{R}^2$ and $\mathbb{R}^3$ are shown in these pictures.



Any vector in $\mathbb{R}^2$ can be expressed in terms of the standard basis vectors $\boldsymbol{e_1}$ and $\boldsymbol{e_2}$. For example, the red star is at the tip of the vector $\boldsymbol{v} = \begin{bmatrix}-2\\1\end{bmatrix}$ and

$$\boldsymbol{v} = \begin{bmatrix}-2\\1\end{bmatrix} = (-2)\begin{bmatrix}1\\0\end{bmatrix} + \begin{bmatrix}0\\1\end{bmatrix} = (-2)\,\boldsymbol{e_1} + \boldsymbol{e_2}$$

You can also draw the standard basis in $\mathbb{R}^3$.



These three vectors live in $\mathbb{R}^3$ and are linearly independent, making them the basis for $\mathbb{R}^3$. Any vector $\boldsymbol{v}$ in $\mathbb{R}^3$ can be expressed as a linear combination of these vectors in one and only one way.

> **Key Point** The columns of matrix A tell exactly where the standard basis vectors will go under the linear transformation that A represents.

This **Key Point** makes sense when you realize that the product A times $\boldsymbol{e_i}$ equals the i-th column of A. For example, suppose $A = \begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix}$. Then $\boldsymbol{e_1}$ is sent to $A\,\boldsymbol{e_1}$ – A's first column (the red vector on the right); $\boldsymbol{e_2}$ goes to $A\,\boldsymbol{e_2}$, A's second column (the blue vector on the right).

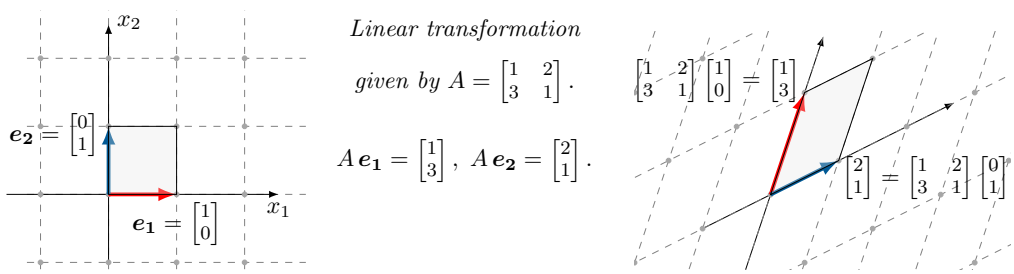


Figure B.1: A sends standard basis vectors to its own columns.

Remark Incidentally, note that the determinant of this A is negative, so the vectors change orientation under the transformation: the red $\boldsymbol{e_1}$ vector is to the right of the blue $\boldsymbol{e_2}$ vector (if you are looking in the same direction the vector points), but when A acts on them, $A\,\boldsymbol{e_1}$ ends up to the left of $A\,\boldsymbol{e_2}$.

B.2 More Examples of Linear Transformations

`//TODO:` after the break

B.3 Exercises

Exercise B.1. Prove that if T is a linear transformation (whose definition is on page 139), then $T(\mathbf{0}) = \mathbf{0}$.

That is, under any linear transformation, the zero vector maps to the zero vector. (Prove this by using the definition alone.)

C Determinants

The determinant of a square matrix A is a single number associated with A. Denoted by $\det(A)$, the determinant can be computed from the entries of A and tells us whether or not A is invertible: If $\det(A) = 0$, then A is *not* invertible, and if $\det(A) \neq 0$, then A is invertible.

The need to compute determinants came up when we needed to find A's eigenvalues by setting A's characteristic polynomial to zero to find its roots. The lecture slides are reproduced in Figure C.1 for your reference.

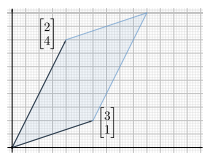
The determinant of a 2x2 matrix

- We define the determinant of the 2x2 matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ to be $\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$
- So, for example, $\det \begin{bmatrix} 2 & 3 \\ 4 & 1 \end{bmatrix} = (2)(1) - (4)(3) = -10$.
- Notice that if you exchange the columns, the sign of the determinant will flip.

$$\det \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix} = (3)(4) - (2)(1) = 10 .$$

- *The determinant tells you the signed area of the parallelogram spanned by the two column vectors of the matrix.*

The sign depends on the order in which you put the vectors into the matrix.



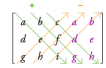
The shaded area is 10 units², which is equal to the determinant of the matrix whose columns are these two vectors.

Q: Think about the row space of *A*. The row vectors span a parallelogram also. Draw them. What is the area those vectors span?

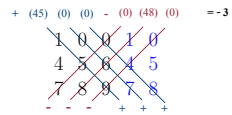
The determinant of a 3x3 matrix

- We define the determinant of the 3x3 matrix $\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$ to be

$$\det \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = (aei + bfg + cdh) - (gec + hfa + idb).$$



- Granted, at first, this is not pretty to look at. But there is a useful mnemonic if you want to remember it. Take the matrix, append to it its first two columns as shown, and look which terms go into the formula with a plus sign and which ones with a minus sign. The terms fall neatly on the lines shown.



$$\det \begin{bmatrix} 1 & 0 & 0 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} = (1 \cdot 5 \cdot 9) + (0 \cdot 6 \cdot 7) + (0 \cdot -4 \cdot 8) - (0 \cdot 5 \cdot 7) - (1 \cdot 6 \cdot 8) - (0 \cdot 4 \cdot 9)$$
$$= 45 + 0 + 0 - 0 - 48 - 0 = -3.$$

The determinant of a 3x3 matrix

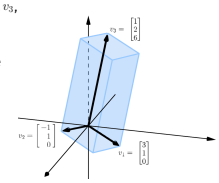
- If you have a square matrix A with columns v_1, v_2, v_3,

$$A = \begin{bmatrix} v_1, v_2, v_3 \end{bmatrix}$$

then the determinant tells you the signed volume of the parallelepiped spanned by those 3 vectors.

- For example, if your matrix A has the columns

$$A = \begin{bmatrix} 3 & -1 & 1 \\ 1 & 1 & 2 \\ 0 & 0 & 6 \end{bmatrix},$$



you can compute its determinant using the formula on the previous slide and get 24:

$$\det \begin{bmatrix} 3 & -1 & 1 \\ 1 & 1 & 2 \\ 0 & 0 & 6 \end{bmatrix} = 24.$$

Computing determinants

- Although there is a neat recursive formula for computing determinants of $n \times n$ matrices, nobody uses it in practice because the time to compute the determinant this way grows on the order of $n!$. Computing the determinant of a 50×50 matrix using that formula takes longer than the lifetime of the universe.

This method takes $O(n^3)$ time.

- A faster way to compute determinants is to appeal to elementary row operations (EROs).
- EROs may change the determinants, but we can keep track of the changes. If we row-reduce our matrix to the row-echelon form, then we'll be able to easily compute the determinant, because it will be the product of the diagonal entries.

Example: The determinant of a triangular matrix is the product of its diagonal entries.

$$\det \begin{bmatrix} -1 & 2 & 3 & 4 \\ 0 & 5 & 11 & \pi \\ 0 & 0 & 7 & e^2 \\ 0 & 0 & 0 & 2 \end{bmatrix} = -70.$$

An upper-triangular matrix.

- If you swap two rows of a matrix, its determinant changes sign.
- If you multiply any row by a constant c, the determinant changes by a factor of c.
- Adding a multiple of one row to another does not change the determinant at all.

This is how the three row operations affect determinants.

Figure C.1: Lecture slides on determinants.